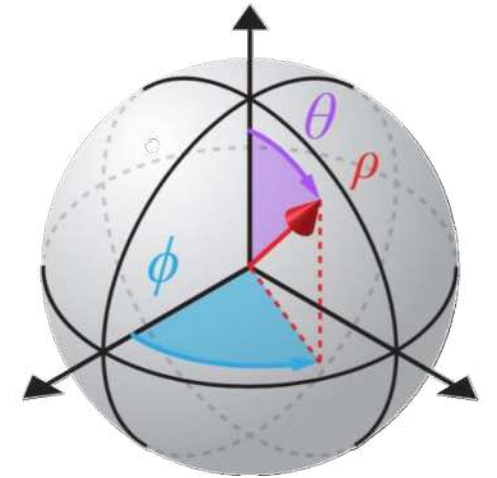
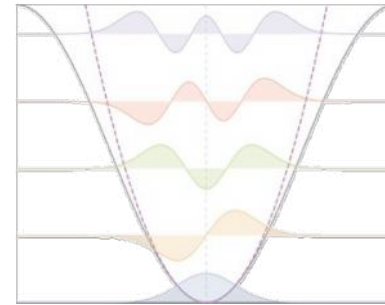
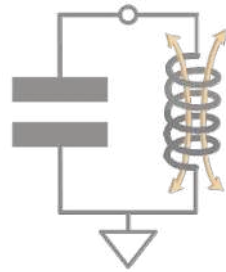
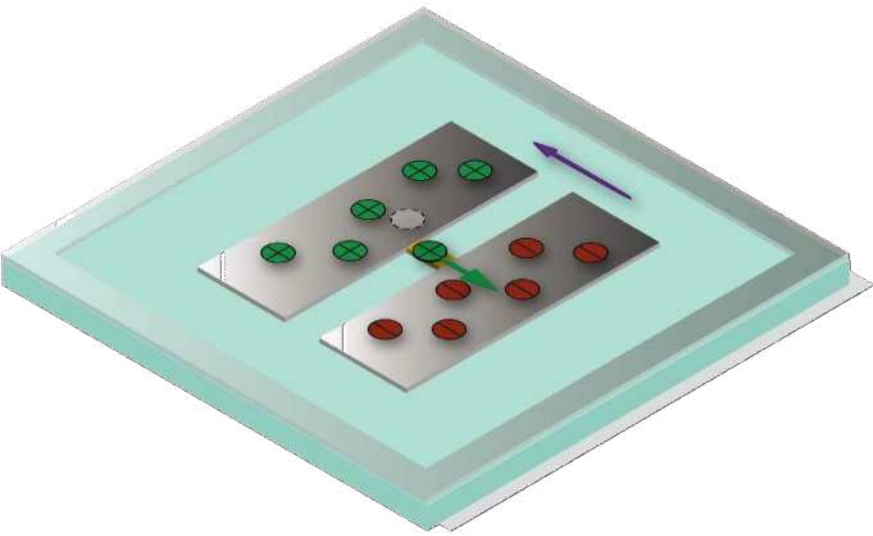


To design quantum hardware

Energy-participation-ratio (EPR)
and Qiskit Metal



Zlatko K. Minev

IBM Quantum

IBM T.J. Watson Research Center, Yorktown Heights, NY



@zlatko_minev



zlatko-minev.com

nature

International journal of science

Letter

Published: 03 June 2019

To catch and reverse a quantum jump mid-flight

Z. K. Minev , S. O. Mundhada, S. Shankar, P. Reinhold, R. Gutiérrez-Jáuregui, R. J. Schoelkopf, M. Mirrahimi, H. J. Carmichael & M. H. Devoret 

Nature **570**, 200–204 (2019) | [Download Citation](#) 



Le Monde

theguardian



The CHRISTIAN SCIENCE
MONITOR

GIZMODO

...

WIRED

physicsworld

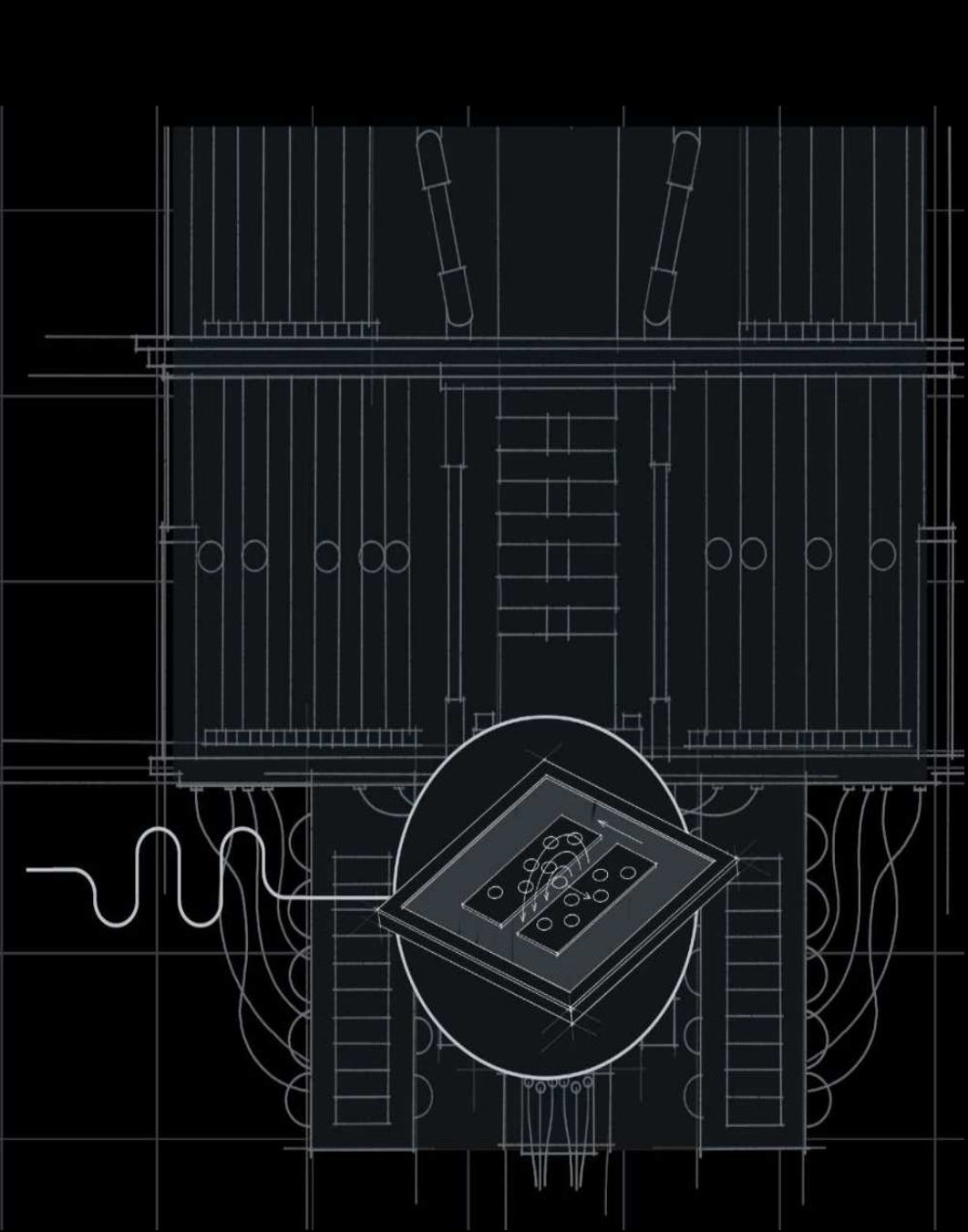
Популярная Механика

Discover

SCIENCE FOR THE CURIOUS



Quanta magazine



Today

Quantum device design overview

Energy-participation ratio (EPR)

Minev et al. arXiv: 1902.10355 & 2010.00620 (Yale)

Qiskit | quantum device
design

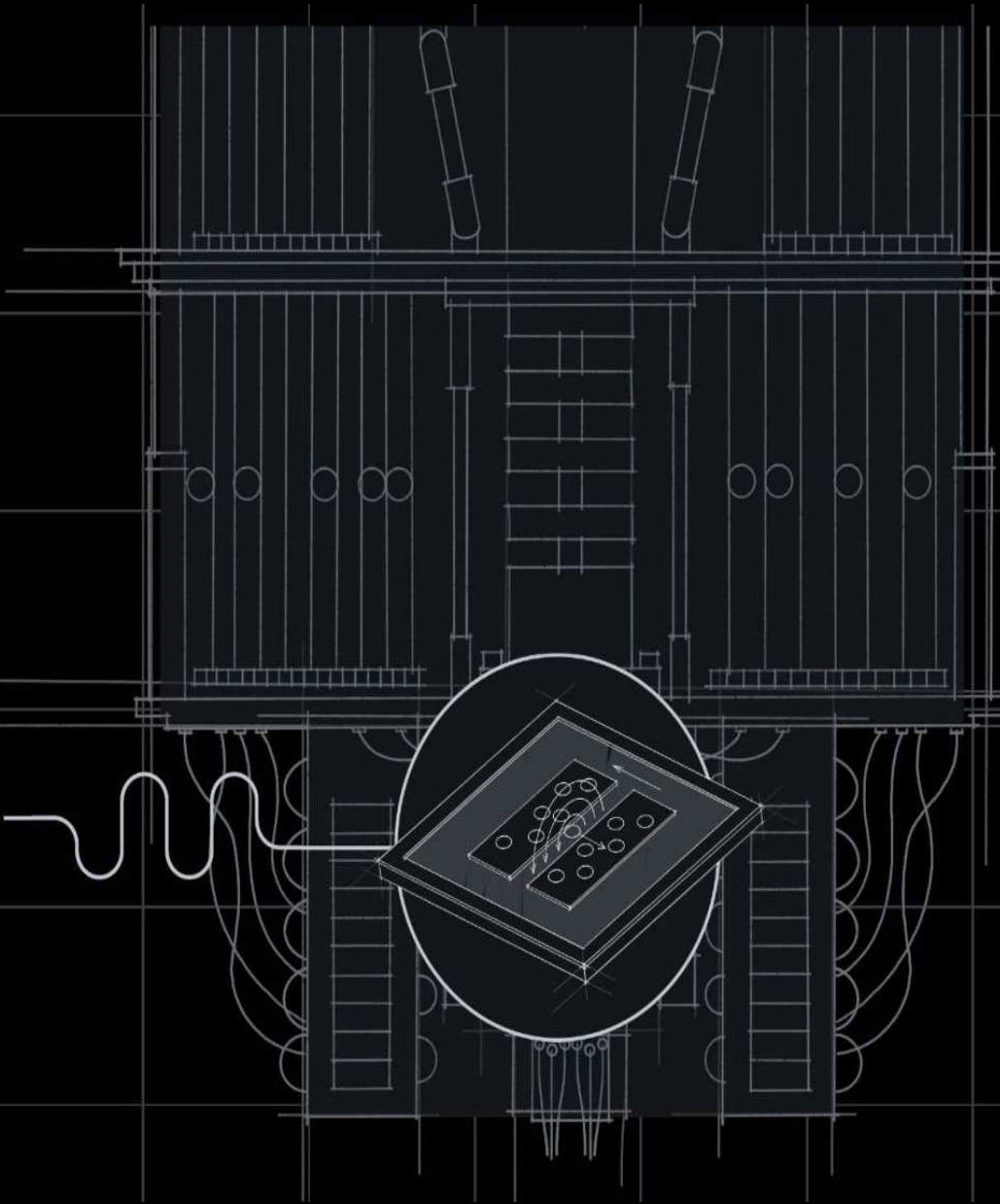
open source Q-EDA and analysis (IBM)

qiskit.org/metal

Modular lumped-oscillator method

Minev, Gambetta et al. (IBM)

Quantum device design in a nutshell



Quantum in the cloud or lab

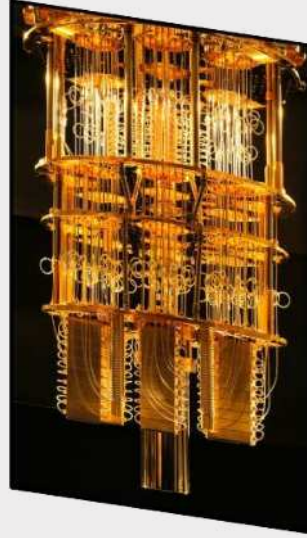
Superconducting qubits



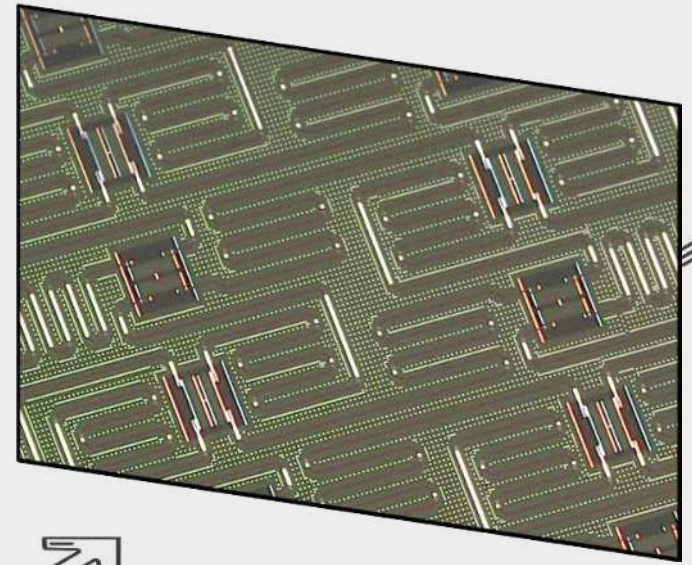
Quantum community



Lab / cloud facility



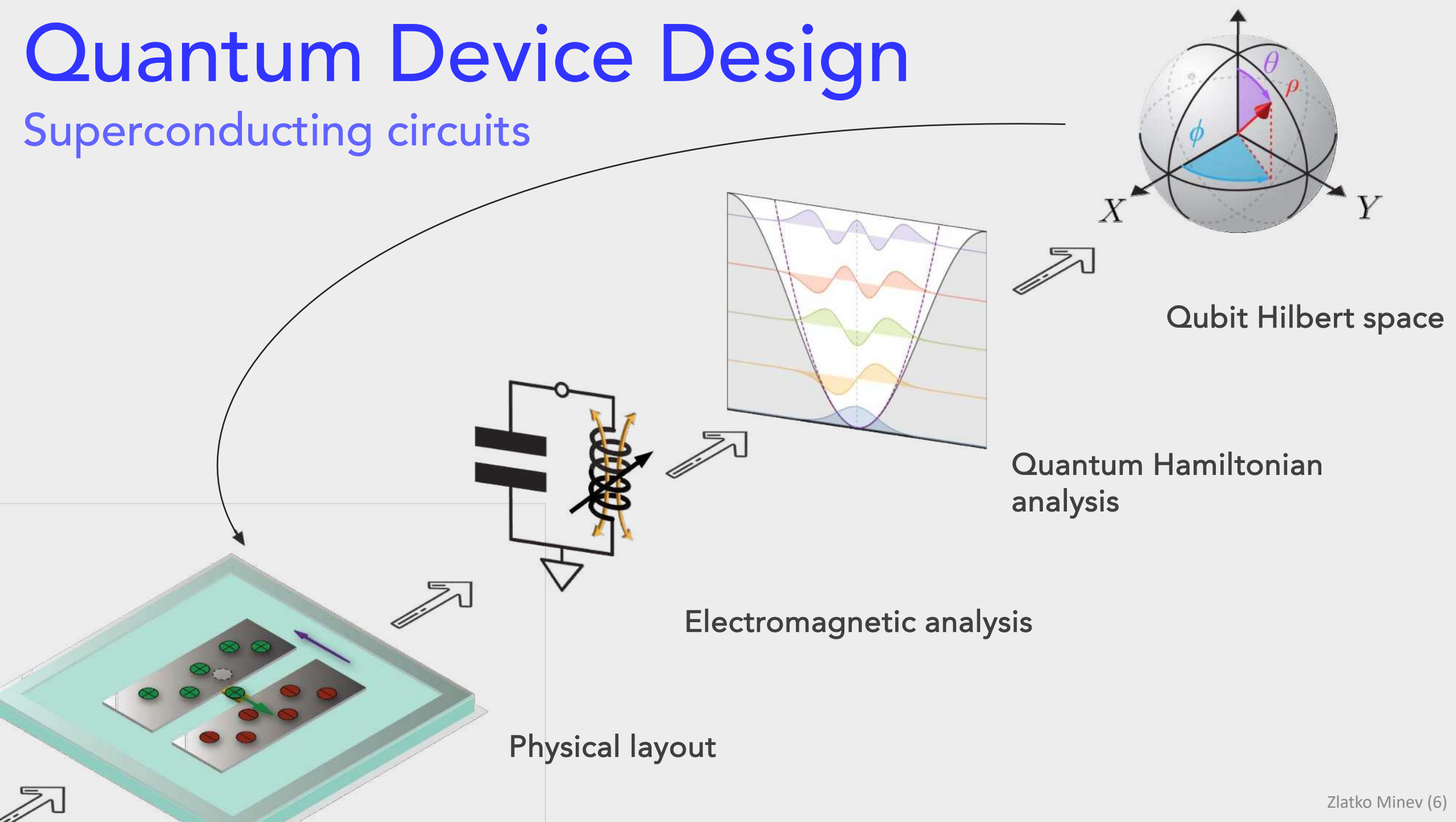
Cryogenic environment

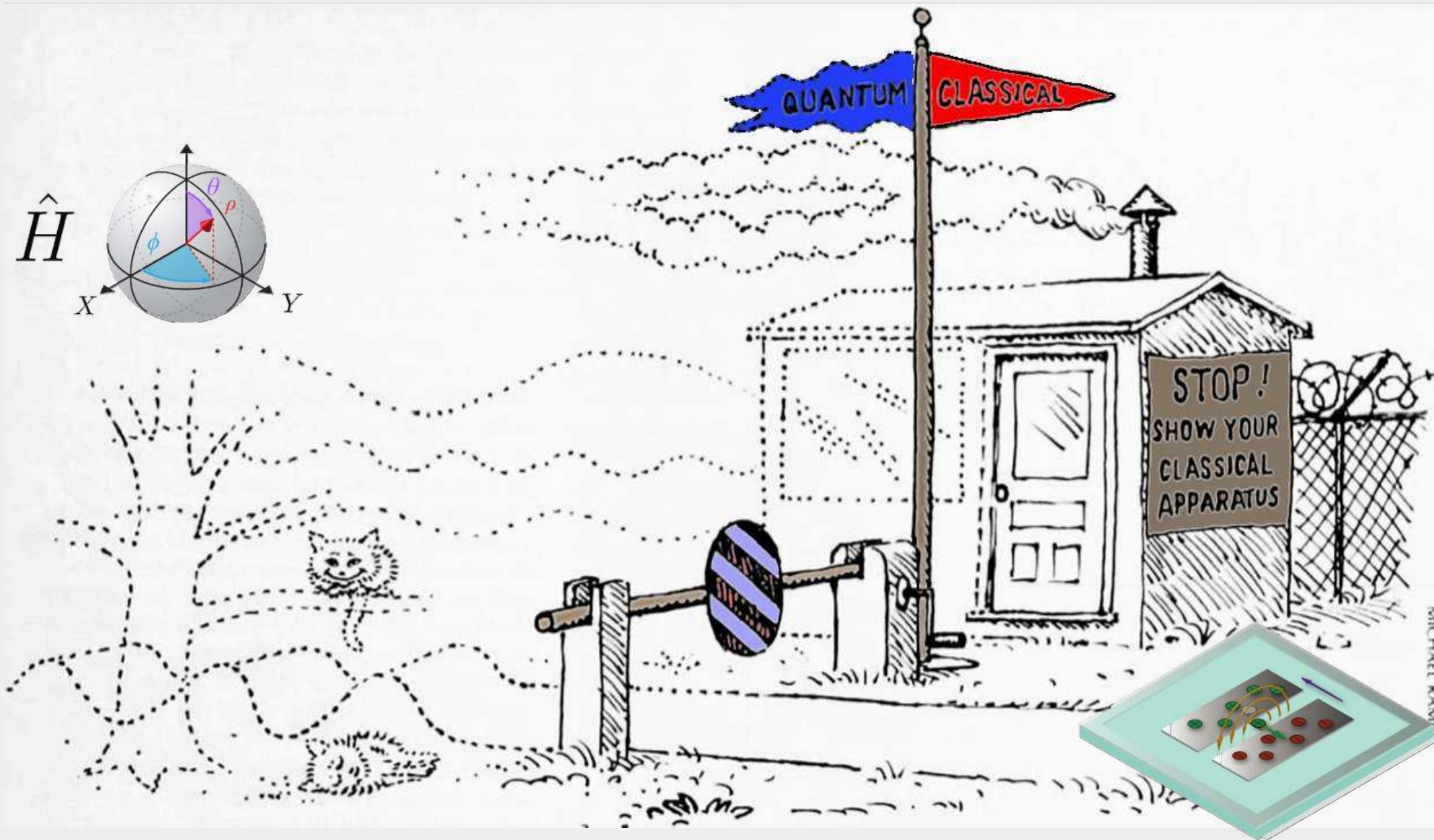


Quantum-device chip

Quantum Device Design

Superconducting circuits





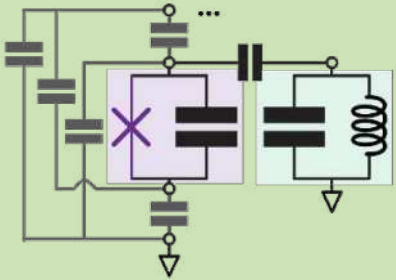
Drawing: Zurek, Physics Today (1991)

Quantization landscape

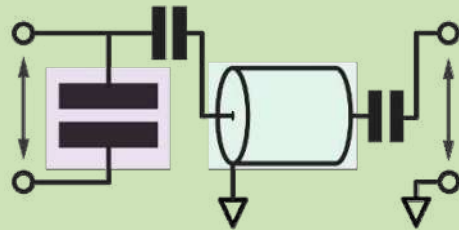
quasi-static

full-wave

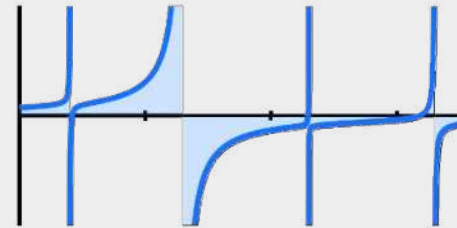
lumped



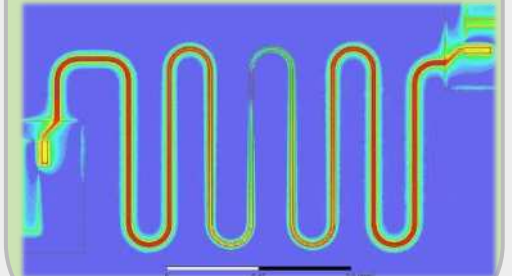
semi-analytic



impedance



eigenmode



Captures more information, complexity, accuracy

Yurke & Denker (1984), Devoret (1997), Burkard *et al.* (2004), Koch *et al.* (2007), Minev *et al.* (2021), ...

Malekakhlagh *et al.* (2017, 2019), Gely *et al.* (2019), Parra-Rodriguez *et al.* (2019), ...

Nigg *et al.* (2012), Bourassa *et al.* (2012), Solgun *et al.* (2014, 2015, 2017) ...

Minev (2019)
Minev *et al.* (2020)

A few introductory reviews

And many more... check online or ask us for specific topic

Mineev, Z. Lec. 16-22, *Introduction to Quantum Computing and Quantum Hardware* (2020) qiskit.org/learn/intro-qc-qh

Blais, A., Grimsmo, A. L., Girvin, S. M., & Wallraff, A. (2020) *Circuit Quantum Electrodynamics* (arXiv:2005.12667)

Kjaergaard, M., Schwartz, ... Oliver, W. D. (2020) *Superconducting Qubits: Current State of Play*
Annual Reviews of Condensed Matter Physics 11, 369-395

Krantz, P., Kjaergaard, M., Yan, F., ... & Oliver, W. D. (2019) A quantum engineer's guide to superconducting qubits
Applied Physics Reviews, 6(2), 021318

Corcoles, A. D., Kandala, A., ... Gambetta, J. M. (2019) Challenges and Opportunities of Near-Term Quantum Computing Systems. *Proceedings of the IEEE*, 1–15.

Wendin, G. (2017) Quantum information processing with superconducting circuits. *RPP*, 80(10), 106001

Gambetta, J. M., Chow, J. M., & Steffen, M. (2017) Building logical qubits in a superconducting quantum computing system. *Npj Quantum Information*, 3(1), 2

Girvin, S. M. (2011) Circuit QED: superconducting qubits coupled to microwave photons. *Quantum machines: measurement and control of engineered quantum systems*, 113, 2.

Clerk, A. A., Girvin, S. M., Marquardt, F., & Schoelkopf, R. J. (2010) Introduction to quantum noise, measurement, and amplification
Reviews of Modern Physics, 82(2), 1155–1208

Clarke, J., & Wilhelm, F. K. (2008) Superconducting quantum bits. *Nature*, 453(7198), 1031–1042

Devoret, M. H. (1997) Quantum Fluctuations in Electrical Circuits.
In *Fluctuations Quantiques/Quantum Fluctuations* (p. 351)

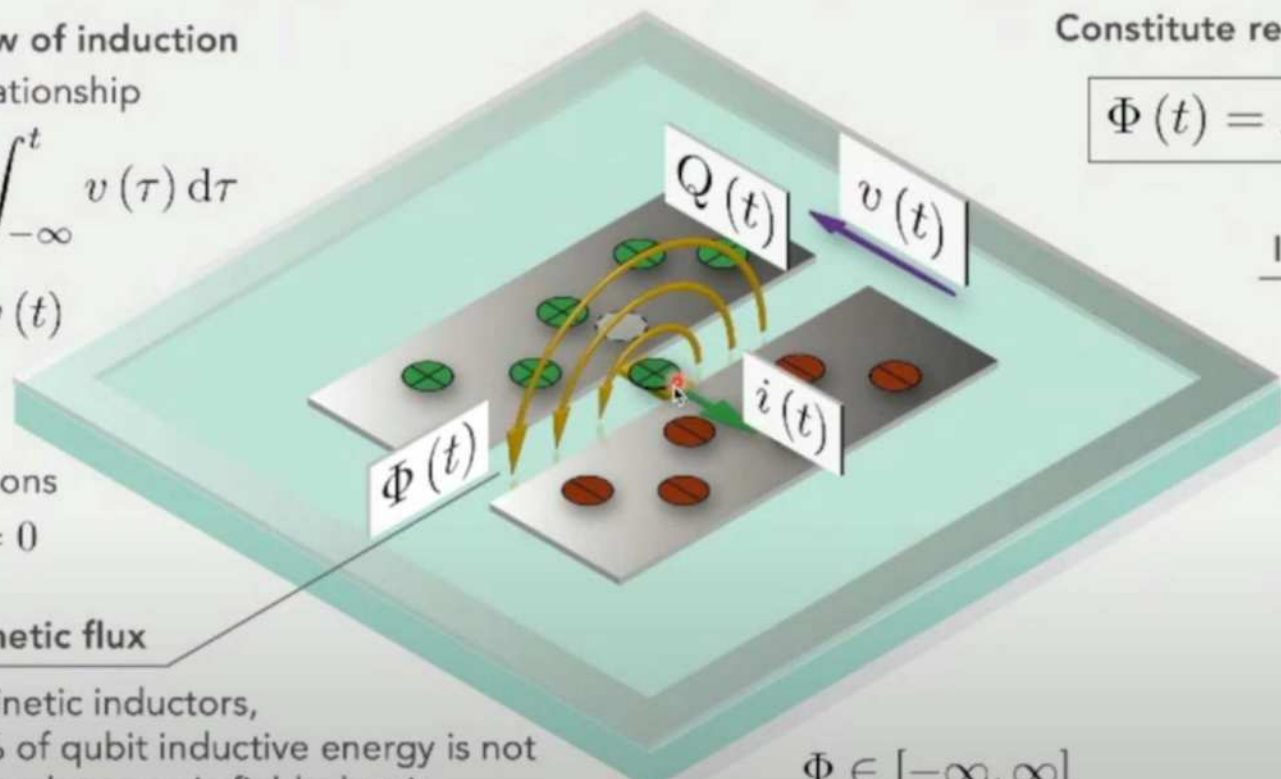
...

Magnetic flux and inductance

Faraday's law of induction
Universal relationship

$$\Phi(t) = \int_{-\infty}^t v(\tau) d\tau$$
$$\frac{d}{dt} \Phi(t) = v(t)$$

Initial conditions
 $\Phi(-\infty) = 0$



Constitute relationship

$\Phi(t) = Li(t)$

Inductance

$L \in (0, \infty)$

Magnetic flux

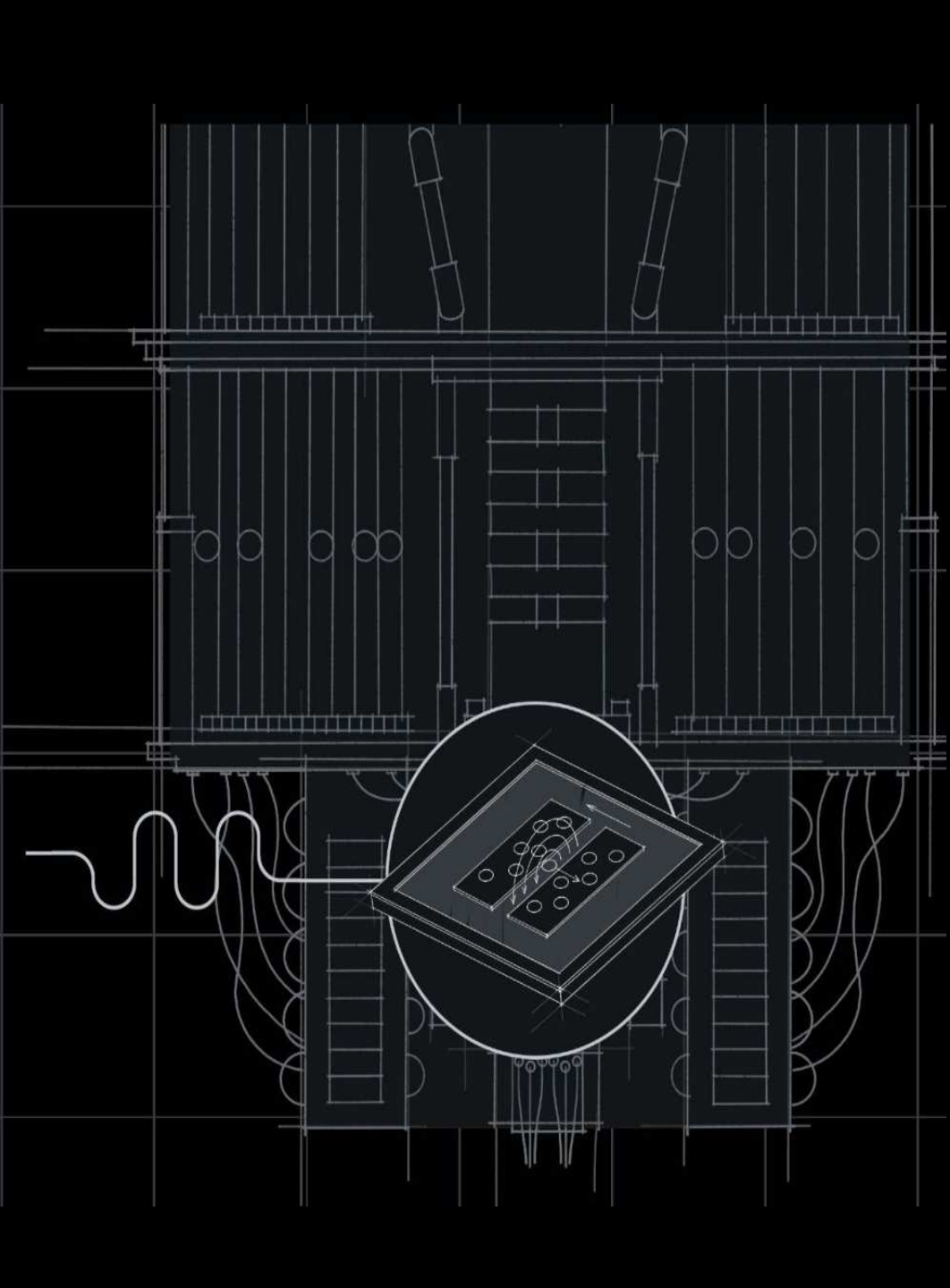
For kinetic inductors,
~98% of qubit inductive energy is not
in stored magnetic fields, but in
kinetic inductance

$\Phi \in [-\infty, \infty]$

Image: Zlatko Minev

which is a superconducting element.



A technical line drawing of a building's structural frame, showing columns and beams. A circular inset provides a magnified view of a specific component, which appears to be a multi-layered electronic or mechanical assembly with internal wiring and components. A wavy line, possibly representing a signal or vibration, enters from the left and connects to the component.

Energy-participation ratio (EPR)

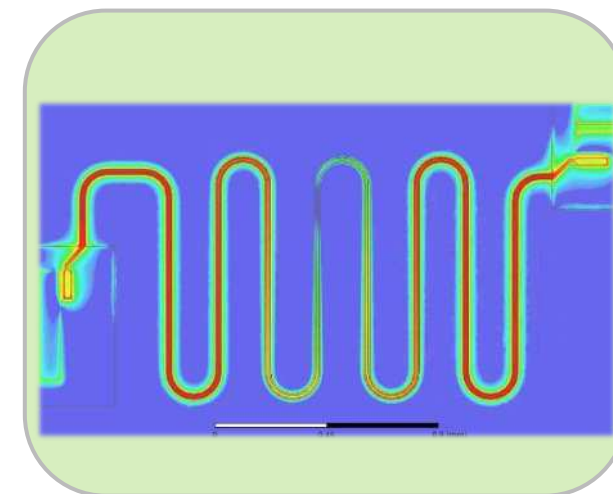
Quantum Physics

[Submitted on 1 Oct 2020]

Energy-participation quantization of Josephson circuits

Zlatko K. Minev, Zaki Leghtas, Shantanu O. Mundhada, Lysander Christakis, Ioan M. Pop, Michel H. Devoret

Superconducting microwave circuits incorporating nonlinear devices, such as Josephson junctions, are one of the leading platforms for emerging quantum technologies. Increasing circuit complexity further requires efficient methods for the calculation and optimization of the spectrum, nonlinear interactions, and dissipation in multi-mode distributed quantum circuits. Here, we present a method based on the energy-participation ratio (EPR) of a dissipative or nonlinear element in an electromagnetic mode. The EPR, a number between zero and one, quantifies how much of the energy of a mode is stored in each element. It obeys universal constraints---valid regardless of the circuit topology and nature of the nonlinear elements. The EPR of the elements are calculated from a unique, efficient electromagnetic eigenmode simulation of the linearized circuit, including lossy elements. Their set is the key input to the determination of the quantum Hamiltonian of the system. The method provides an intuitive and simple-to-use tool to quantize multi-junction circuits. It is especially well-suited for finding the Hamiltonian and dissipative parameters of weakly anharmonic systems, such as transmon qubits coupled to resonators, or Josephson transmission lines. We experimentally tested this method on a variety of Josephson circuits, and demonstrated agreement within several percents for nonlinear couplings and modal Hamiltonian parameters, spanning five-orders of magnitude in energy, across a dozen samples.



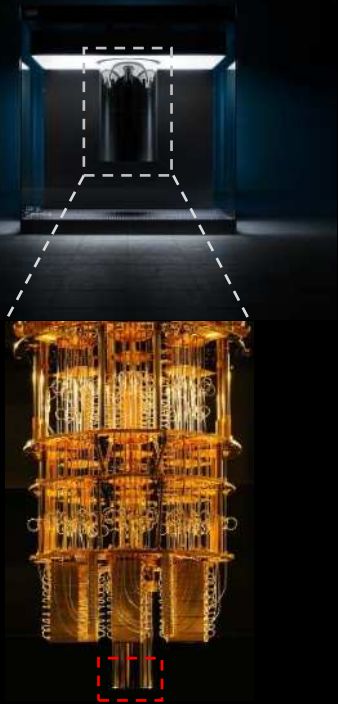
**See also Minev
dissertation Sec. 4.1
(arXiv: 1902.10355)**

Comments: 15 Figures, 39 pages, 4 tables; See also [this http URL](#) Send feedback to zlatko.minev [at] [this http URL](#)

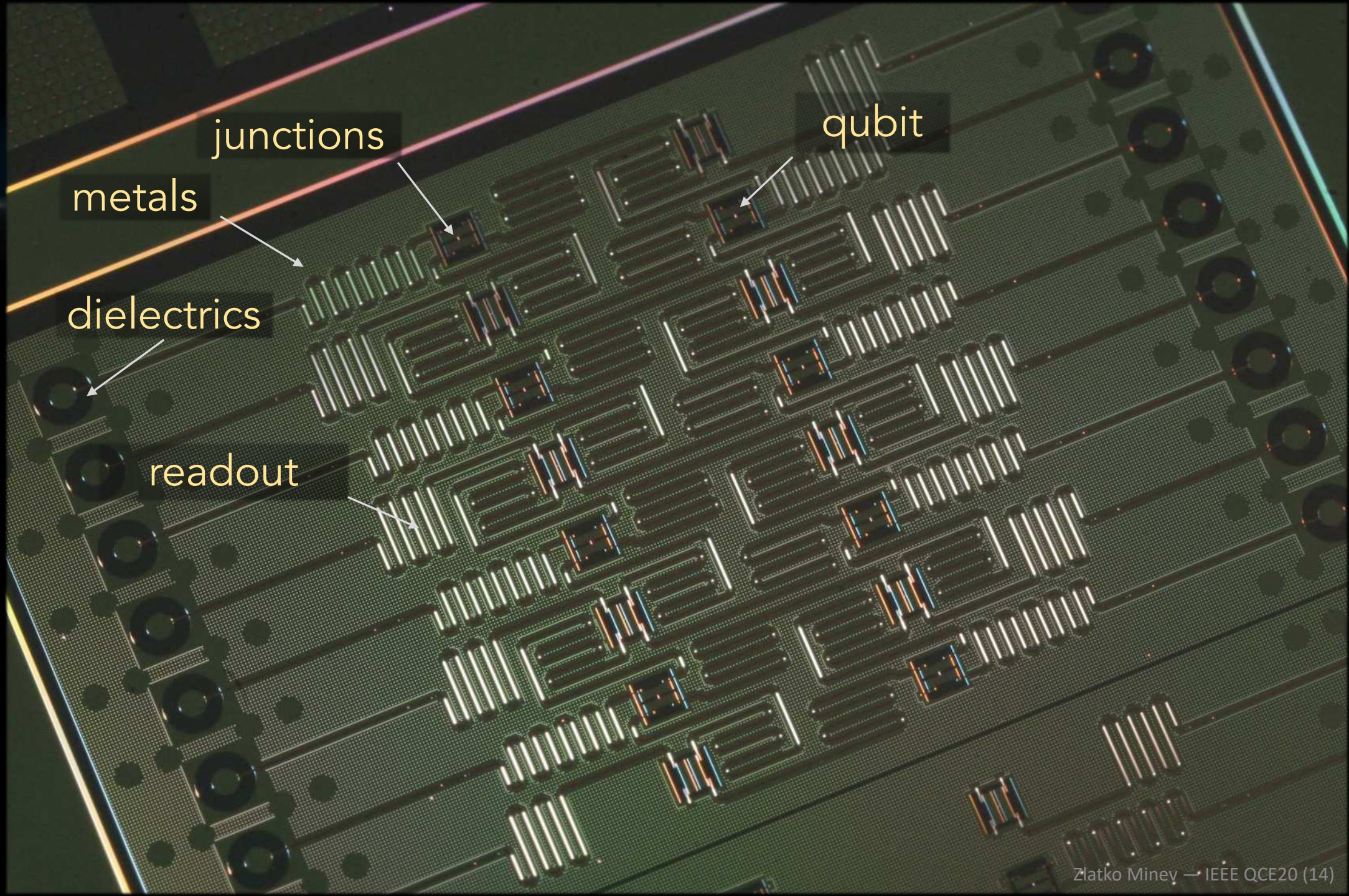
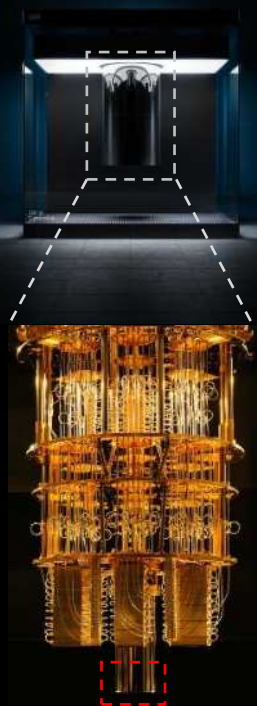
Subjects: **Quantum Physics (quant-ph)**; Superconductivity (cond-mat.supr-con)

Cite as: [arXiv:2010.00620](#) [quant-ph]

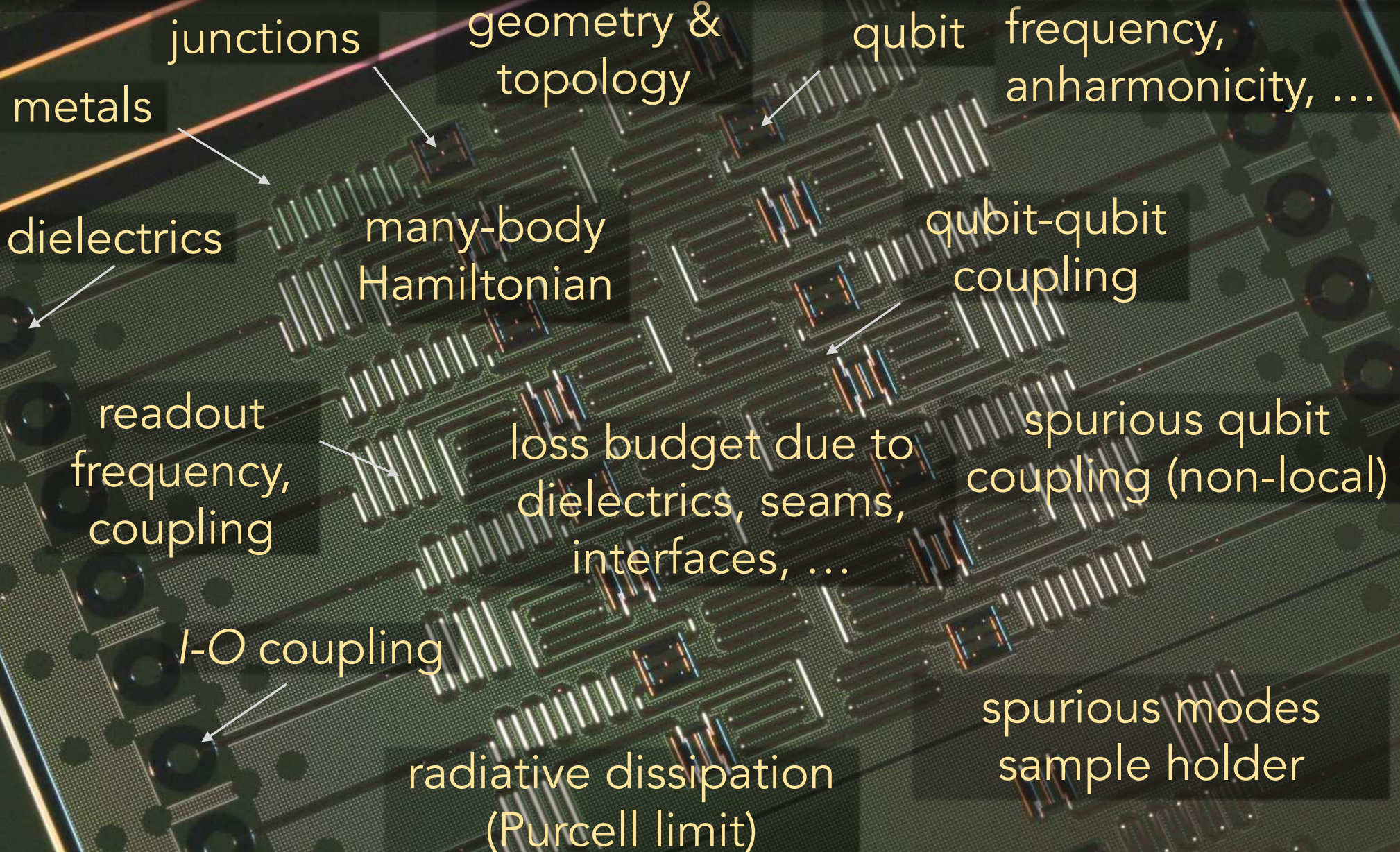
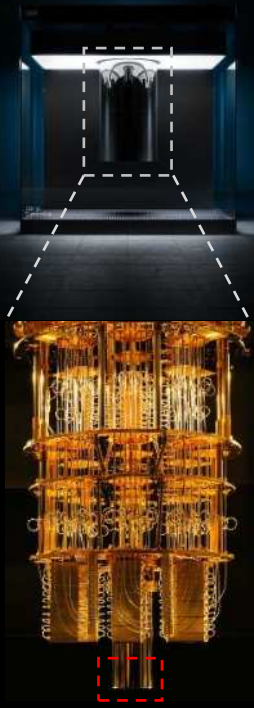
(or [arXiv:2010.00620v1](#) [quant-ph] for this version)



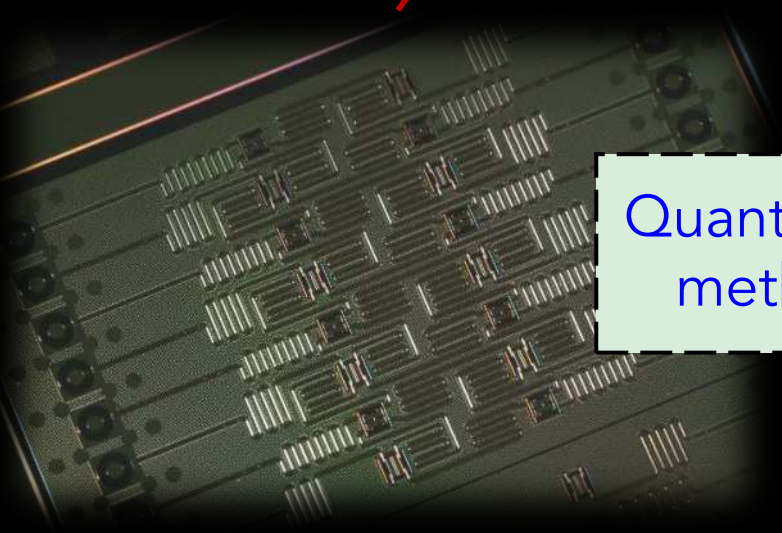
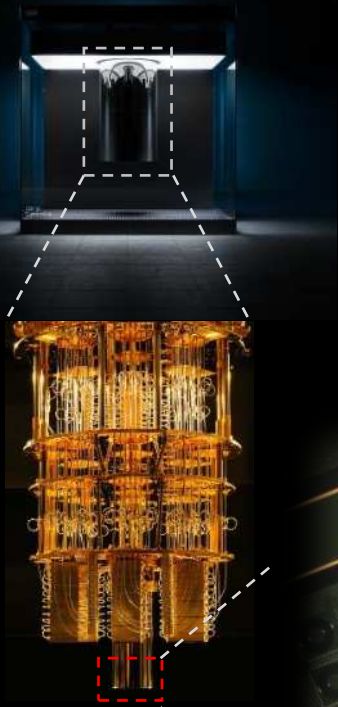
Anatomy of a quantum chip



What needs to be designed?



classical



geometry, circuits, materials,
electromagnetic fields,
...

Quantization
methods

quantum

$\hat{H}_{\text{full}}$

$$= \sum_m (\omega'_m - i\kappa_m) \hat{a}_m^\dagger \hat{a}_m$$

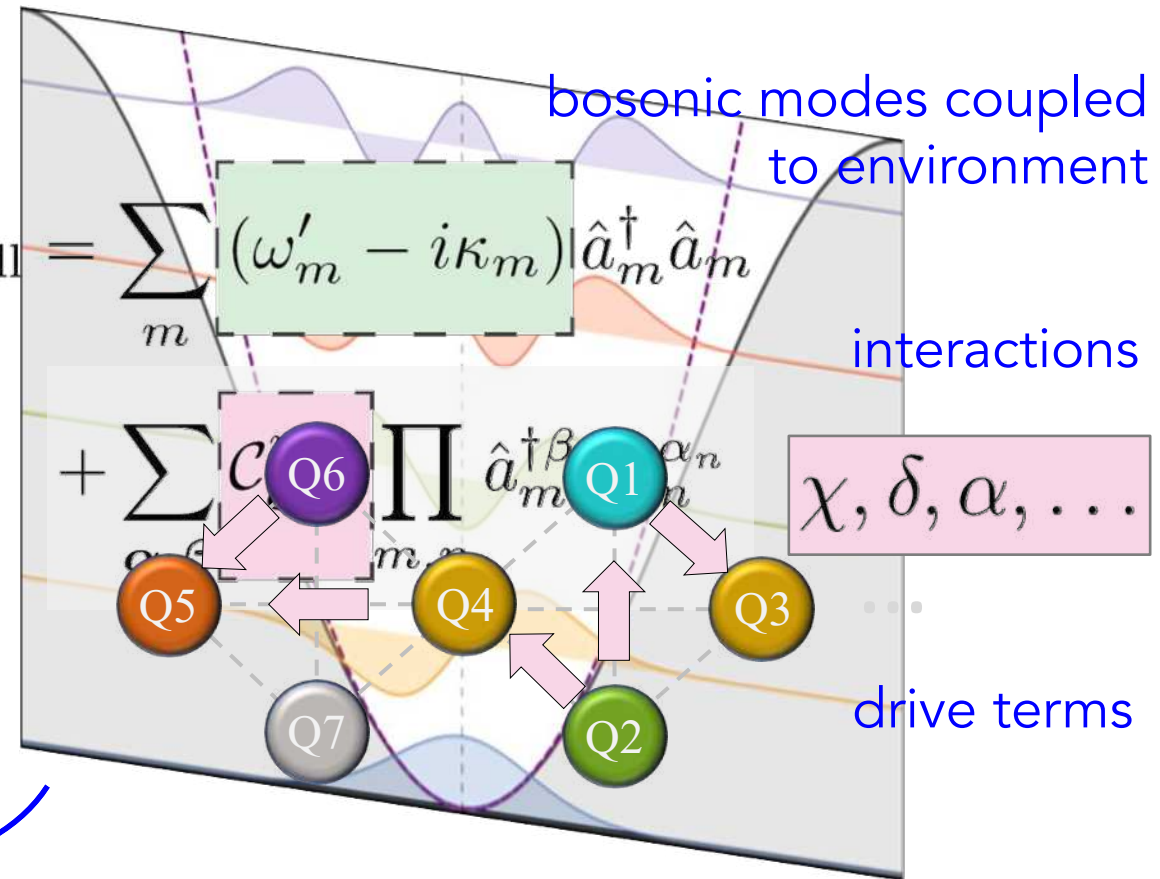
bosonic modes coupled
to environment

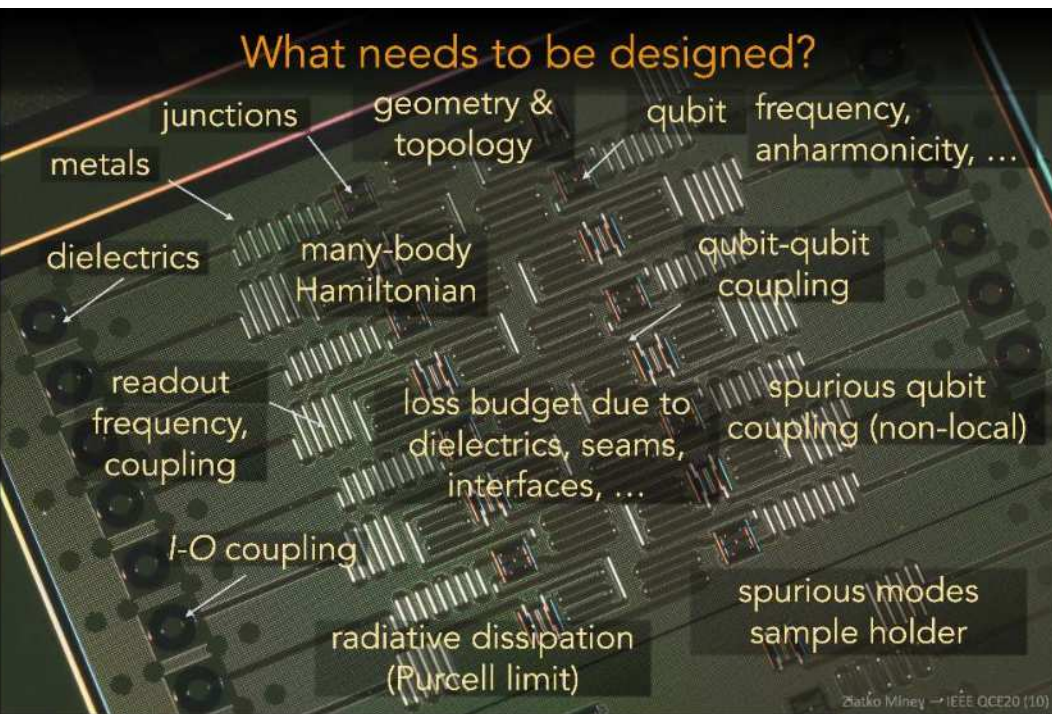
interactions

$$+ \sum_{\alpha, \beta} c_{\alpha\beta} \prod_m \hat{a}_m^{\dagger\beta} \hat{a}_m^{\alpha}$$

$\chi, \delta, \alpha, \dots$

drive terms





The solution of all these questions reduces to:

Where is the energy?

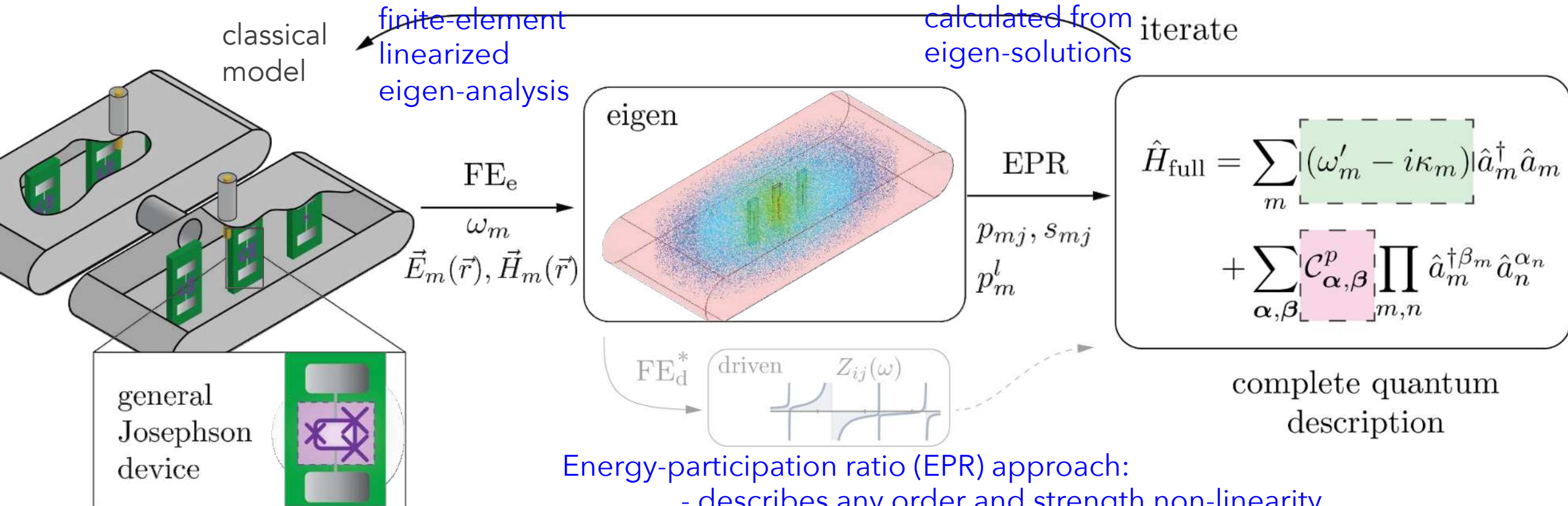
What *fraction* of the energy of a mode p_j (p_l)
is stored in the non-linear (dissipative) element?

$$0 \leq p_j, p_l \leq 1$$


$$\hat{H}_{\text{full}}$$


$$\mathcal{D}[\sqrt{\kappa}\hat{a}]\rho$$

Overview of energy-participation approach



Energy-participation ratio (EPR) approach:

- describes any order and strength non-linearity
- describes arbitrary (composite) non-linear inductive devices
- first-principle derivation
- zero approximations (aside from truncation of modes)
- fully automated in python (github.com/zlatko-minev)

Practical limits: Fock and mode basis truncation due to computing power

* Nigg, Paik, *et al.*, PRL (2012),
Bourassa *et al.* (2012),
Solgun *et al.* (2014, 2015, 2017), ...

A simple example:

transmon qubit coupled to resonator

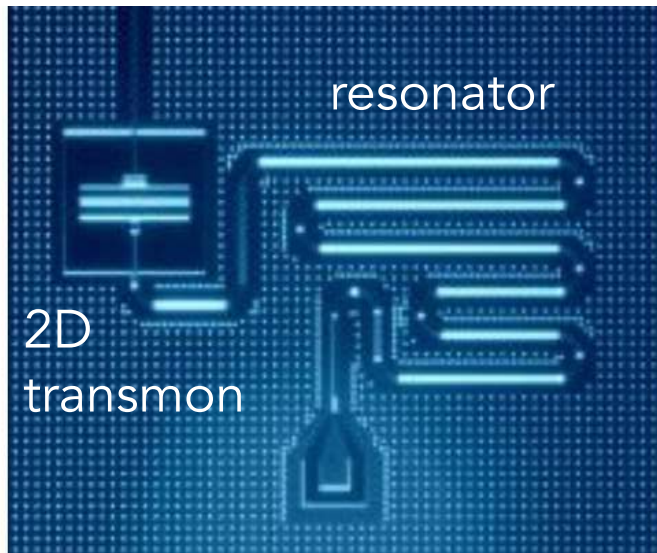
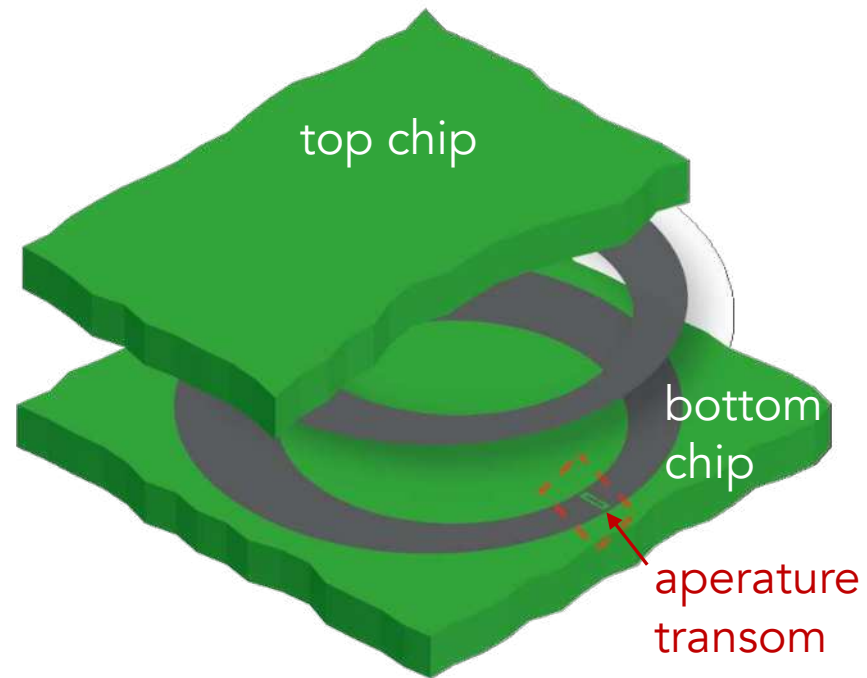
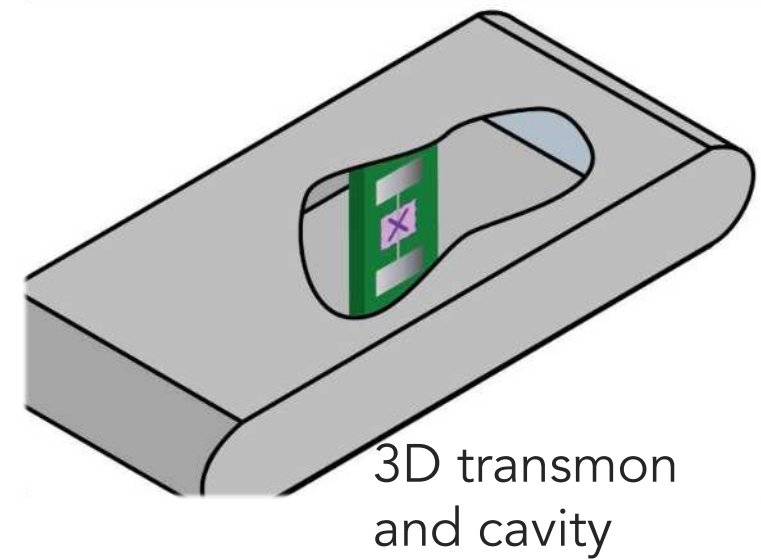


Image: Gambetta, Chow, and Steffen (2017)

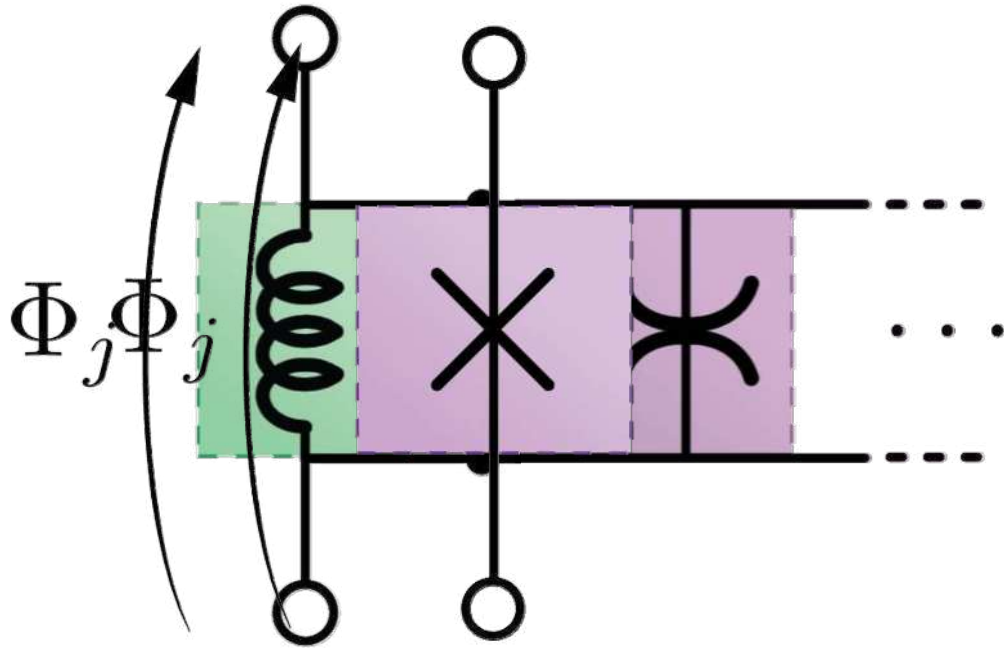


Minev *et al.* (2013, 2016)



Paik *et al.* (2011), ...

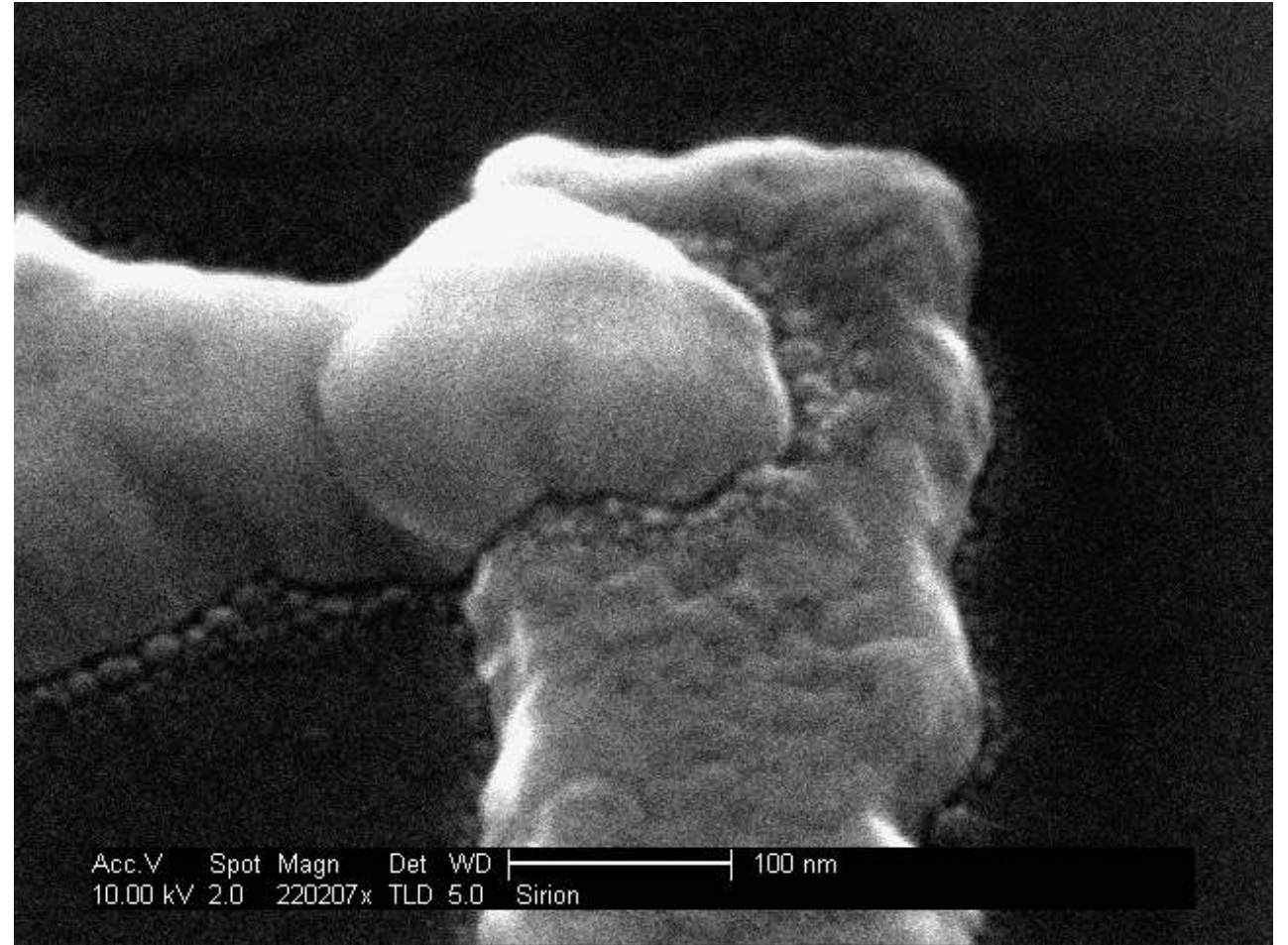
Josephson tunnel junction



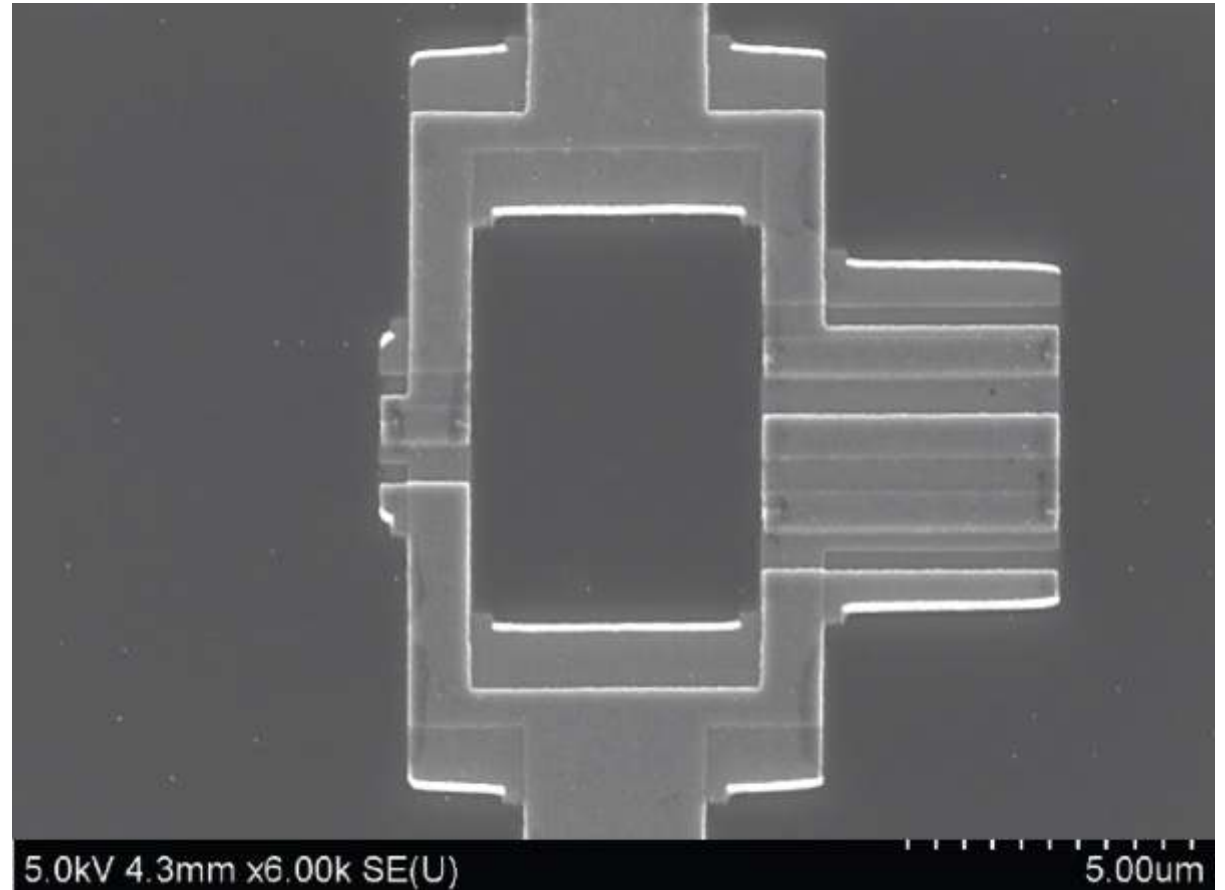
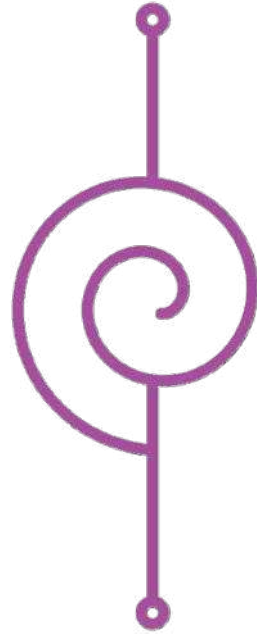
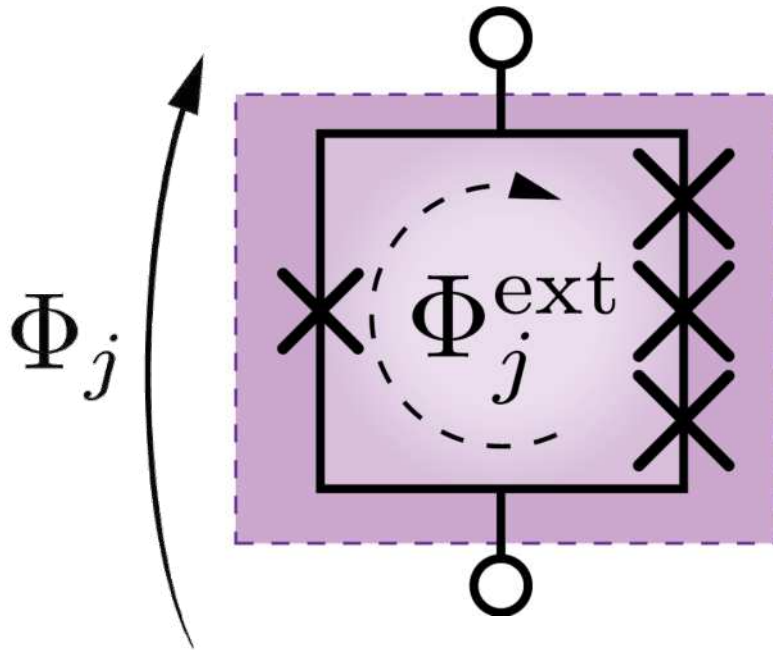
$$\mathcal{E}_j(\Phi_j) = E_j (1 - \cos(\Phi_j/\phi_0))$$

$$\phi_0 \equiv \hbar/2e \Phi_j + \mathcal{E}_j^{\text{nl}}(\Phi_j)$$

$$= \frac{E_j}{2} \left(\frac{\Phi_j}{\phi_0} \right)^2 - \frac{E_j}{4!} \left(\frac{\Phi_j}{\phi_0} \right)^4 + \mathcal{O}(\Phi_j^6)$$



Composite Josephson device: example

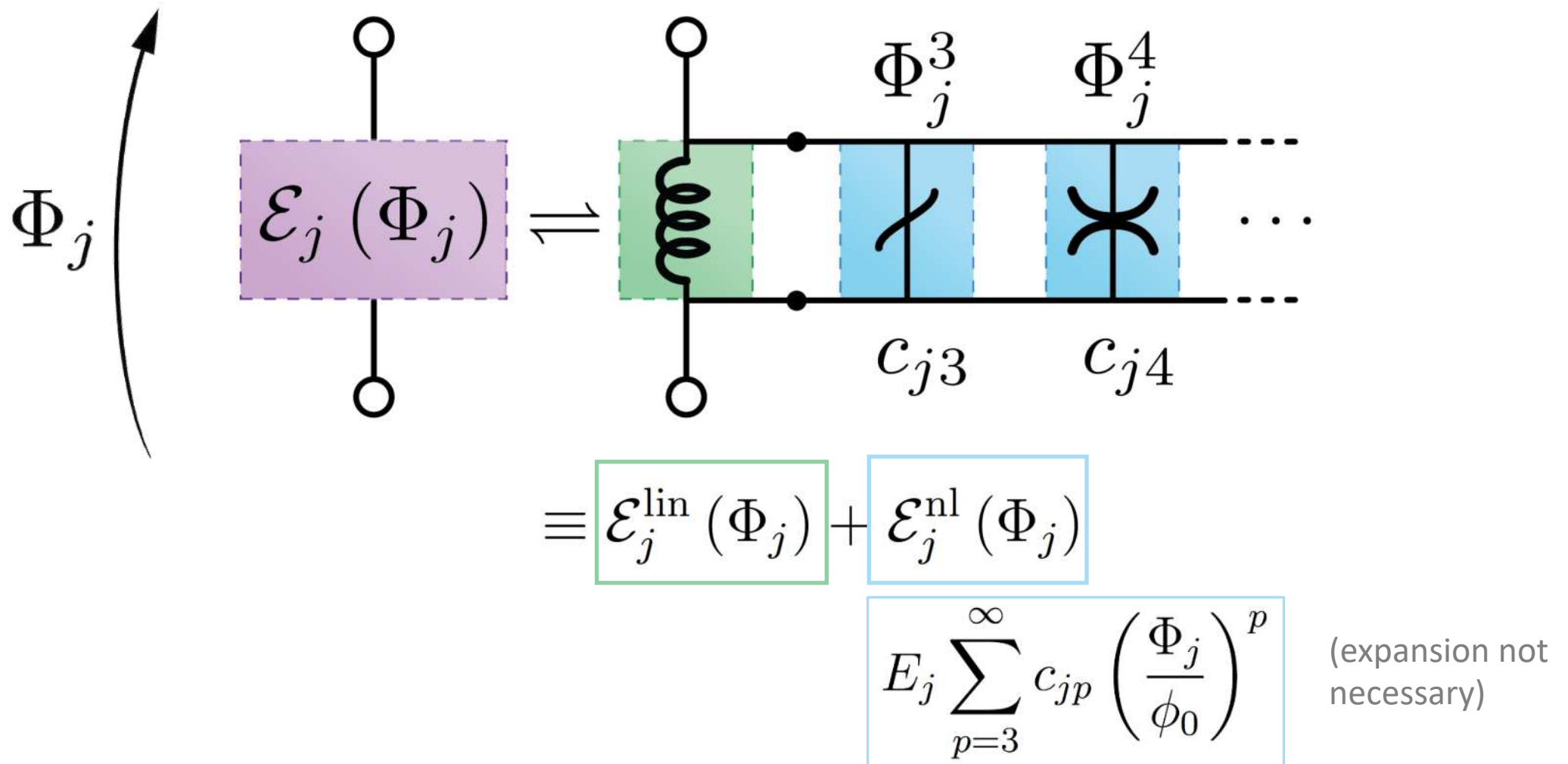


Energy function

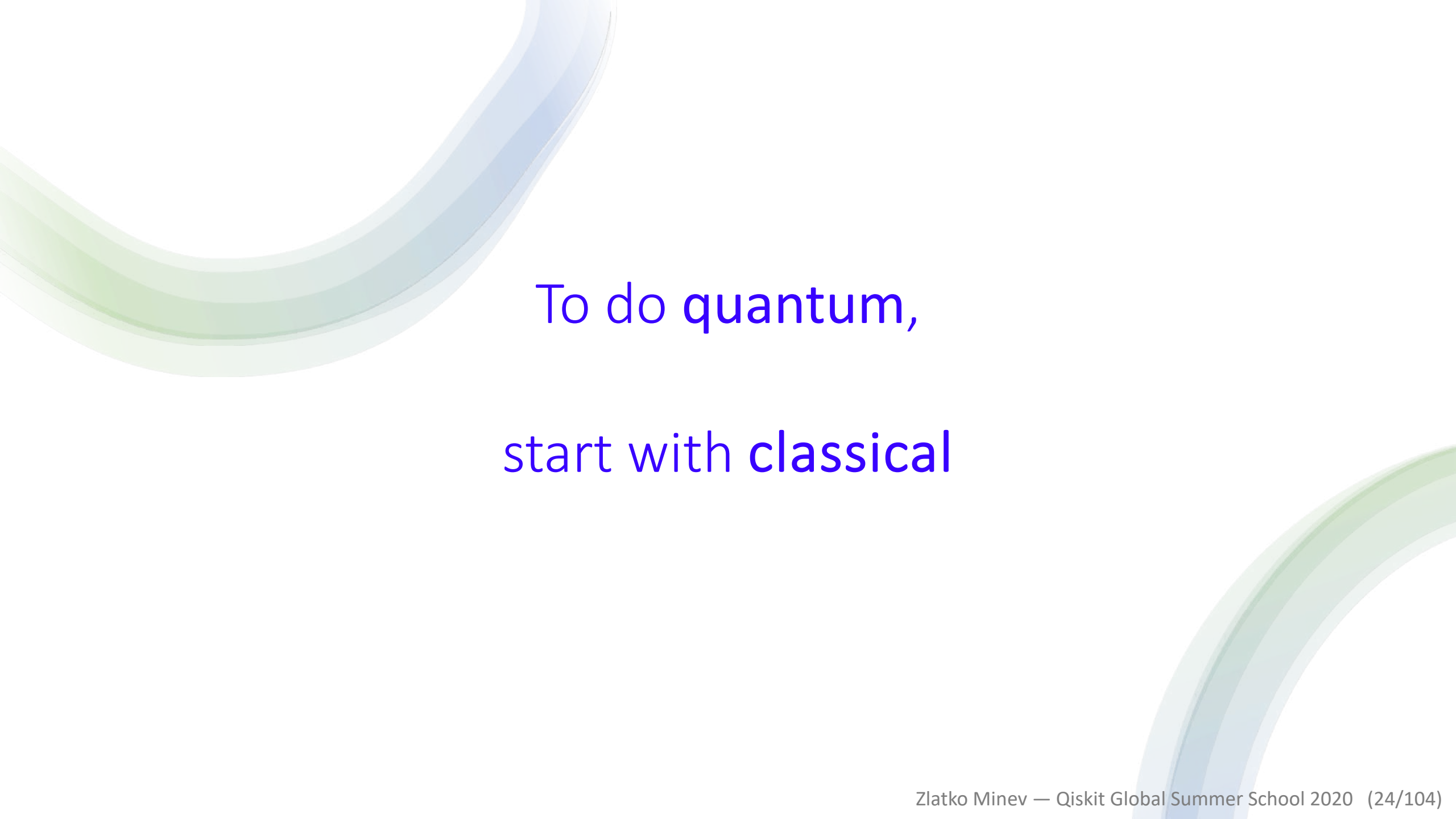
$$\mathcal{E}_j(\Phi_j; \Phi_j^{\text{ext}}) = \mathcal{E}_j^{\text{lin}}(\Phi_j) + \mathcal{E}_j^{\text{nl}}(\Phi_j)$$

SEM image: Frattini & Sivak; see APL (2017)

General Josephson device

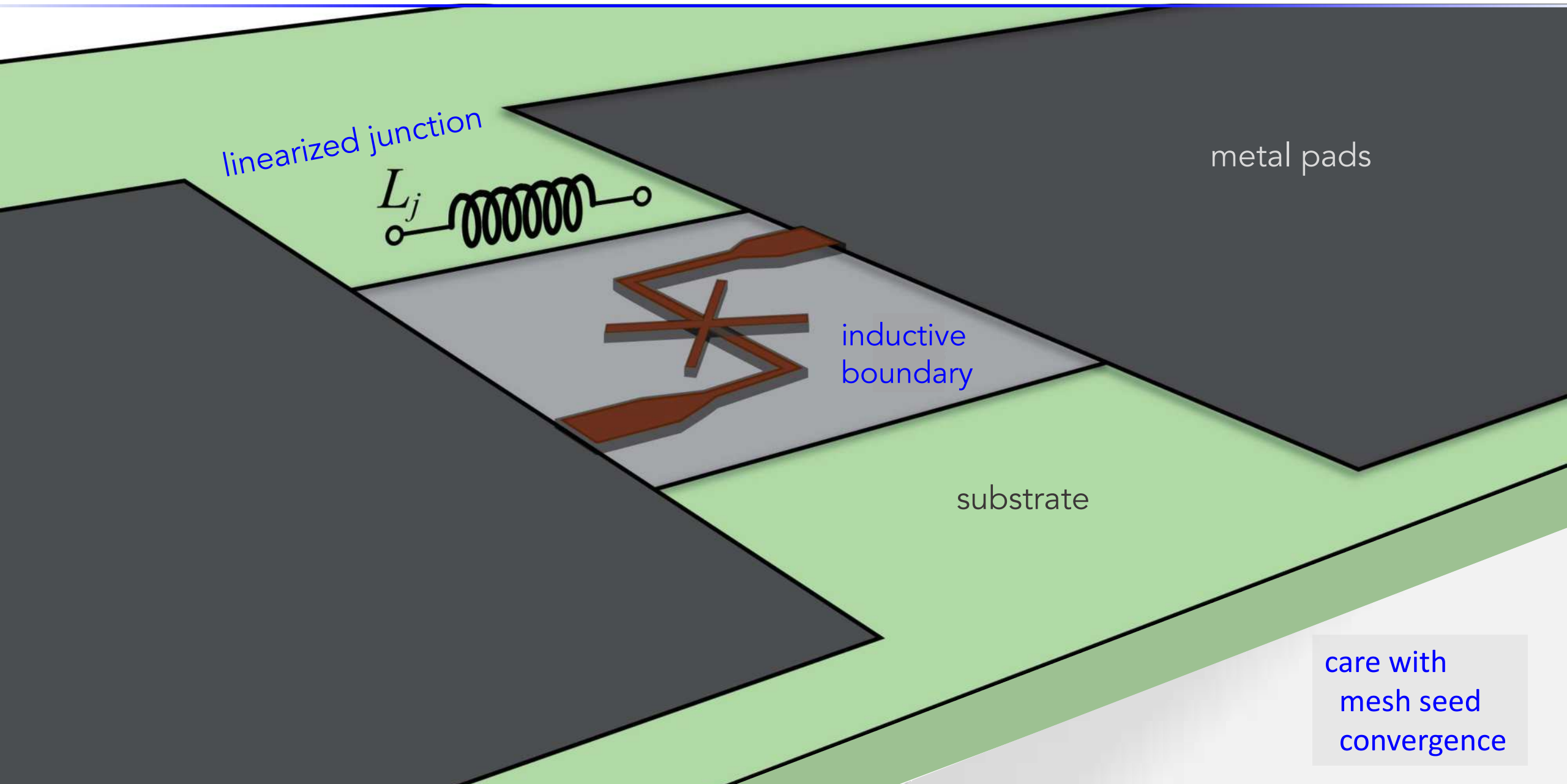


series with respect to operational equilibrium point

The slide features two decorative curved lines. One line, in shades of green and blue, starts from the left edge and curves upwards and to the right. Another similar line starts from the bottom right edge and curves upwards and to the left. The text is centered between these two curves.

To do quantum,
start with classical

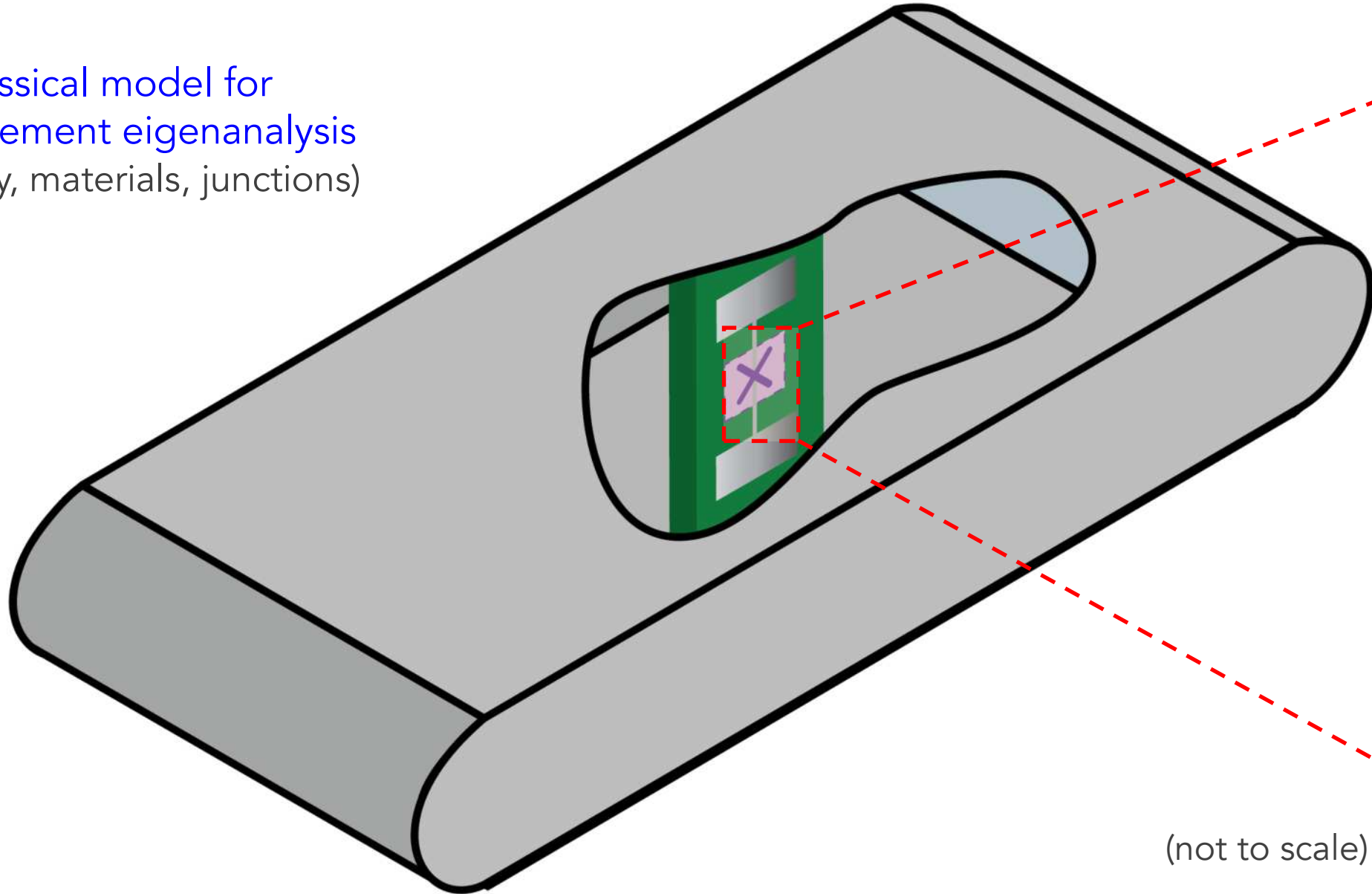
Finite-element model of linearized Junction



care with
mesh seed
convergence

Transmon qubit coupled to cavity

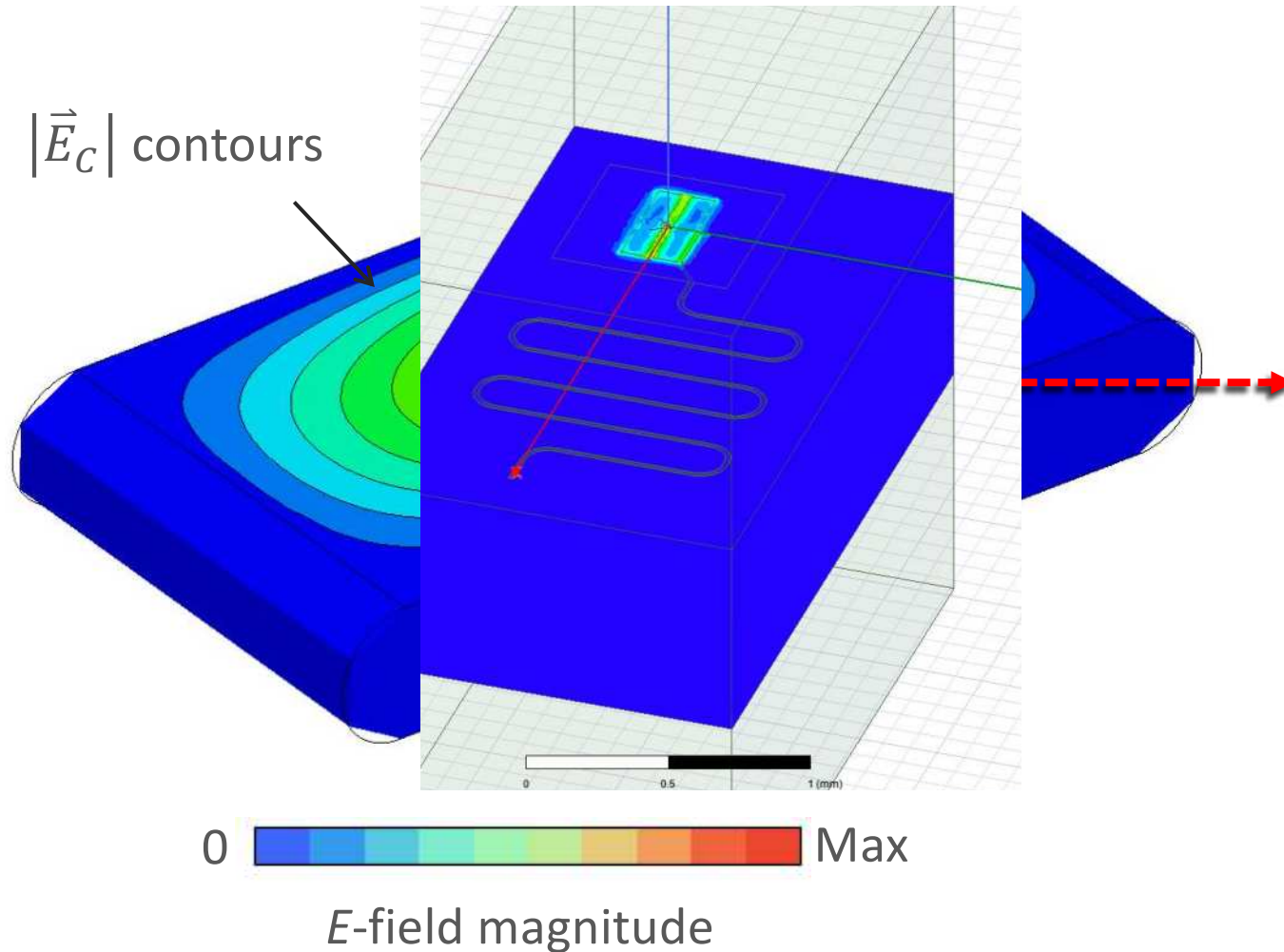
Classical model for
finite-element eigenanalysis
(geometry, materials, junctions)



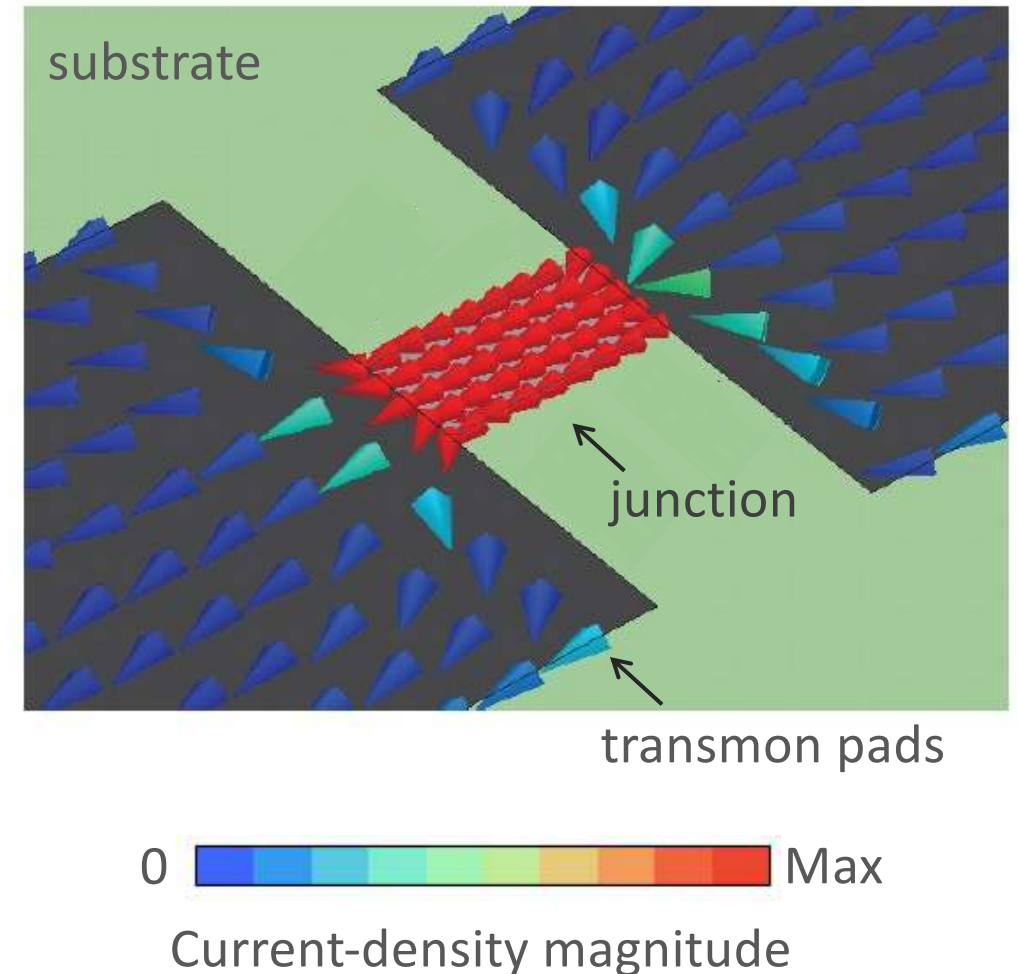
(not to scale)

$\mathcal{H}_{\text{lin}}$ eigen modes

Cavity mode (7.0 GHz)



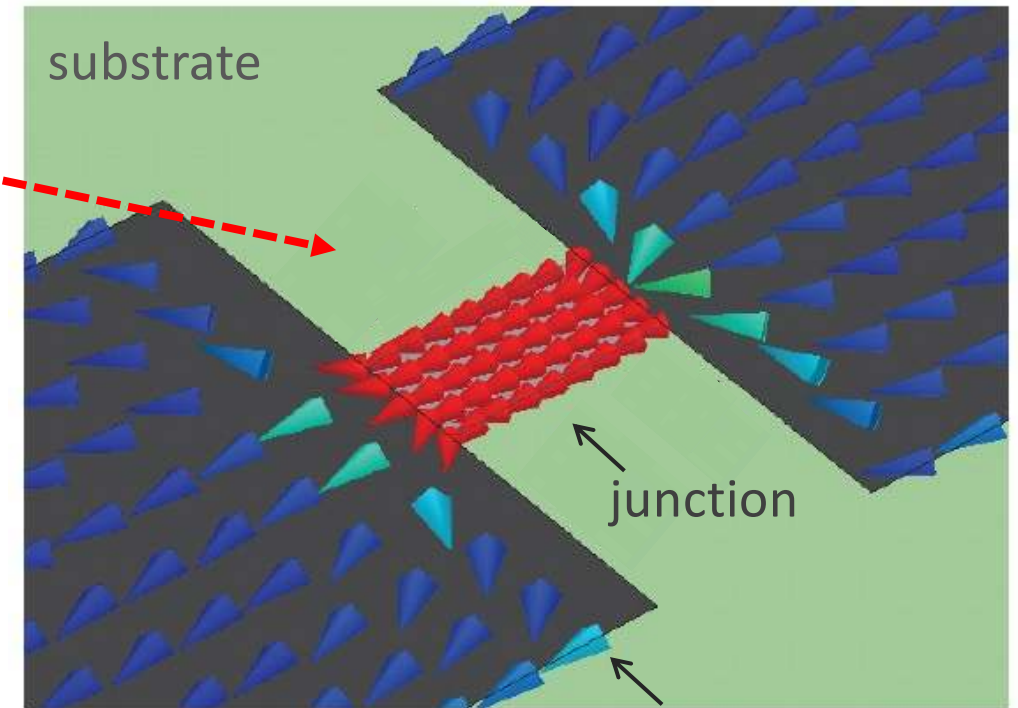
Qubit mode (linearized, 5 GHz)



Energy participation of the junction

$$p_m = \frac{\text{Energy stored in junction}}{\text{Inductive energy stored in mode } m}$$
$$p_m = \frac{\frac{1}{2} I_{cd}^2}{E_{ind,m}}$$

Qubit mode (linearized, 5 GHz)

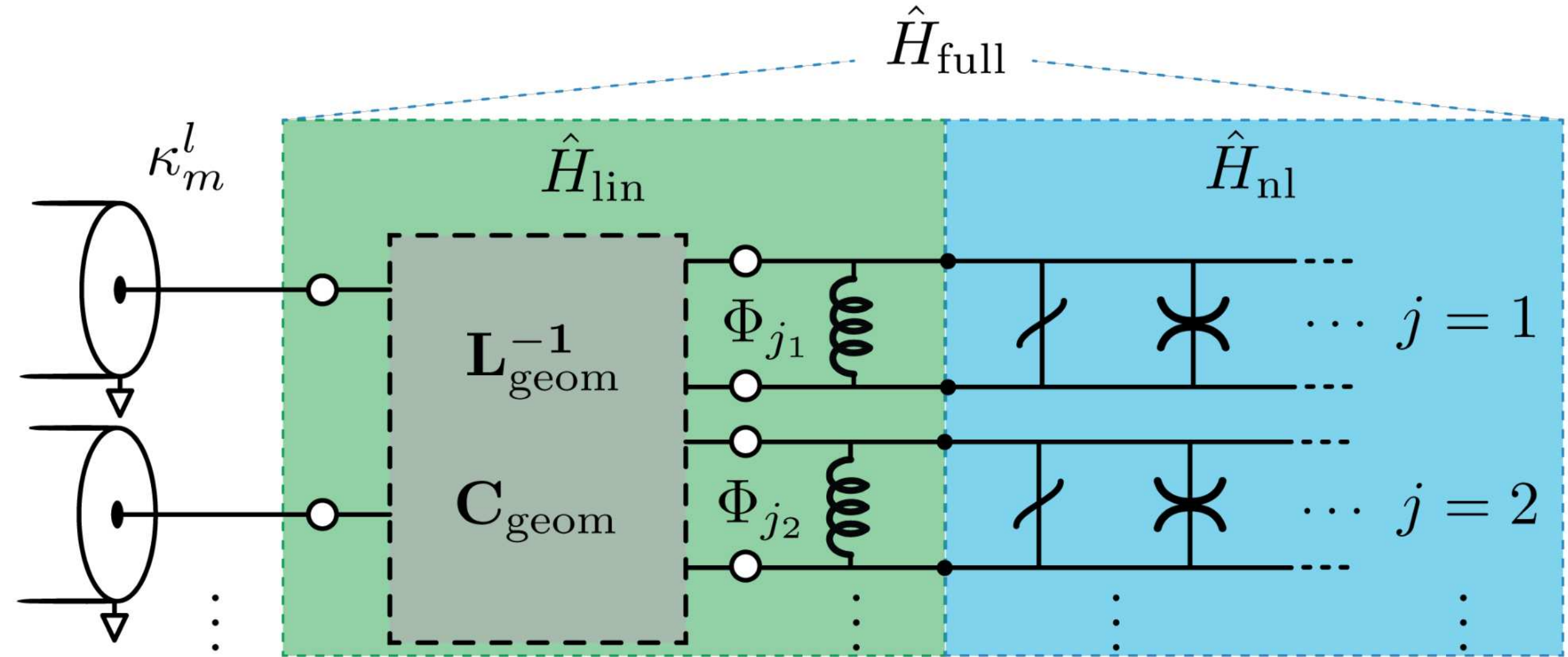


transmon pads

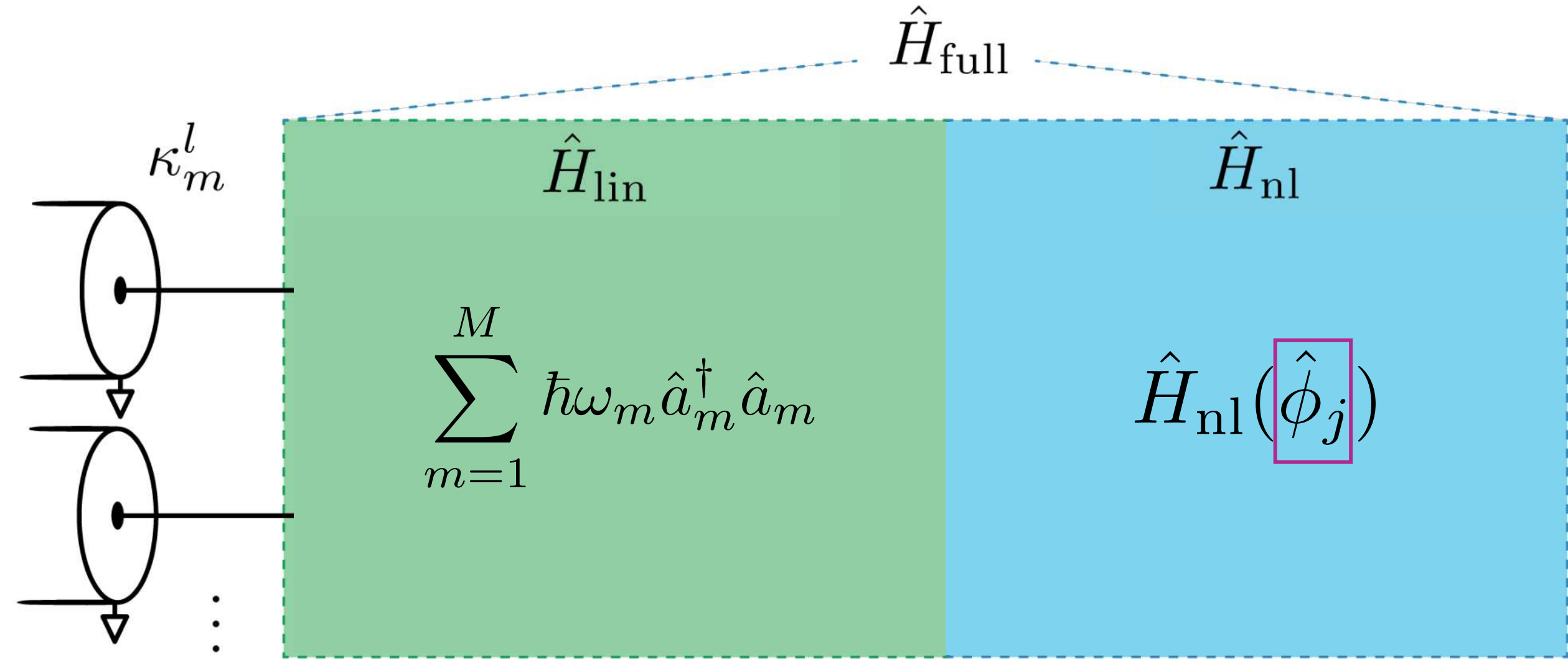


Current-density magnitude

Decomposition of a general circuit



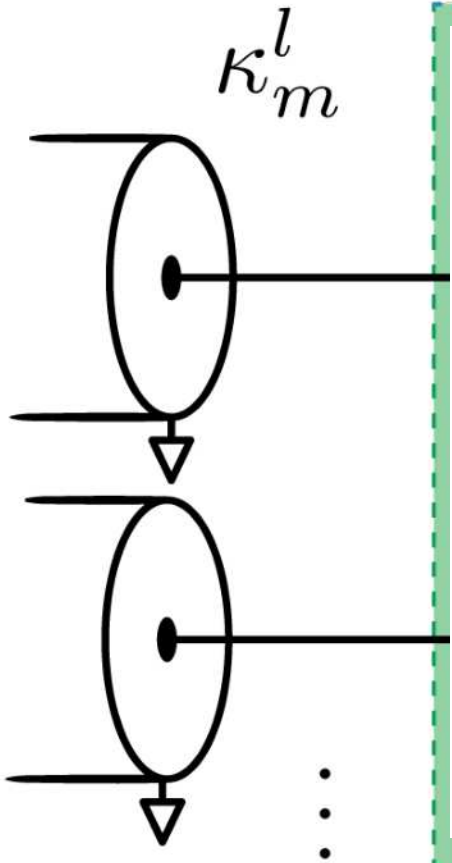
Decomposition of a general circuit



second quantization in eigen basis of linearized circuit

Decomposition of a general circuit

$$\hat{H}_{\text{full}}$$



What is zero-point fluctuation of the phase of junction j due to mode m ?

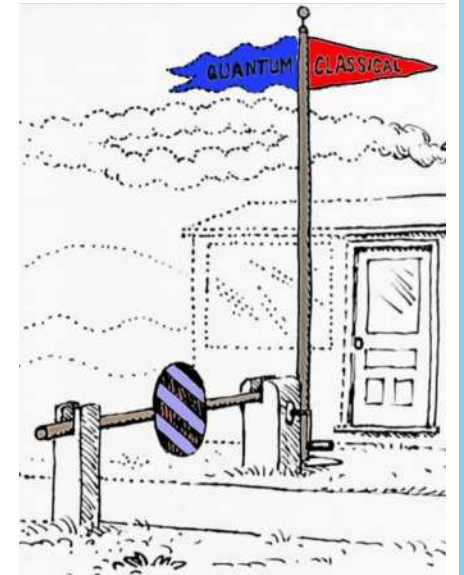
$$\hat{\phi}_j = \sum_{m=1}^M \phi_{mj} (\hat{a}_m^\dagger + \hat{a}_m)$$

Decomposition of a general circuit

$$\hat{H}_{\text{full}}$$

What fraction of the energy of mode m is stored in junction j ?

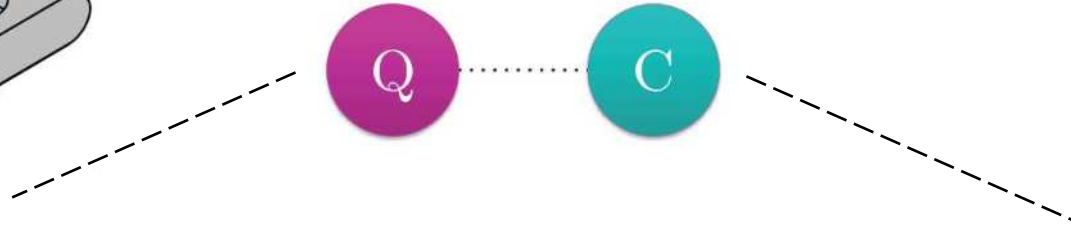
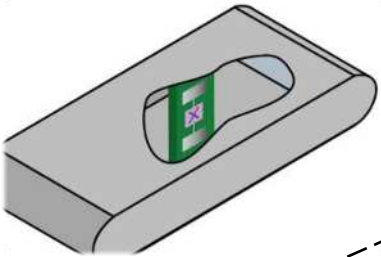
$$\frac{1}{\hbar} \phi_{mj}^2 = p_{mj} \frac{\omega_m}{2E_j}$$



for $j > 1$, root requires sign bit $s_{mj} = \pm 1$

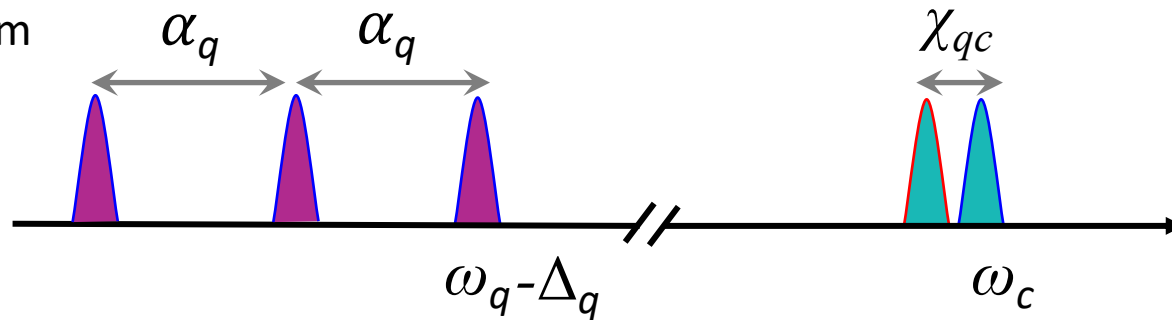
Drawing: Zurek, Physics Today (1991)

Quantum Hamiltonian from the EPR



$$\hat{H}_{\text{eff}} = (\omega_q - \Delta_q) \hat{n}_q + (\omega_q \hat{n}_q + \omega_c \hat{n}_c - \chi_{qc} \hat{n}_q \hat{n}_c - \frac{1}{2} \alpha_q \hat{n}_q (\hat{n}_q - 1) - \frac{1}{2} \alpha_c \hat{n}_c (\hat{n}_c - 1) ,$$

Transition spectrum



for simplicity, showing up to $\mathcal{O}(\phi^6)$ in RWA

Qubit/cavity anharmonicity

$$\alpha_{q/c} = p_{q/c}^2 \frac{\hbar \omega_{q/c}^2}{8 E_J}$$

Qubit-cavity dispersive shift

$$\chi_{qc} = p_q p_c \frac{\hbar \omega_q \omega_c}{4 E_J}$$

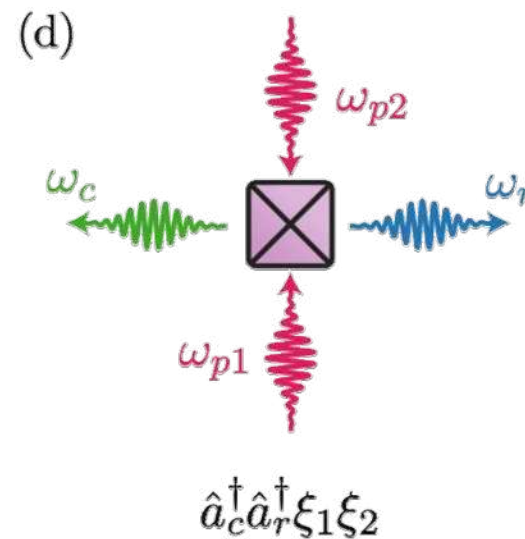
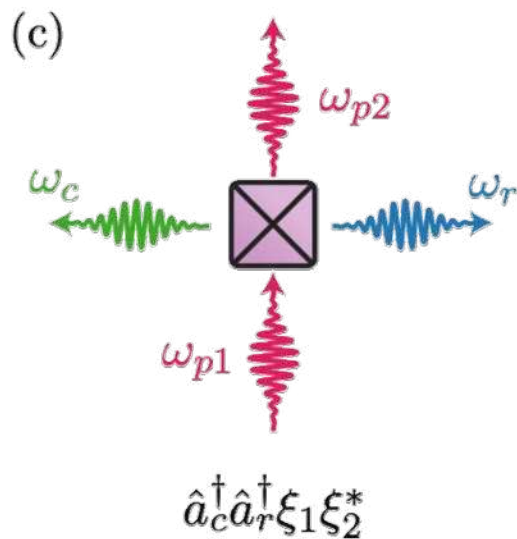
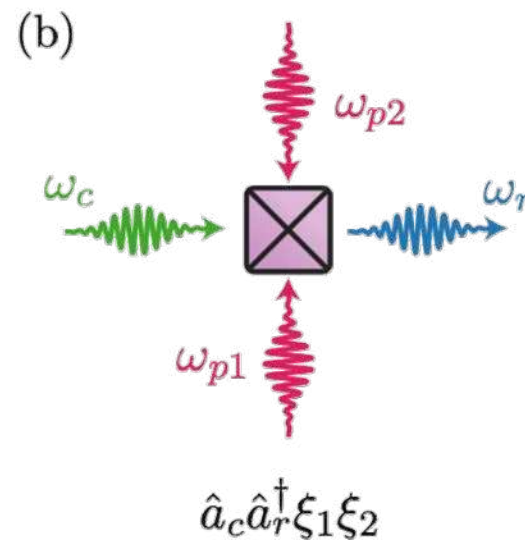
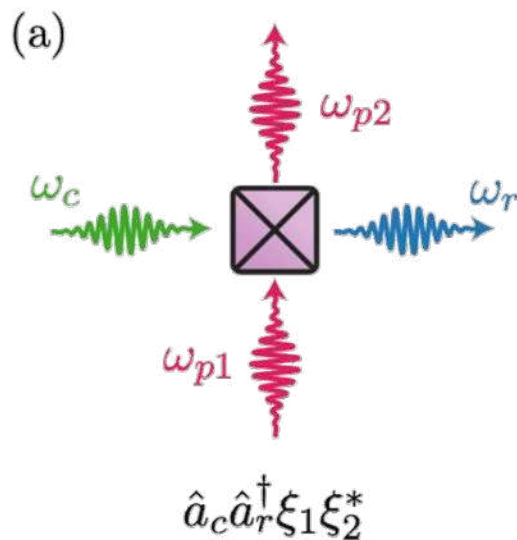
Qubit Lamb shift

$$\Delta_q = \alpha_q - \frac{1}{2} \chi_{qc}$$

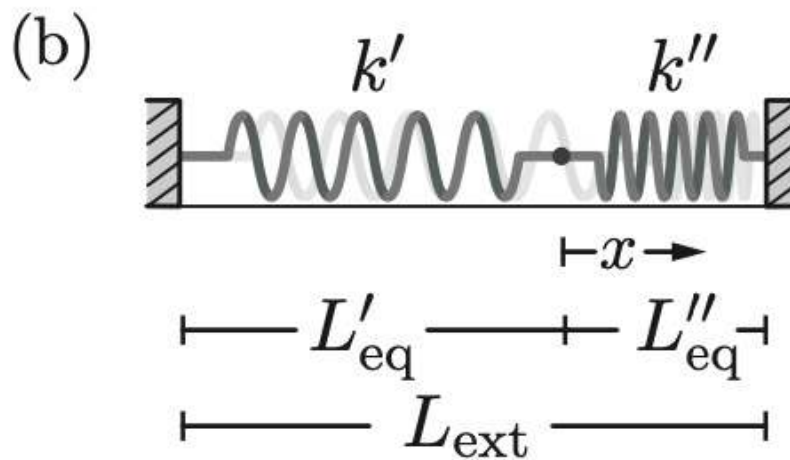
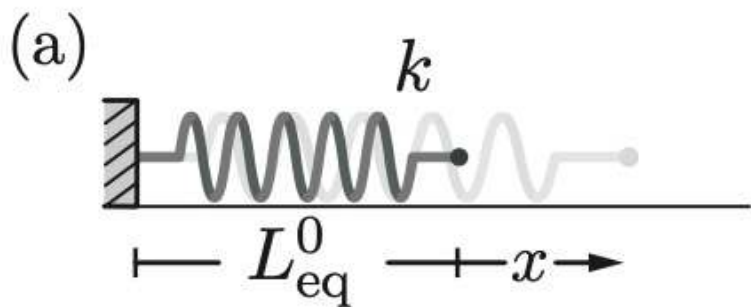
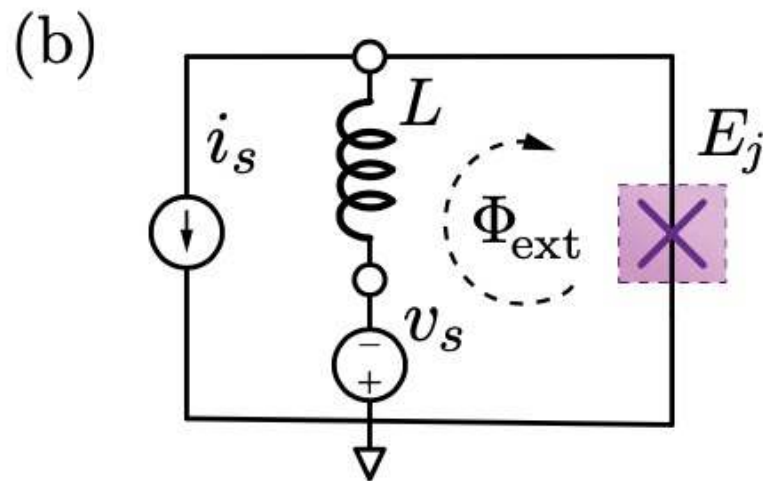
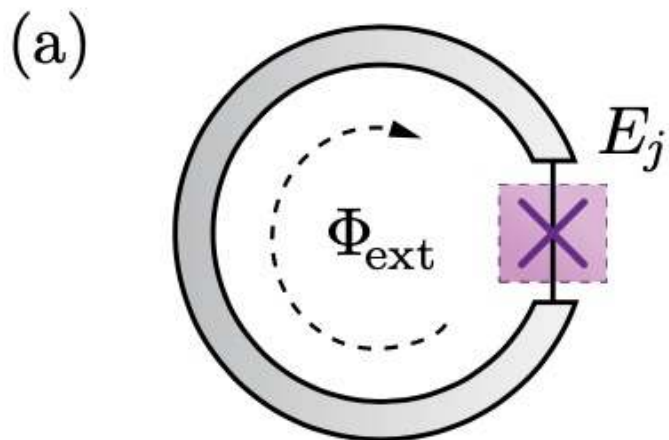
Reconstruct full Hamiltonian

$$\sum_{m=1}^M \hbar \omega_m \hat{a}_m^\dagger \hat{a}_m$$

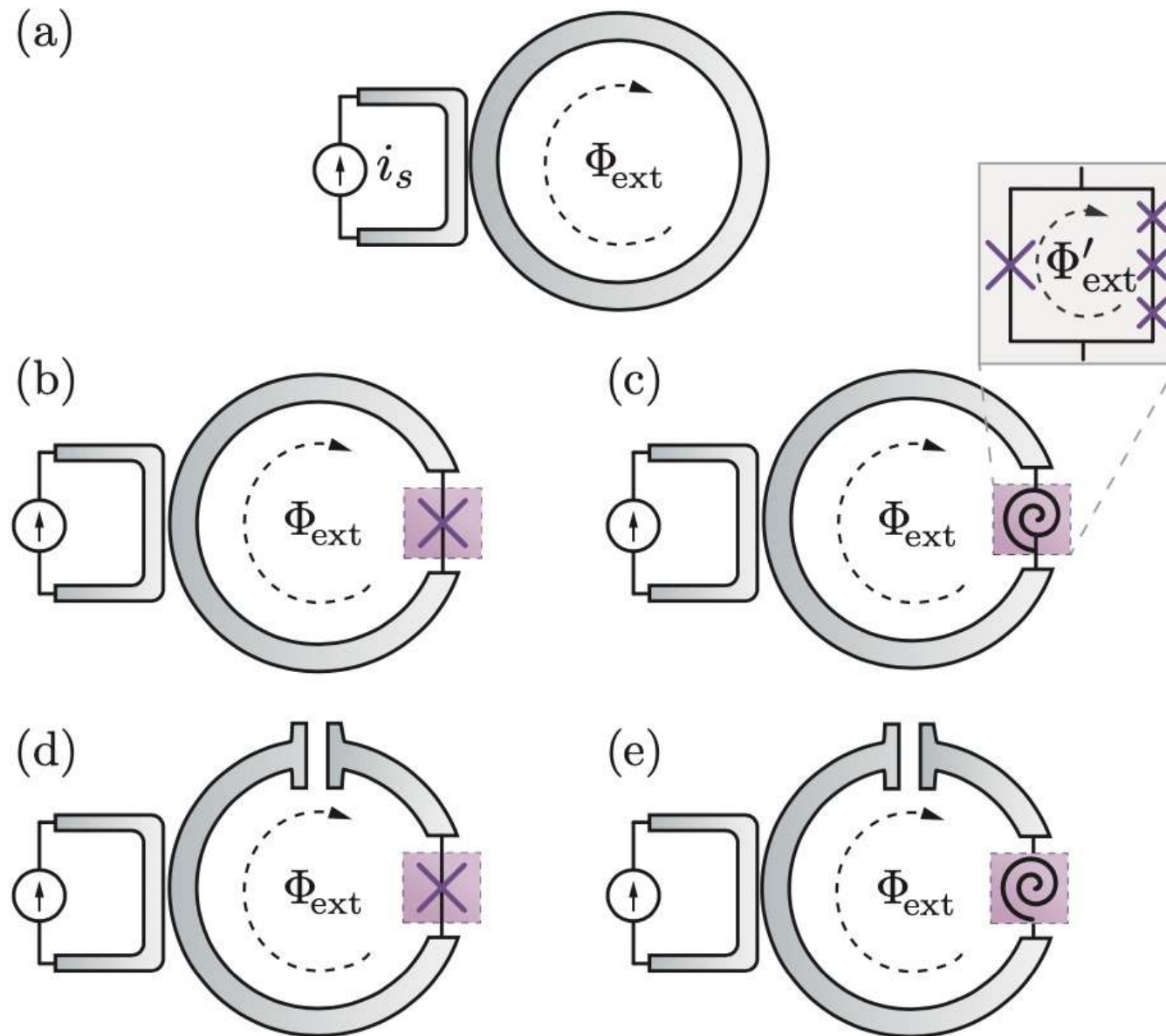
$$+ \hat{H}_{nl}(\phi_j)$$



Flux conditions



Flux



Universal EPR properties

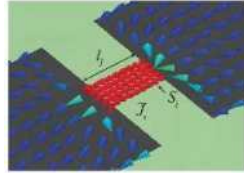
ZPF are not independent of each other

The monogamy of EPR

Commandments of EPR

1. All modes and all dipole are created equal
2. Each dipole has exactly 1 EPR it must dilute among modes
3. Each mode receives exactly 1 EPR units

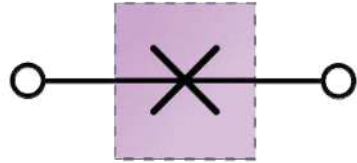
EPR



$$0 \leq p_{mj} \leq 1$$

EPR is bounded

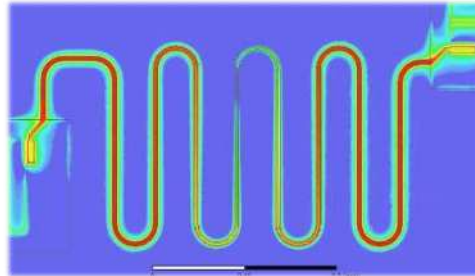
Non-linear dipole



$$\sum_{m=1}^M p_{mj} = 1 \quad \text{for } j \in \{1, \dots, J\}$$

Every dipole comes w/ 1 unit of EPR

Mode



$$0 \leq \sum_{j=1}^J p_{mj} \leq 1 \quad \text{for } m \in \{1, \dots, M\}$$

Every mode accepts at most 1 total unit of EPR

Orthogonality

$$\sum_{m=1}^M s_{mj} s_{mj'} \sqrt{p_{mj} p_{mj'}} = 0 \quad \text{for } j \neq j'$$

For a dipole, its EPRs across all modes are orthogonal

Dissipation budget & EPR

Coherence in superconducting circuits

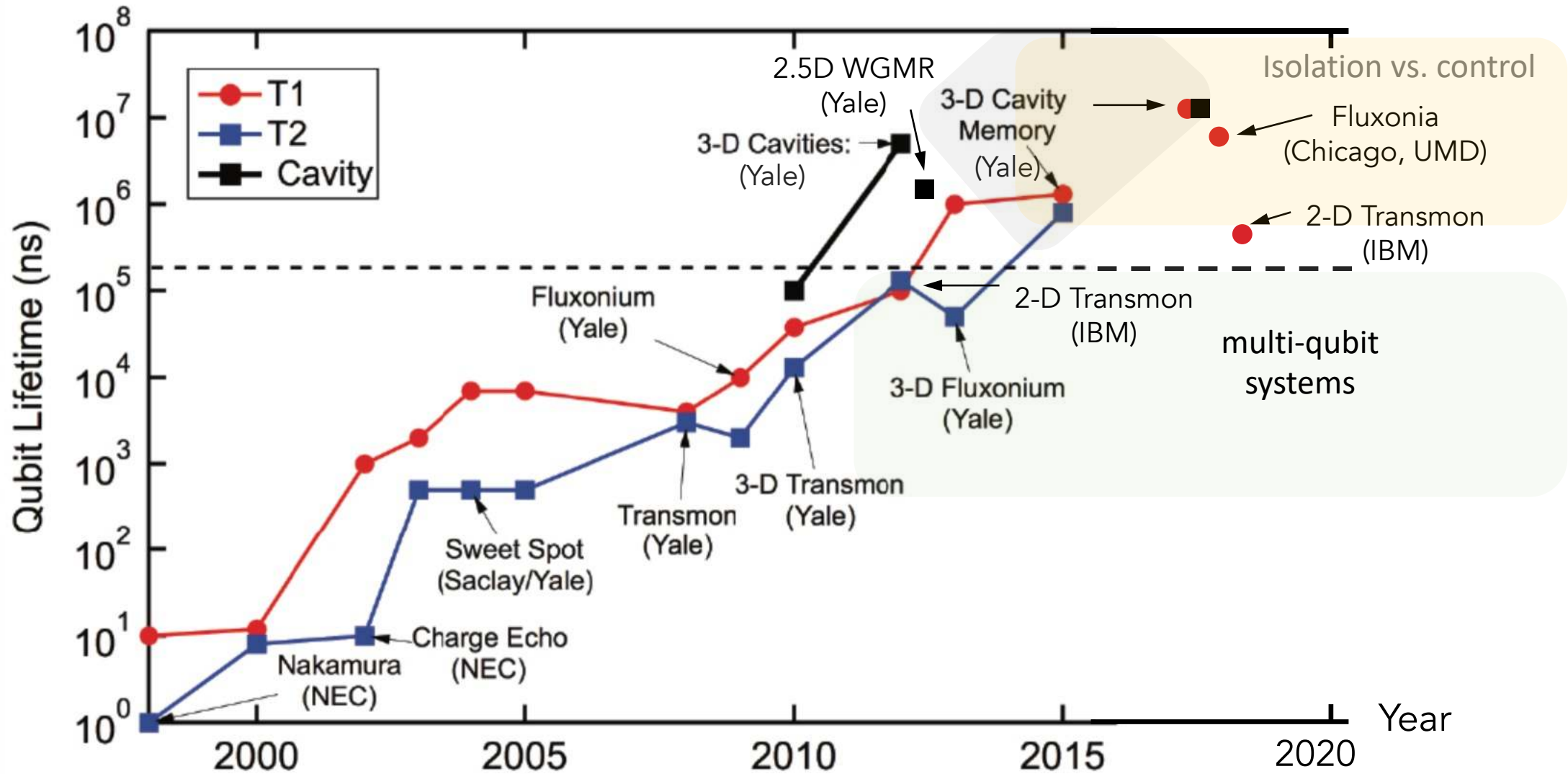
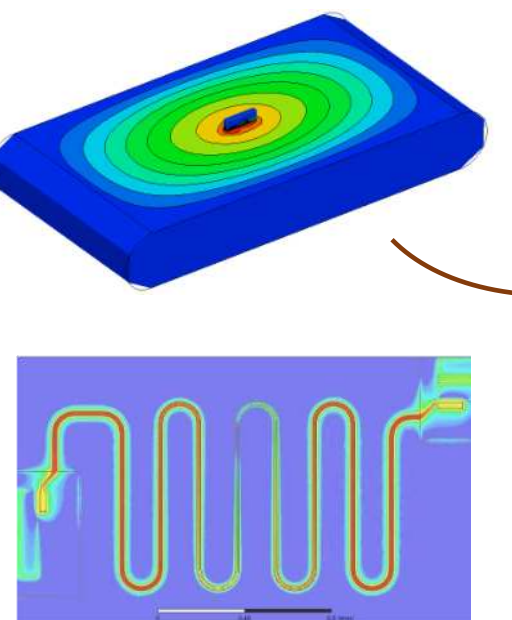


Image reproduced Reagor (2015), an update of Devoret and Schoelkopf (2013), and updated

Dissipation budget and the energy-participation ratio (EPR)



Quality factor of mode m

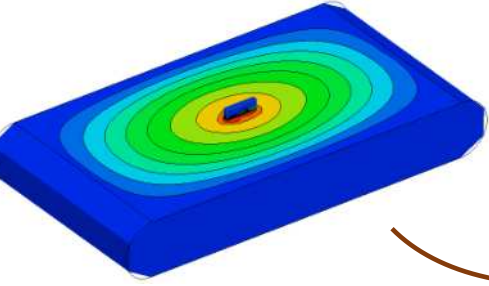
EPR of lossy element l in mode m (design)

Quality of lossy element l (material property)

$$Q_{\text{total},m}^{-1} = \sum_l p_{ml} Q_l^{-1}$$
$$= \frac{1}{Q_{\text{cap}}} + \frac{1}{Q_{\text{ind}}} + \frac{1}{Q_{\text{rad}}}$$

Pozar (1990), Martinis *et al.* (2005), Gao (2008), Werner, *et al.* (2011), Zmuidzinas (2012), Megrant *et al.* (2012), Geerlings (2013), Quintana *et al.* (2014), Minev *et al.* (2013), Brecht *et al.* (2015), Wang *et al.* (2015), Dial *et al.* (2016), ...

Lossy energy-participation ratios (EPRs)



Capacitive



bulk

$$p_{ml}^{\text{cap}} = \frac{1}{\mathcal{E}_{\text{elec}}} \frac{1}{4} \Re \int_{V_l} \vec{E}_{\text{max}}^* \overleftrightarrow{\epsilon} \vec{E}_{\text{max}} dv ,$$

surface

$$p_{ml}^{\text{cap,surf}} = \frac{1}{\mathcal{E}_{\text{elec}}} \frac{t_l \epsilon_l}{4} \Re \int_{\text{surf}_l} \left| \vec{E}_{\text{max}} \right|^2 ds ,$$

Inductive



bulk

$$p_{ml}^{\text{ind,surf}} = \frac{1}{\mathcal{E}_{\text{mag}}} \frac{\lambda_0 \mu_l}{4} \Re \int_{\text{surf}_l} \left| \vec{H}_{\text{max},\parallel} \right|^2 ds ,$$

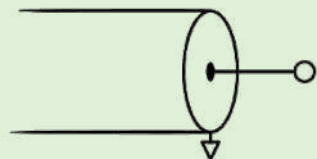
surface

$$p_{ml}^{\text{ind,bulk}} = \frac{1}{\mathcal{E}_{\text{mag}}} \frac{1}{4} \Re \int_{V_l} \vec{H}_{\text{max}}^* \overleftrightarrow{\mu} \vec{H}_{\text{max}} dv .$$

seam

$$p_{ml}^{\text{ind,seam}} = \frac{1}{\mathcal{E}_{\text{mag}}} \frac{\lambda_0 t_l \mu_l}{4} \Re \int_{\text{seam}_l} \left| \vec{H}_{\text{max},\perp} \right|^2 dl ,$$

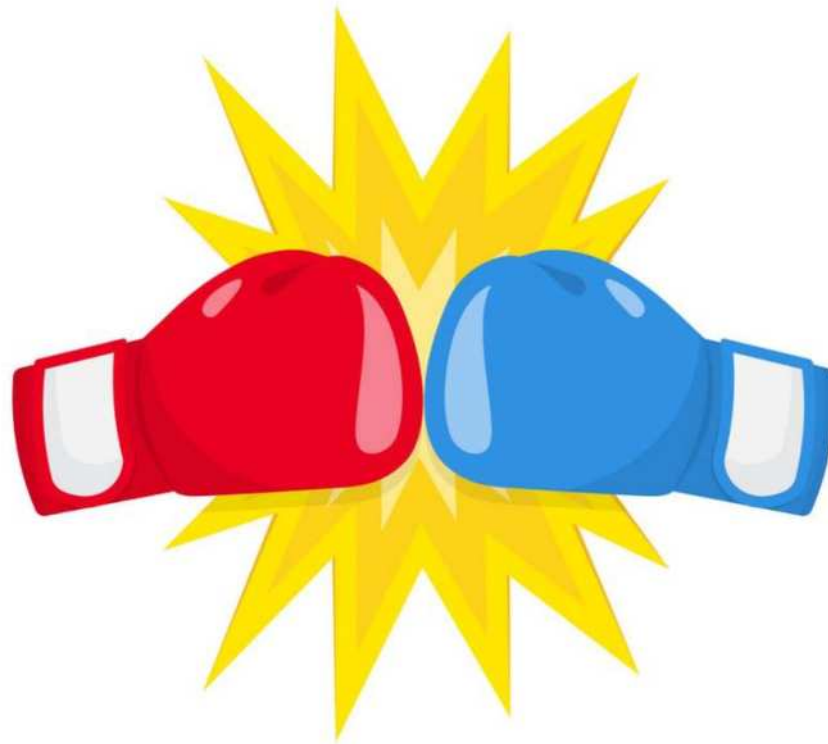
Radiative



external

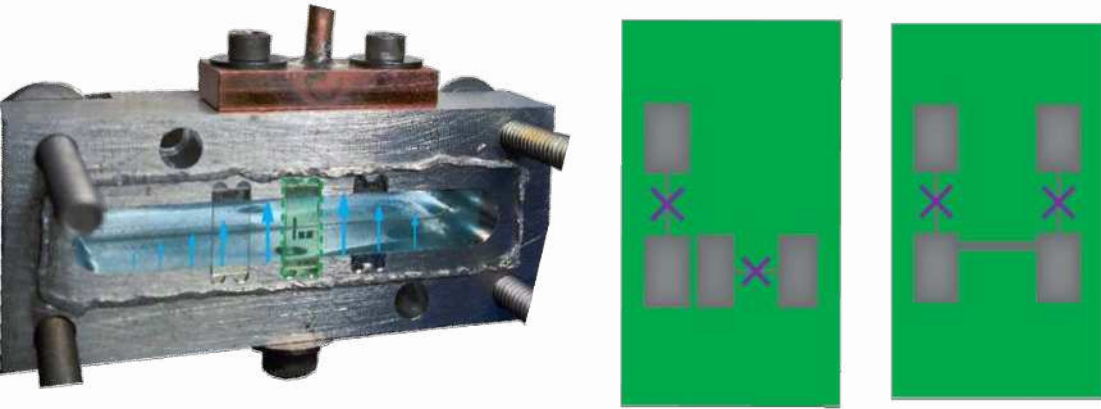
$$Q_{mp} = \frac{\omega_m \mathcal{E}_m(0)}{\frac{1}{2} R I_{mp}^2} .$$

EPR theory vs. experiment

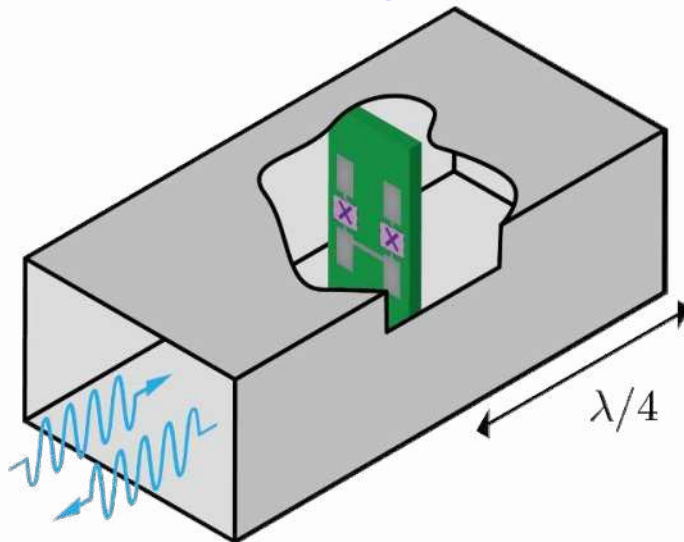


First set of measured architectures and devices

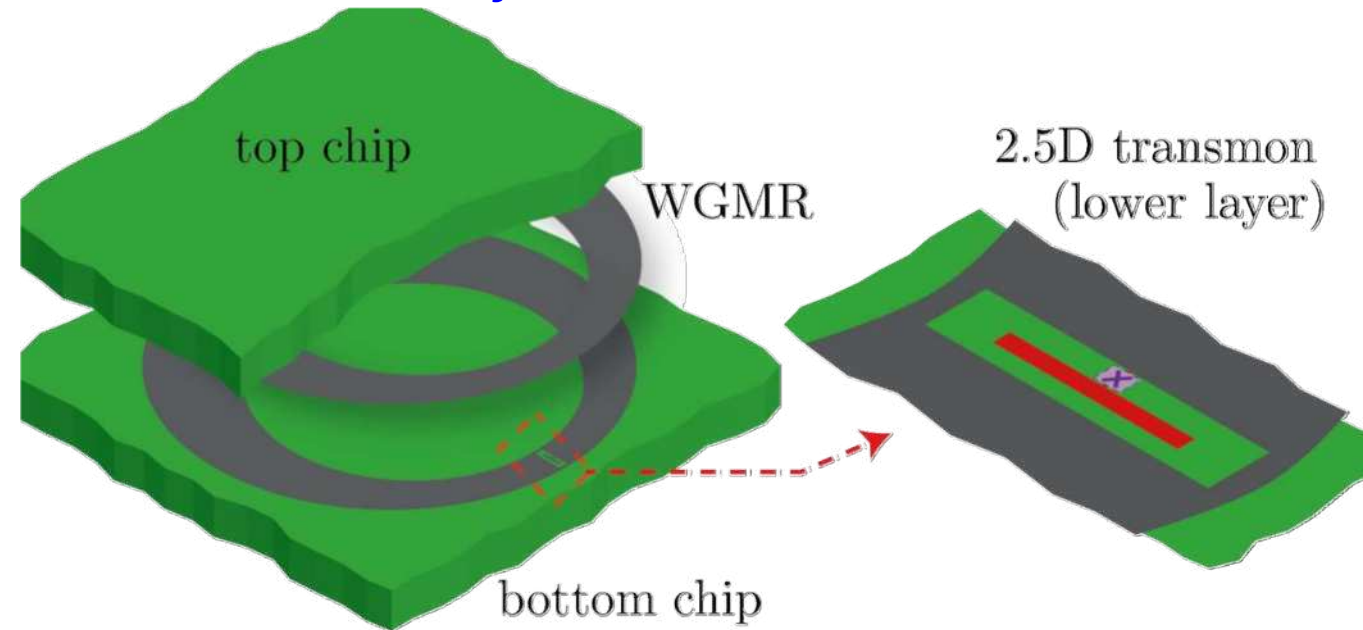
Three-dimensional (3D) *



Waveguide



Multilayer (2.5D) cQED



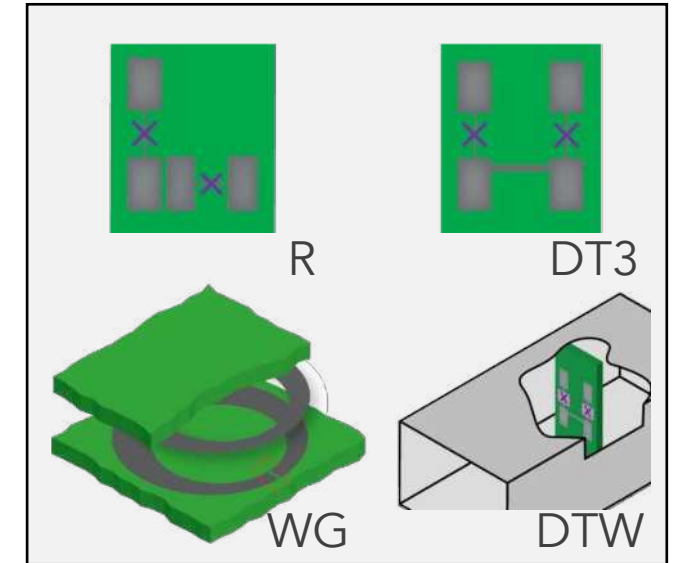
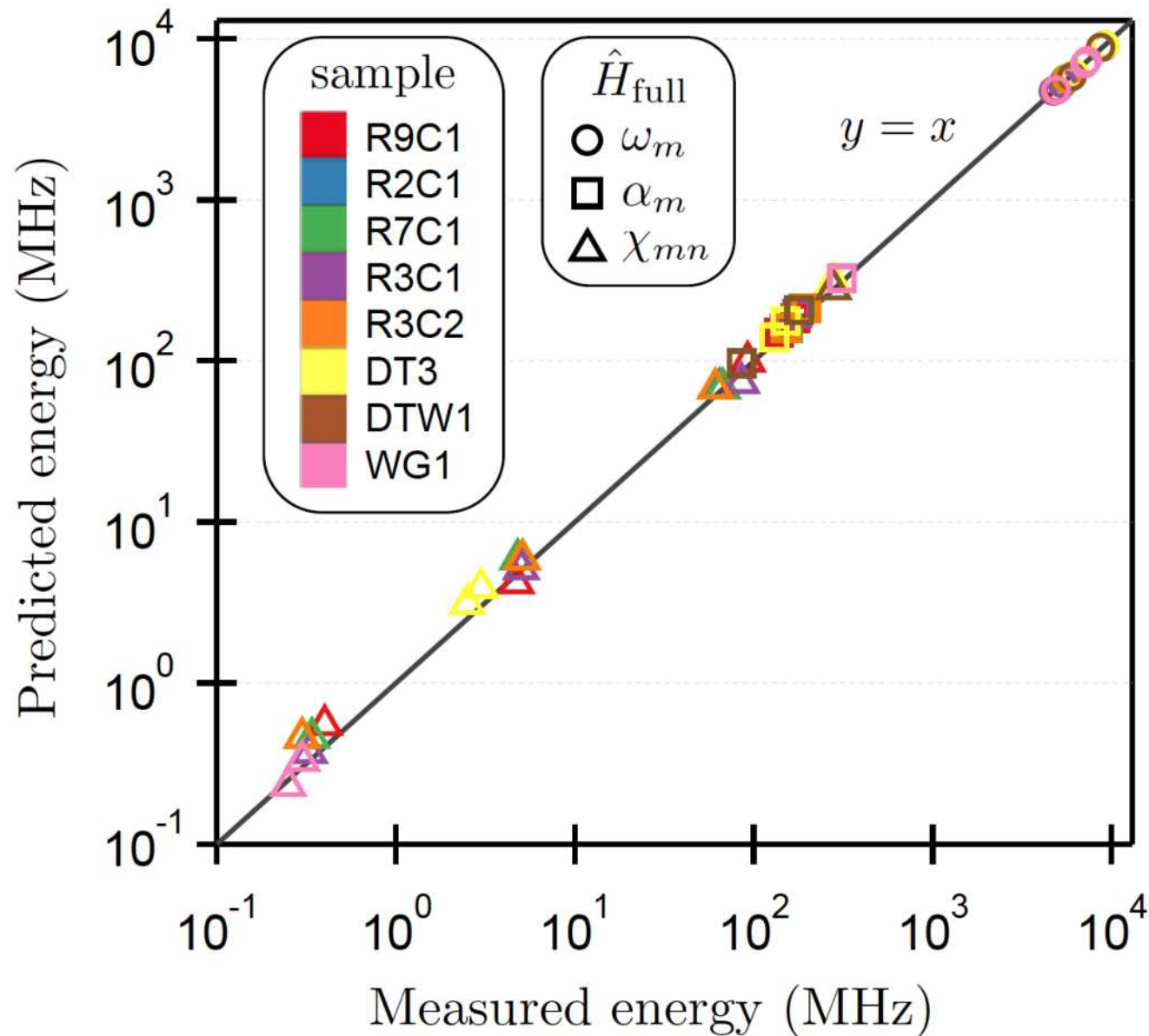
Minev *et al.*, APL (2013)

Minev *et al.*, WO/2016/138395 (2015)

Minev *et al.*, Phys. Rev. App. (2016)

* Minev *et al.*, arXiv:1803.00545 (2018); Related: Gambetta *et al.*, PRL (2011), Srinivasan *et al.*, PRL (2011), Dumur *et al.*, PRB (2015), Zhang *et al.*, Nature JQI (2017) ...

Theory vs. experiment: agreement over 5 orders of magnitude



R: Minev *et al.* (2018)
WG: Minev *et al.* (2013, 2016)
DT3, DTW: Minev *et al.* (2019)

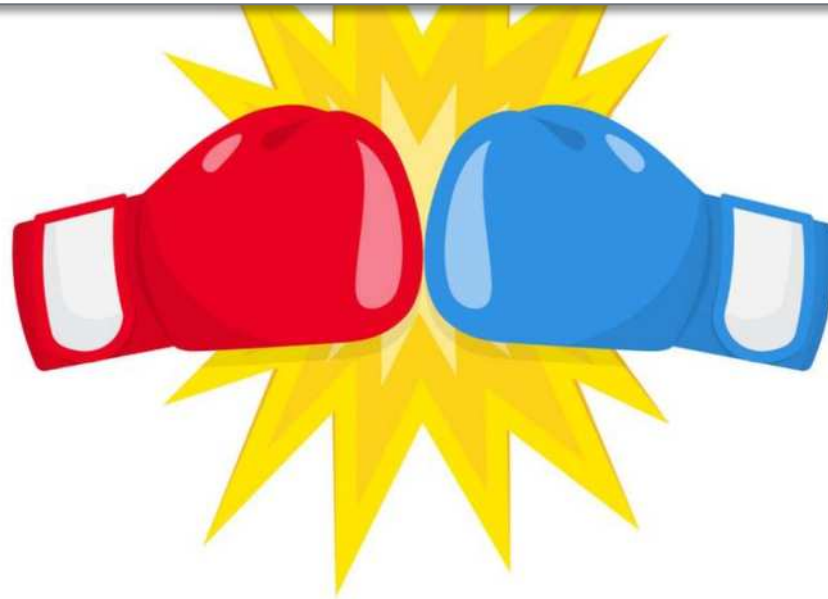
planar devices & comparison to other methods

Quantum Physics

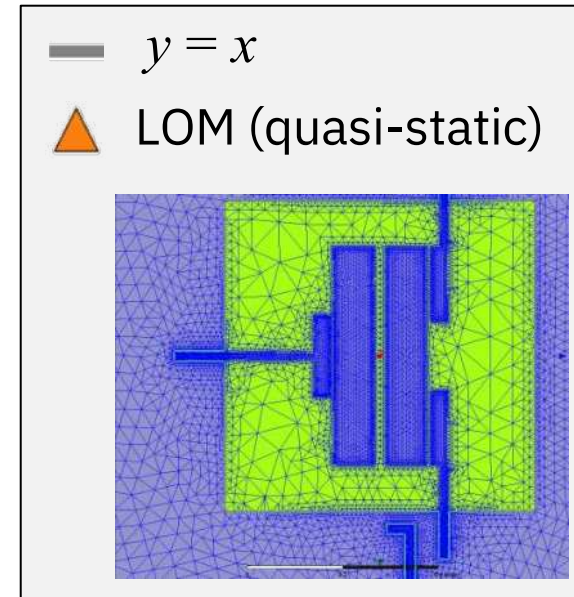
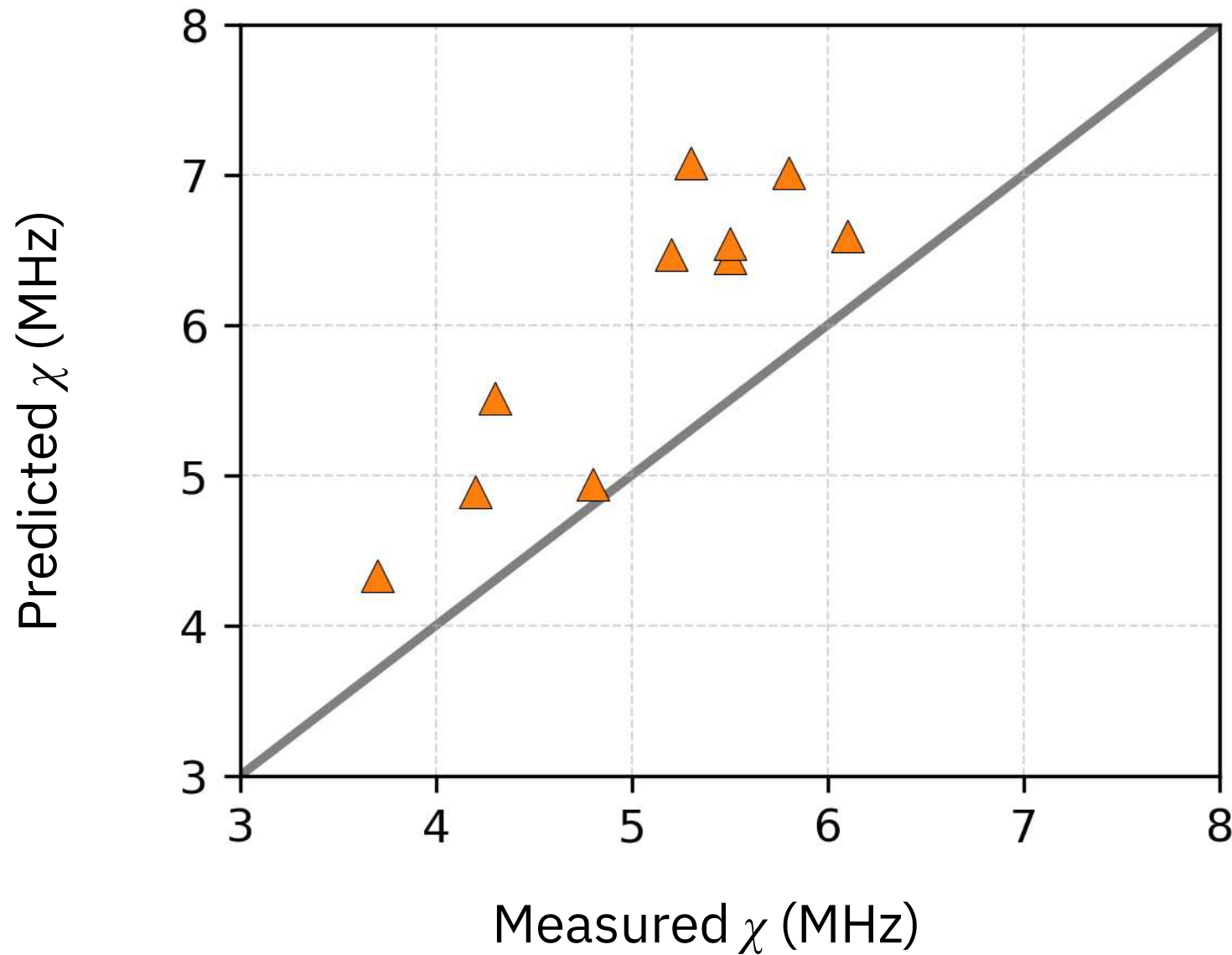
[Submitted on 2 Feb 2021]

Exploiting dynamic quantum circuits in a quantum algorithm with superconducting qubits

Antonio D. Corcoles, Maika Takita, Ken Inoue, Scott Lekuch, Zlatko K. Minev, Jerry M. Chow, Jay M. Gambetta

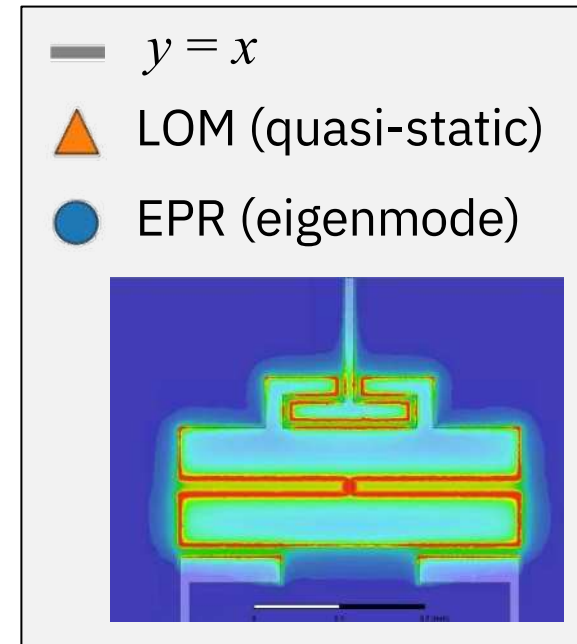
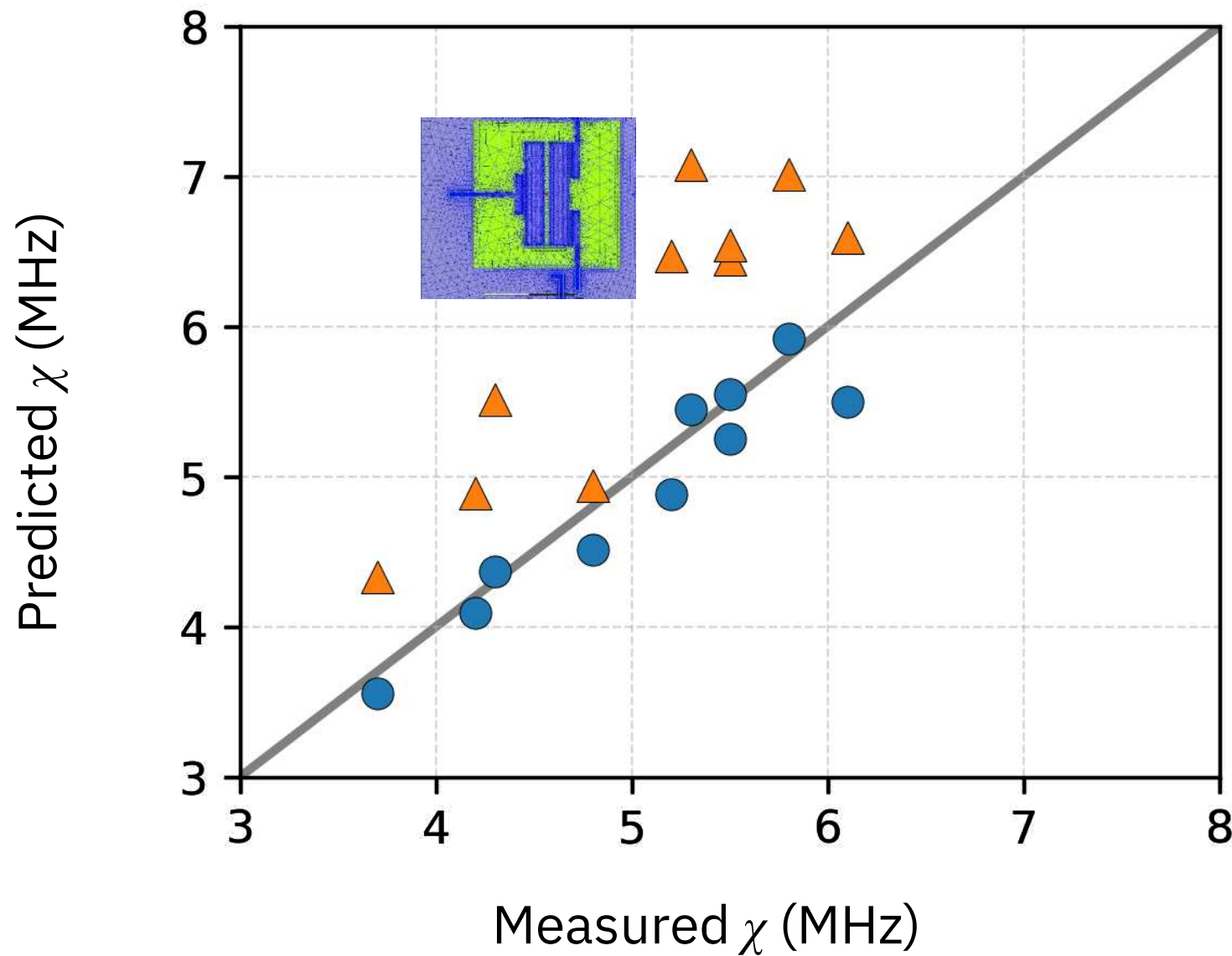


Measured vs. predicted: qubit-readout cross-Kerr



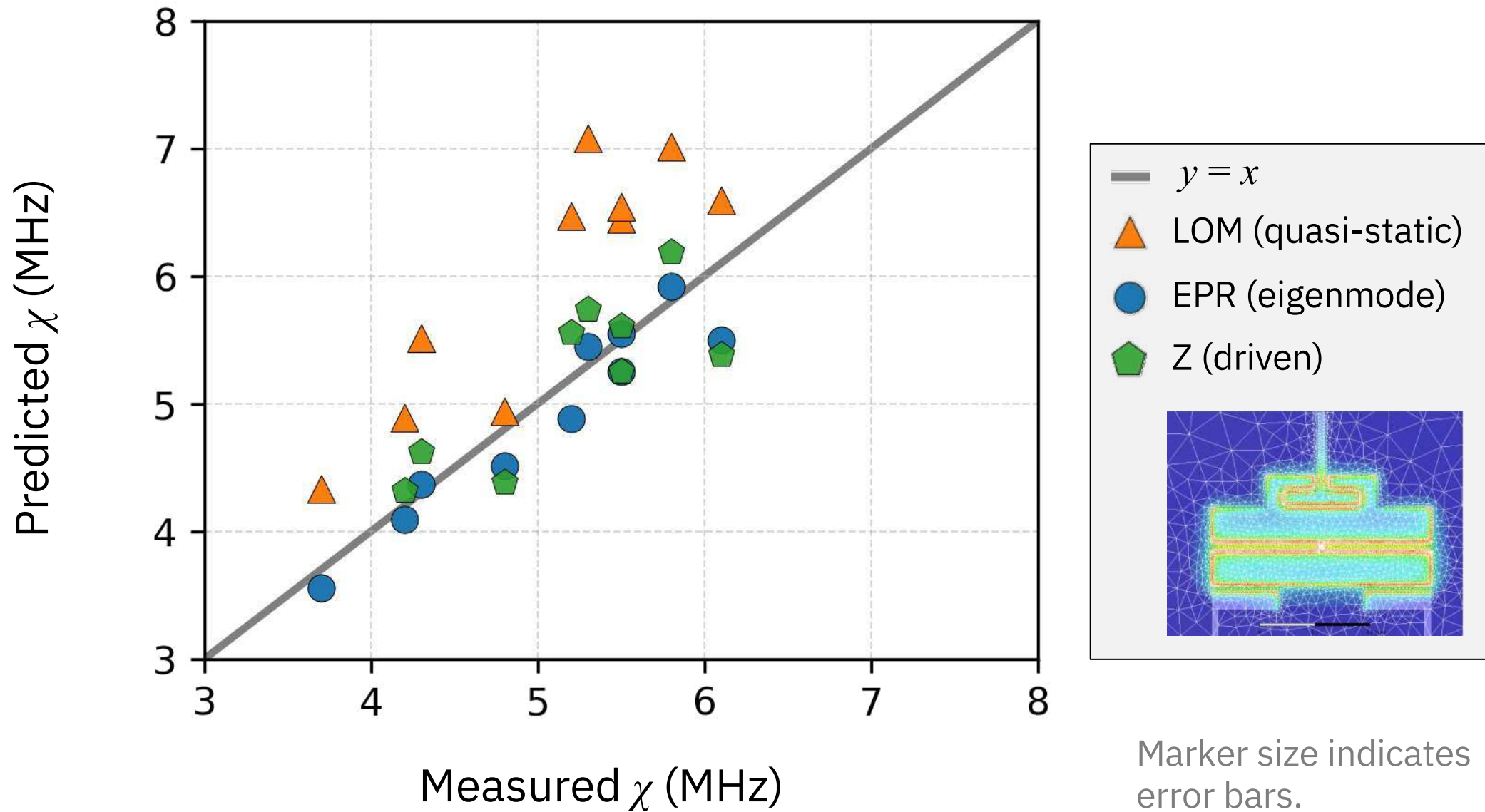
Marker size indicates error bars.

Measured vs. predicted: qubit-readout cross-Kerr



Marker size indicates error bars.

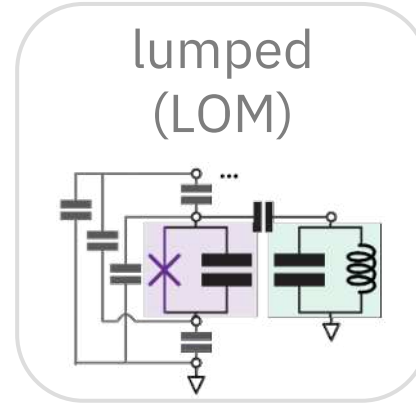
Measured vs. predicted: qubit-readout cross-Kerr



Quantization approaches



Quasi-static



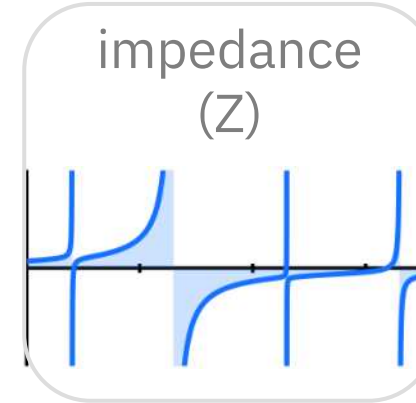
19%

higher

fastest and
cheapest

low

Full-wave

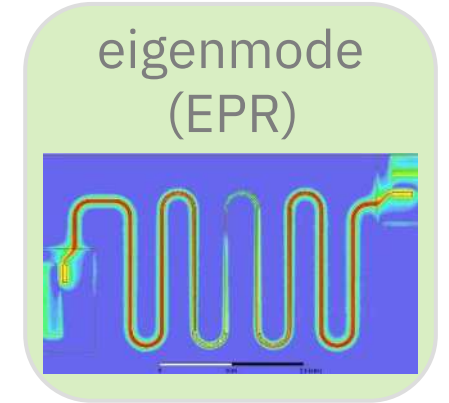


7%

low

moderate

high



4%

low

moderate

highest*

Agreement
for presented results

External inputs

Speed

Generality

* includes dissipative params



Search or jump to...



zlatko-minev / pyEPR

Welcome to pyEPR 🍺! (see [arXiv:2010.00620](#))

Open Source awesome star 78 fork 87 Install with conda pypi package 0.8.4.5

Automated Python module for the design and quantization of Josephson quantum circuits

pyEPR Working group meeting -- Planning for the future of pyEPR

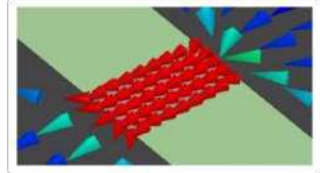
- Please sign-up here: <https://github.com/zlatko-minev/pyEPR/issues/45> or [directly here](#) !! 🍺
- See [pyEPR wiki](#) for notes from first meeting.
- We will schedule a follow-up meeting in 1-2 mo.

Who uses pyEPR?

- Yale University, Michel Devoret lab [QLab](#), CT, USA
- Yale University, Rob Schoelkopf lab [RSL](#), CT, USA
- [IBM Quantum](#)
- [QUANTIC](#) (QUANTUM INFORMATION CIRCUITS), PARISINRIA, ENS, MINES PARISTECH, Groups of Zaki Leghtas and team. France
- [Quantum Circuit Group](#) Benjamin Huard, Ecole Normale Supérieure de Lyon, France
- Emanuel Flurin, CEA Saclay, France
- Ioan Pop group, KIT Physikalisches Institut, Germany
- UC Berkeley, [Quantum Nanoelectronics Laboratory](#), Irfan Siddiqi, CA, USA
- [Quantum Circuits, Inc.](#), CT, USA
- [Seeqc](#) (spin-out of Hypres) Digital Quantum Computing, USA
- Serge [Rosenblum Lab] quantum circuits group (<https://www.weizmann.ac.il/condmat/ro>) Weizmann Institute, Israel
- University of Oxford - LeekLab - Peter [Leek Lab](#), UK
- Britton [Plourde Lab](#), Syracuse University, USA
- Javad [Shabani Lab](#) Quantum Materials & Devices, NYU, NY, USA
- UChicago Dave Schuster Lab, USA
- SQC lab - Shay Hacohen Gourgy, Israel
- Lawrence Berkeley National Lab
- Colorado School of Mines, USA
- Syracuse University, USA
- IPQC, SJTU, Shanghai, China
- Bhabha Atomic Research Centre, India
- Quantum Computing UK
- Alice&Bob, France
- Centre for Quantum Technologies / Qcrew
- Quantum Device Lab ETHZ; Andreas Wallraff
- ... and many more! (Please e-mail zlatko.minev@aya.yale.edu with updates.)

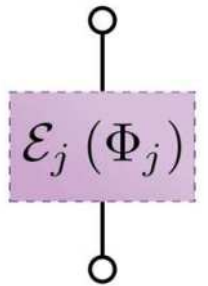
Energy-Participation-Ratio (EPR) Approach

Simple and unified analysis of dissipation and Hamiltonian
single-simulation efficient
obviate driven simulations and notion of $Z_{ij}(\omega)$



General

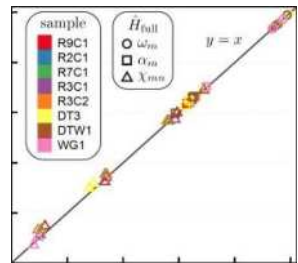
handle arbitrary architectures and non-linear devices
e.g., nanowires, ...



Accuracy

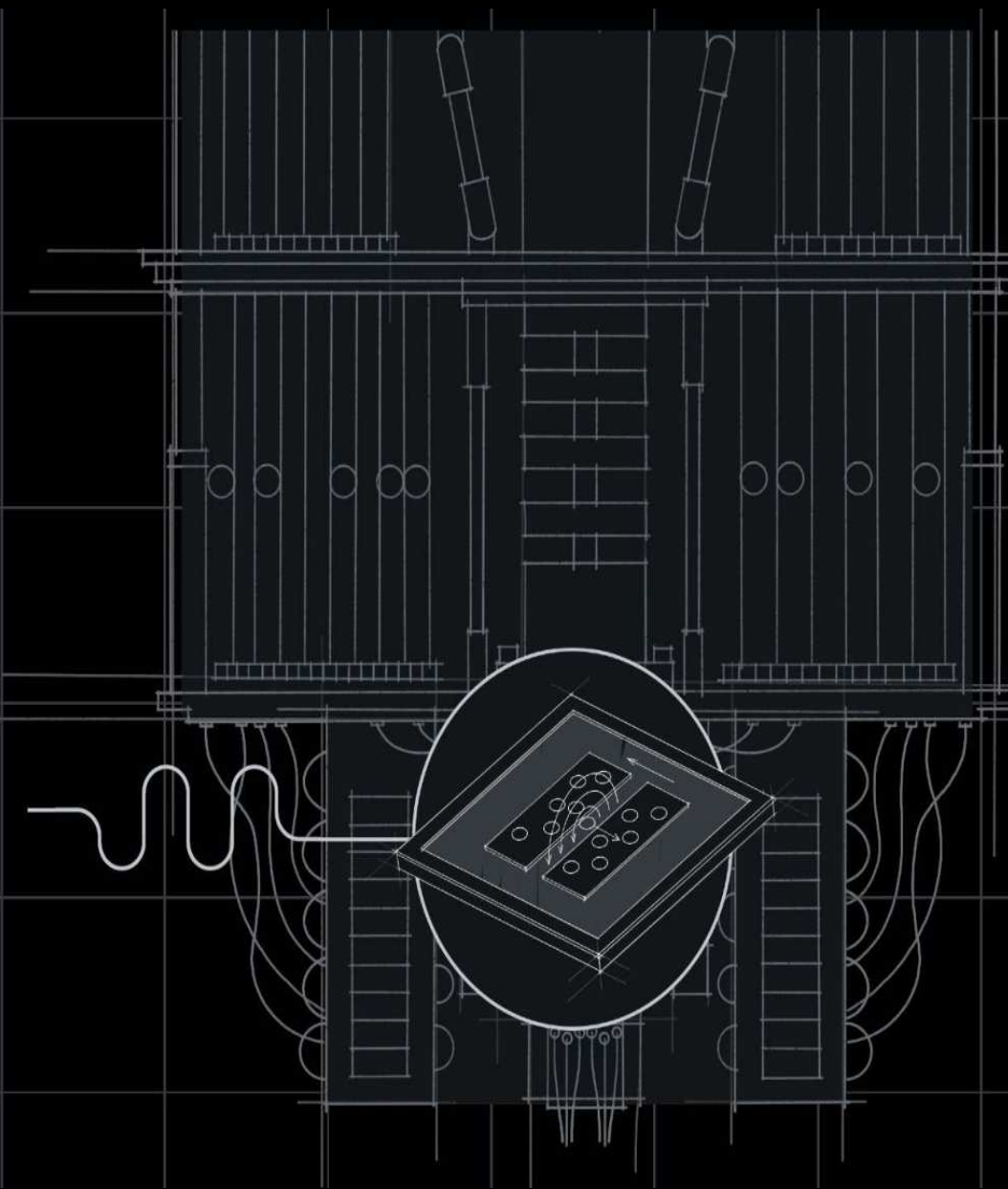
experimental ten to percent level over 5 orders in $\hat{H}_{\text{full}}$

Suited for automated, robust analysis of large systems
fully automated and tested (see GitHub)



arXiv:1902.10355 (Ch. 4);

Application to J and CR gate, ...



Qiskit | quantum device design

Qiskit | quantum device design

github.com/Qiskit/qiskit-metal

slack: [#qiskit-metal-early-access](#)

Project Metal

qiskit.org/metal

Open source
Early-access program (alpha version)



Zlatko Minev
PI & Lead



Thomas McConkey



Grace Harper



Priti Shah



John Blair



Dennis Wang



Marco Facchini



Jeremy Drysdale

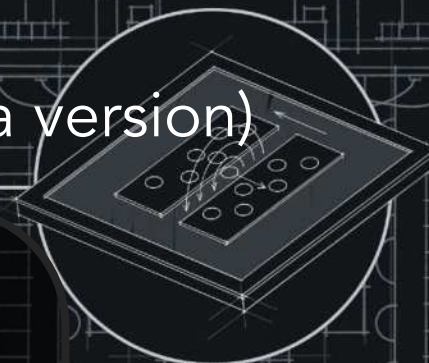


Will Shanks



Olivia Lanes

Power-Ups:

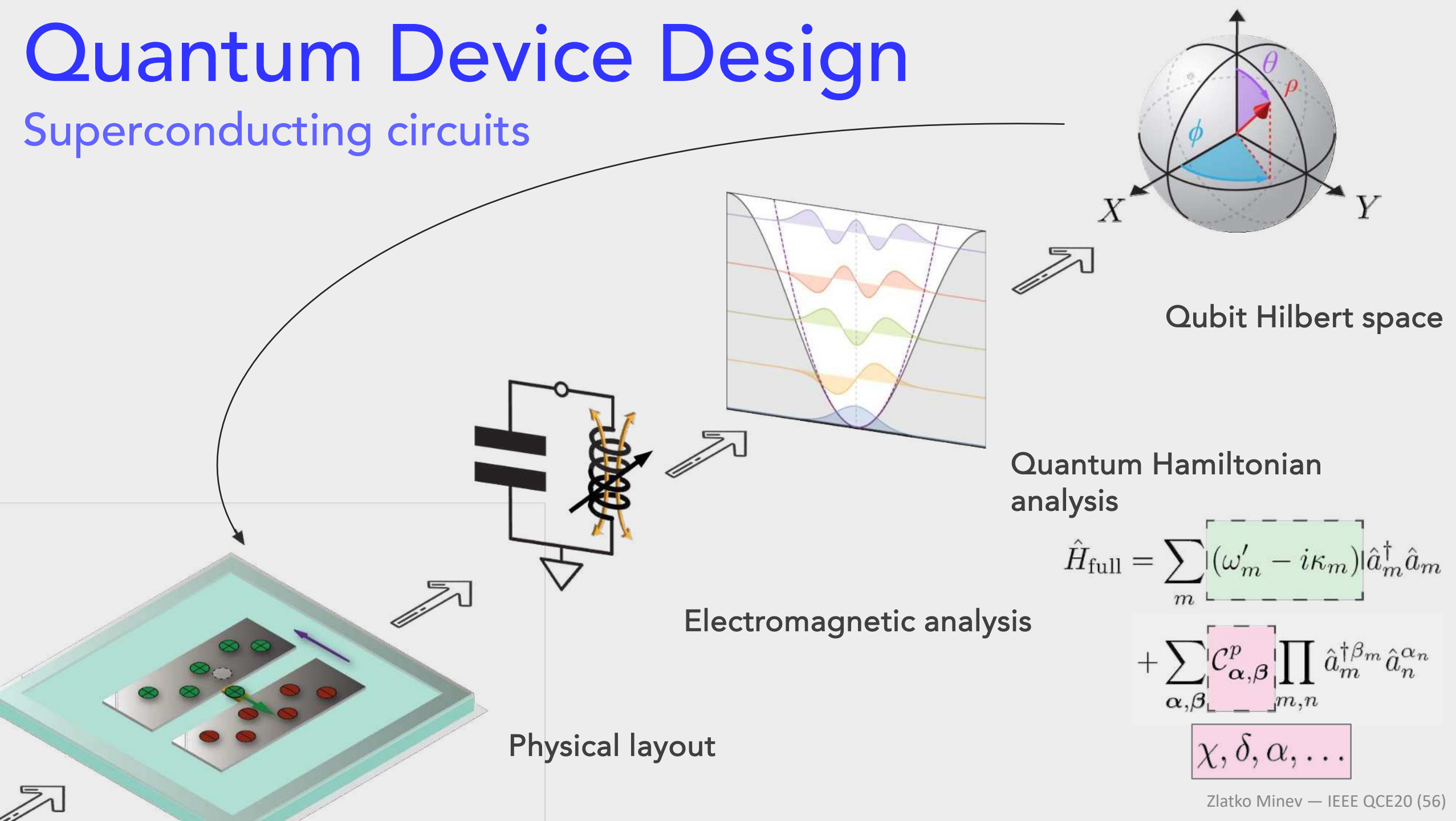


Close collab.:



Quantum Device Design

Superconducting circuits



Let's layout, analyze, and
optimize a four-qubit chip...

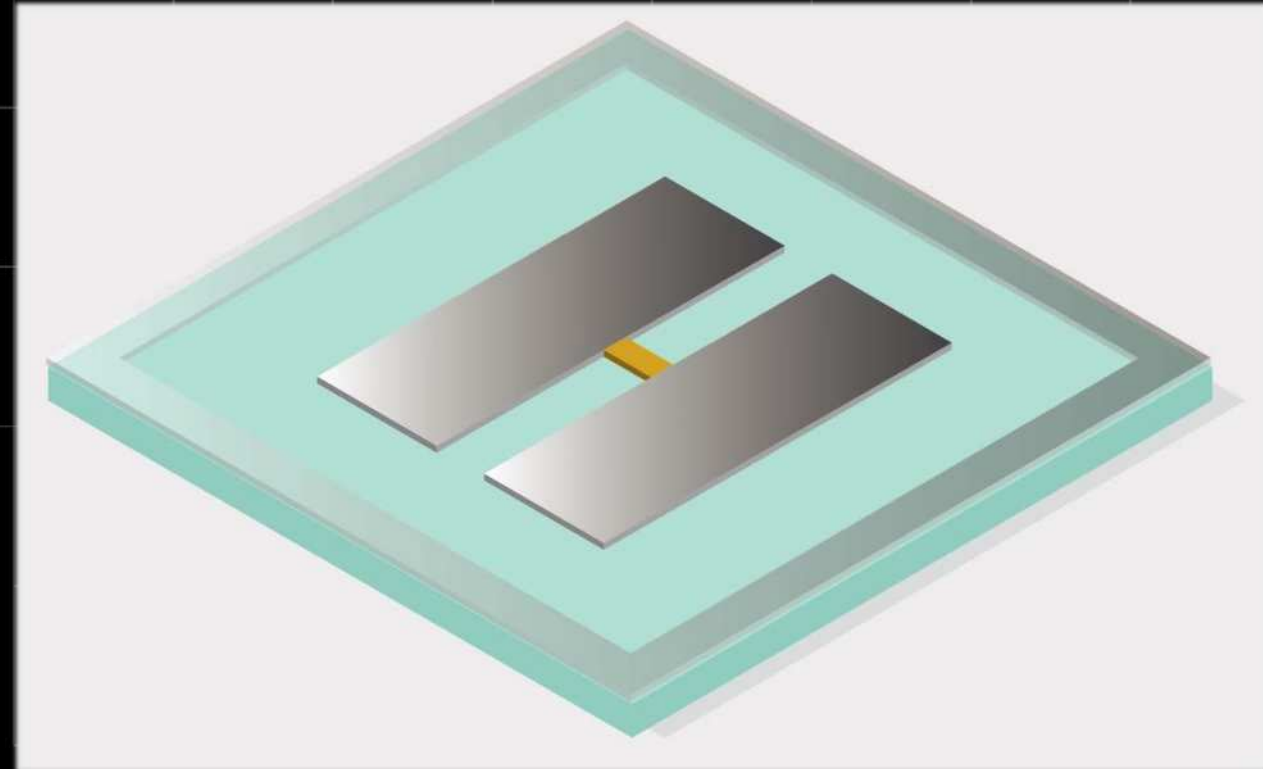
Qubit layout with project Qiskit Metal

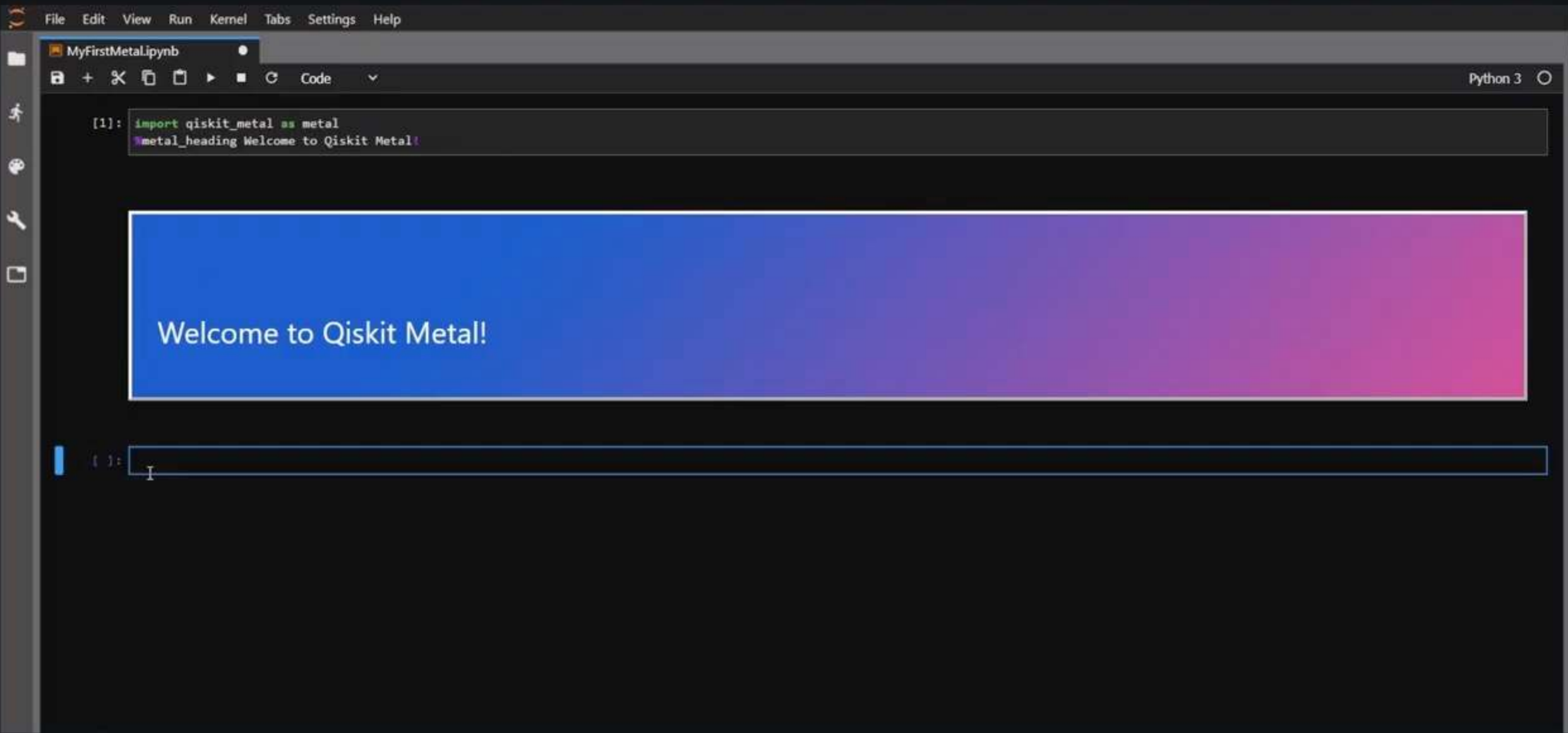
qiskit.org/metal

Create a transmon qubit

```
from qiskit_metal.qlibrary import qubits  
q1 = qubits.TransmonPocket('Q1', options=dict(...))
```

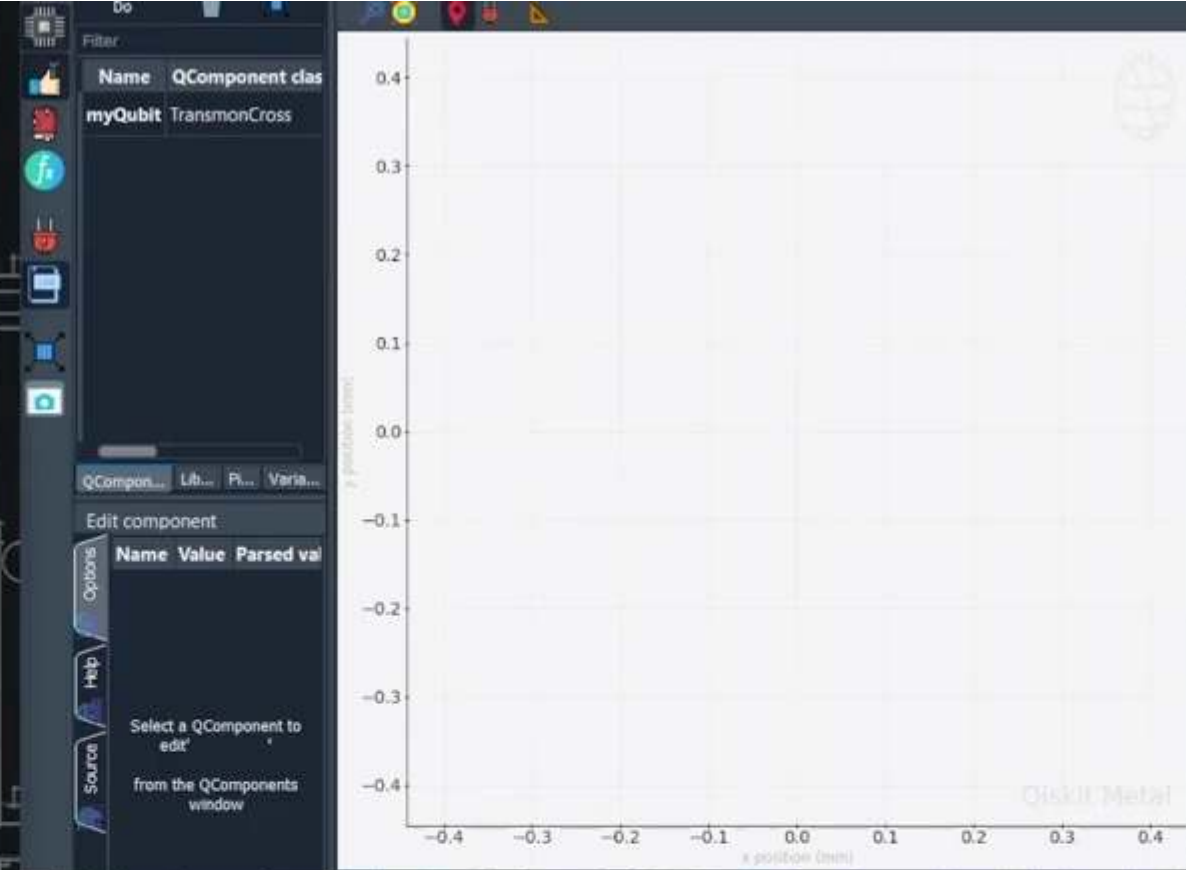
```
from qiskit_metal import MetalGUI  
MetalGUI()
```

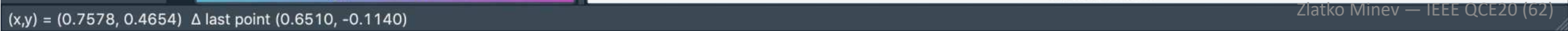


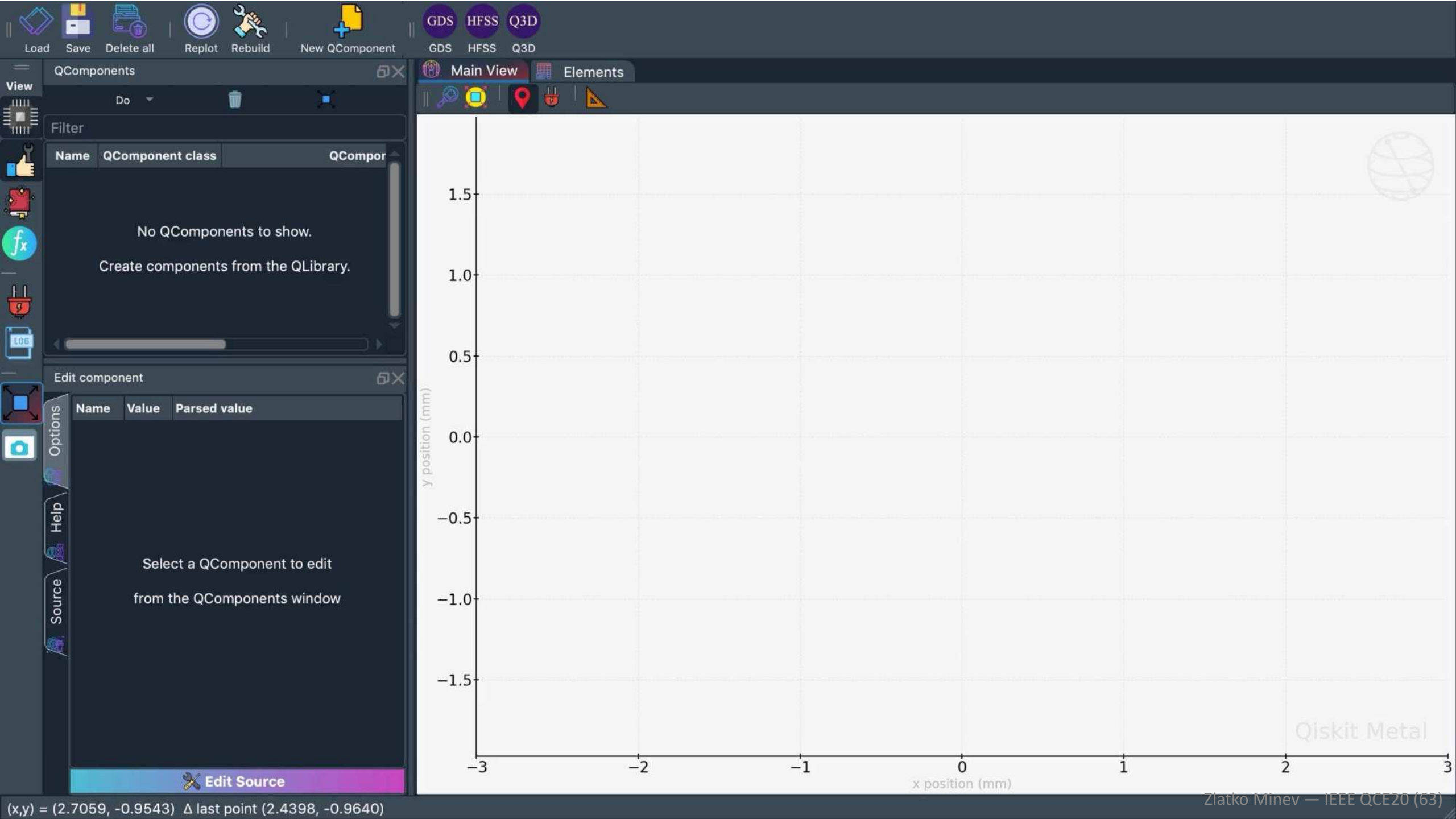


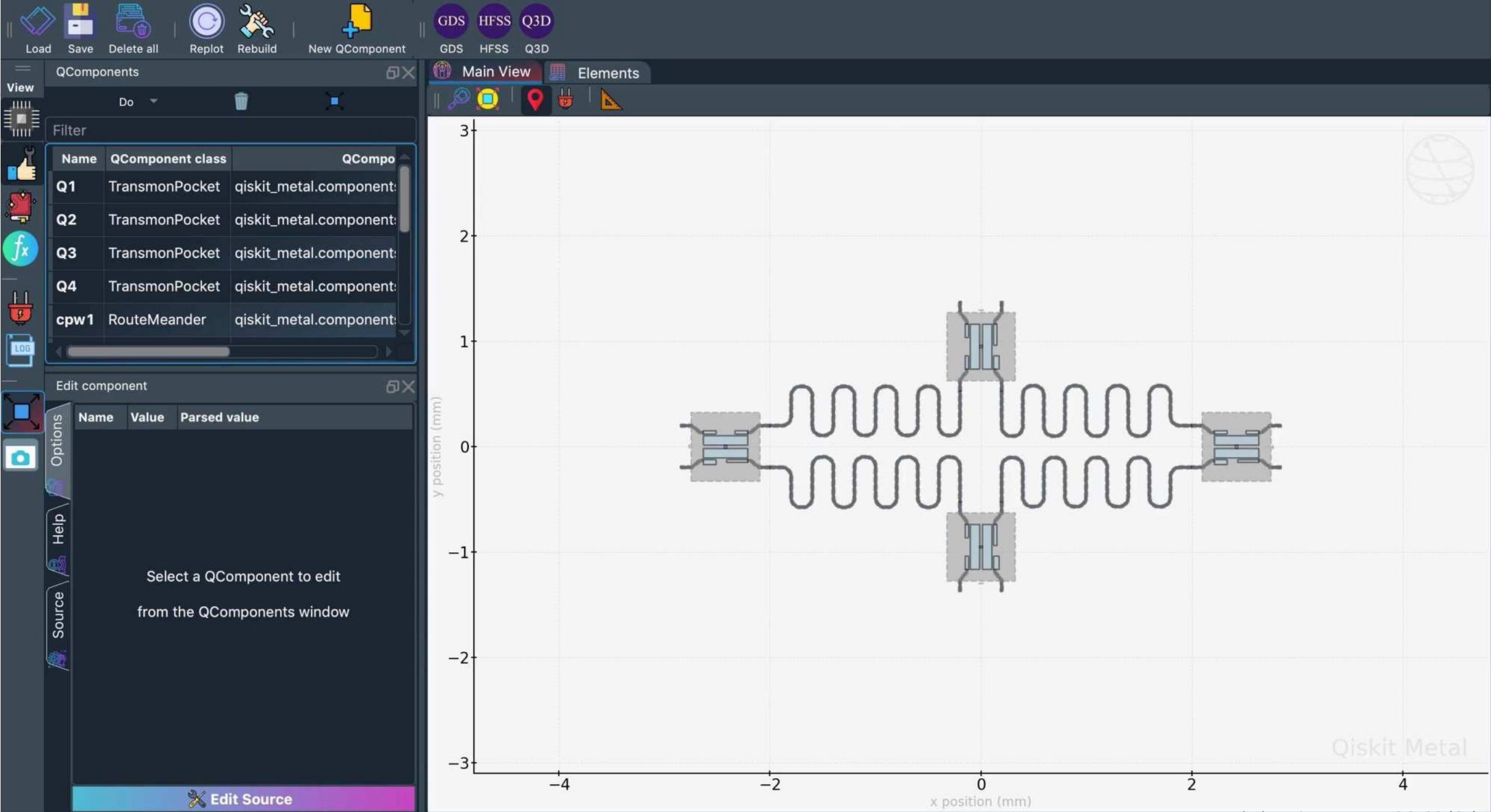
Qiskit | quantum device design

qiskit.org/metal









QComponents

View

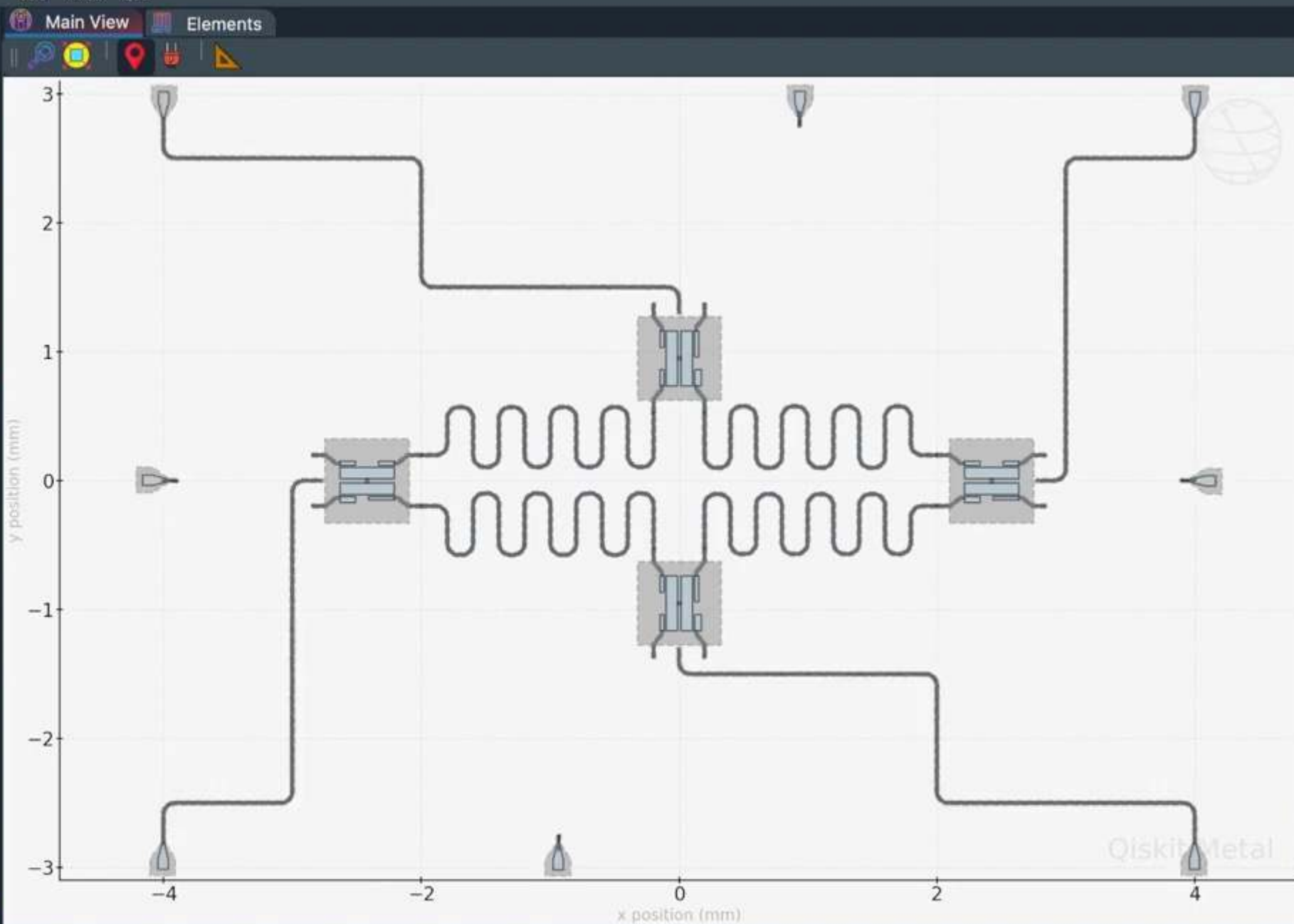
Filter

Name	QComponent class	QCompo
Q1	TransmonPocket	qiskit_metal.component:
Q2	TransmonPocket	qiskit_metal.component:
Q3	TransmonPocket	qiskit_metal.component:
Q4	TransmonPocket	qiskit_metal.component:
cpw1	RouteMeander	qiskit_metal.component:

Edit component

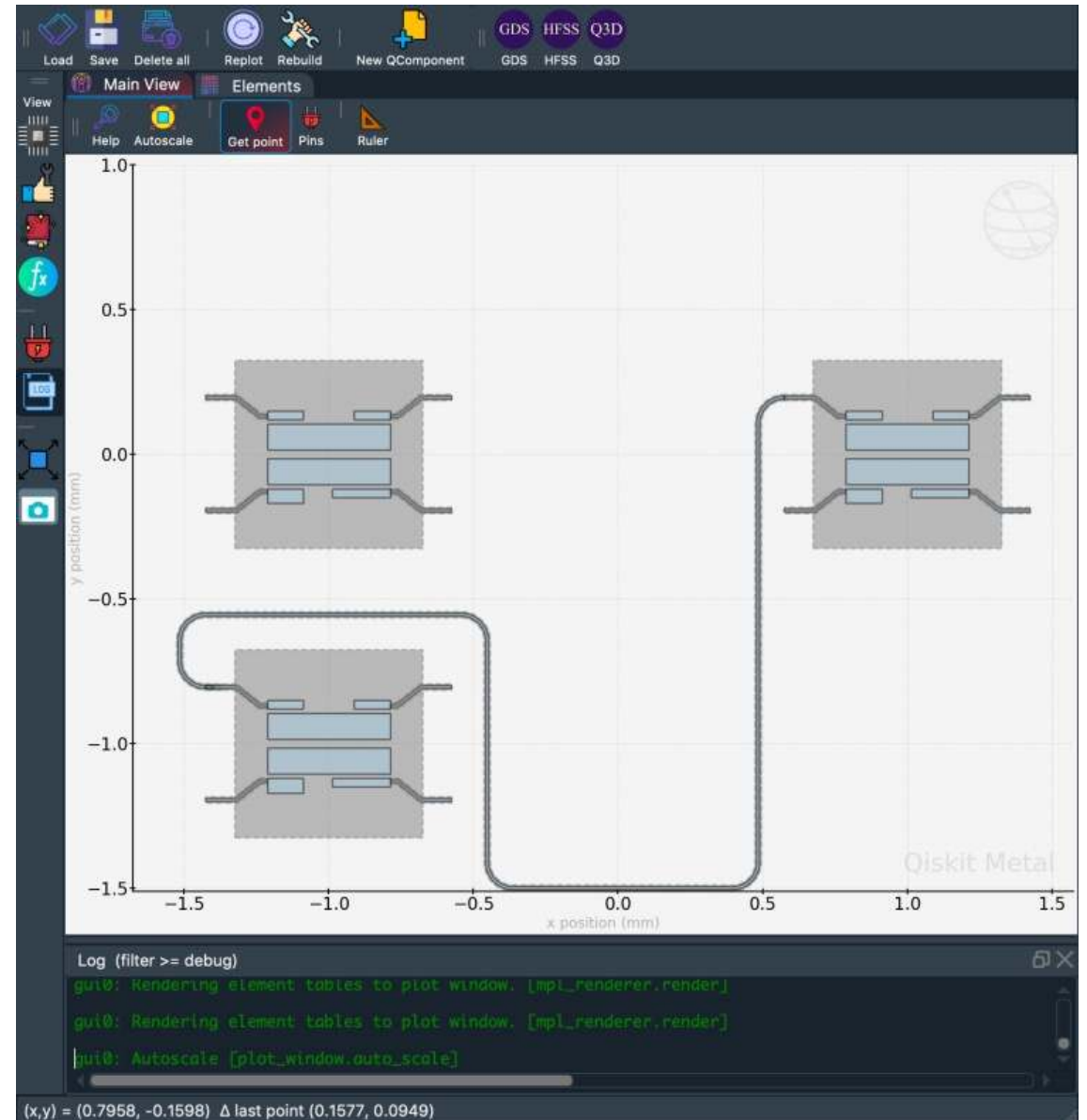
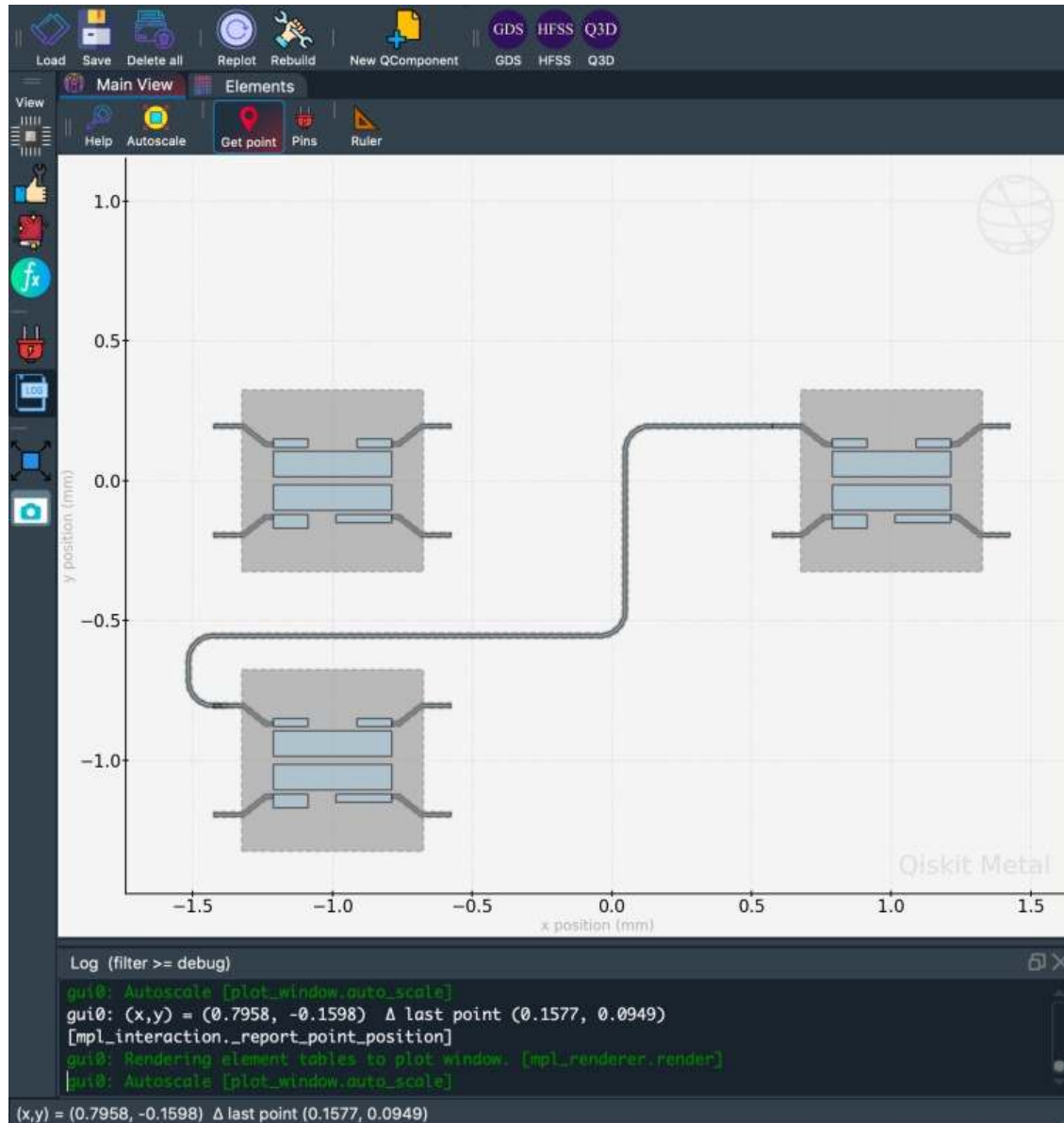
Name	Value	Parsed value
Select a QComponent to edit from the QComponents window		

Edit Source



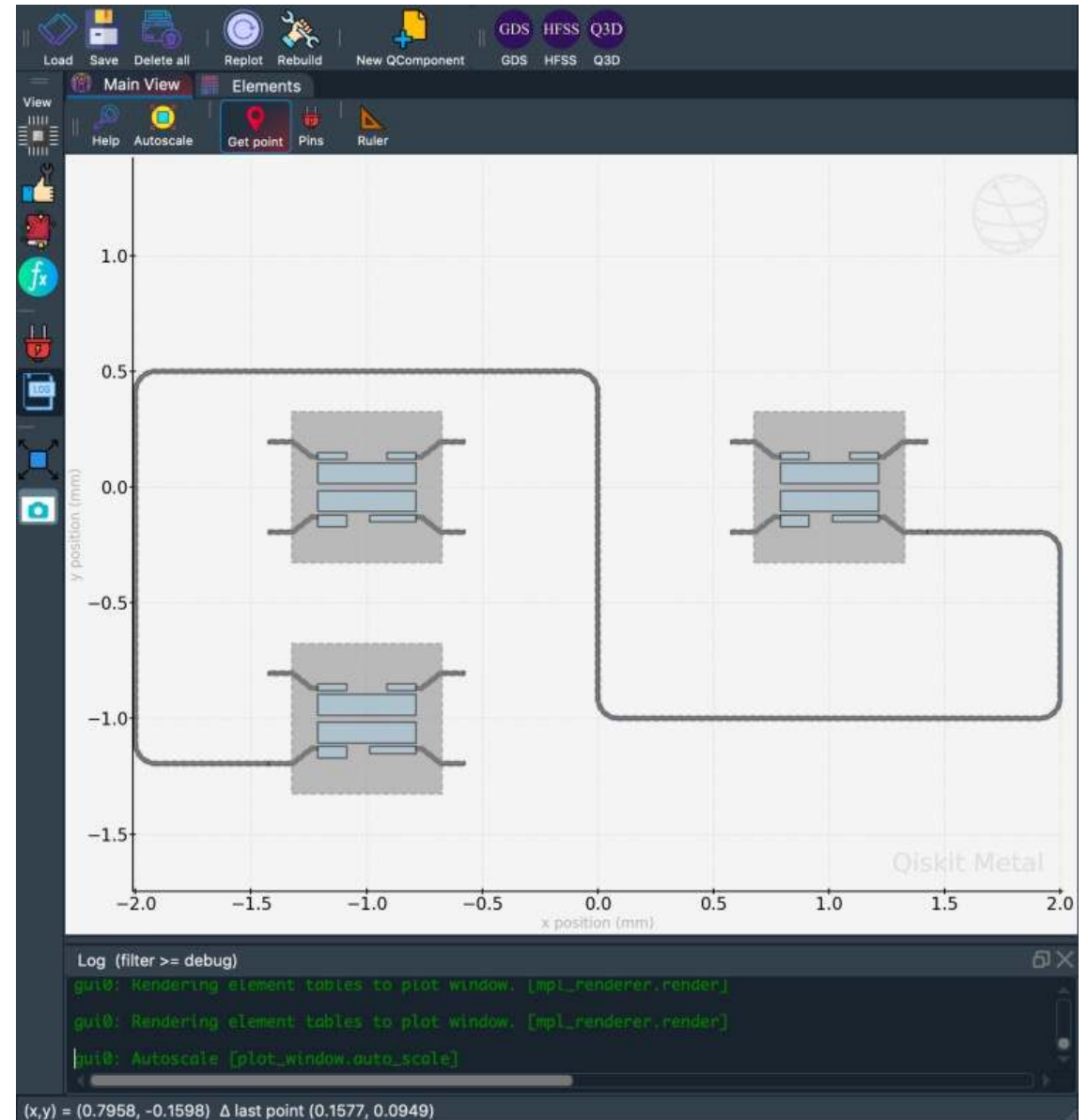
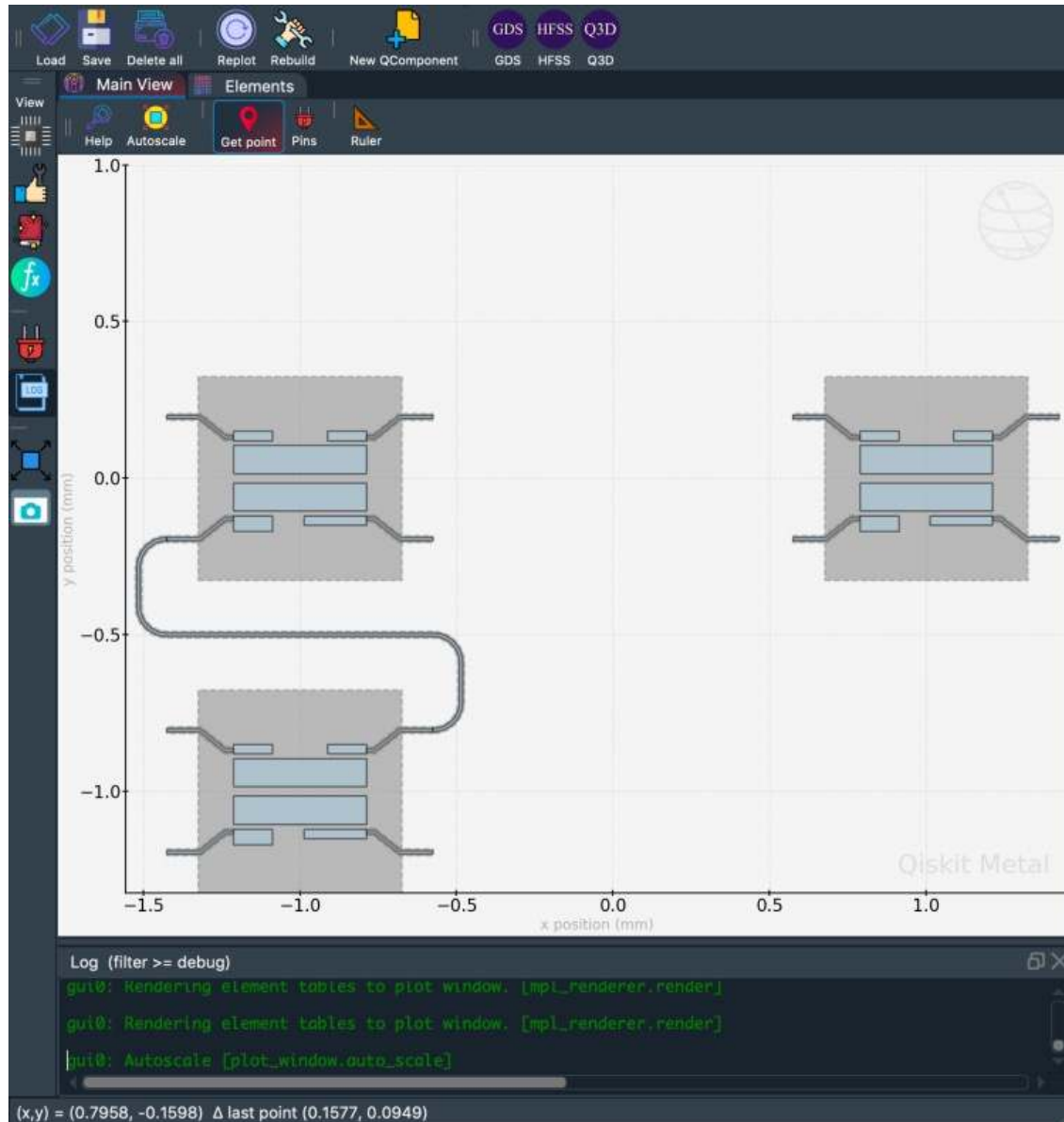
Demo Notebook for Hybrid CPW Algorithm with Anchors

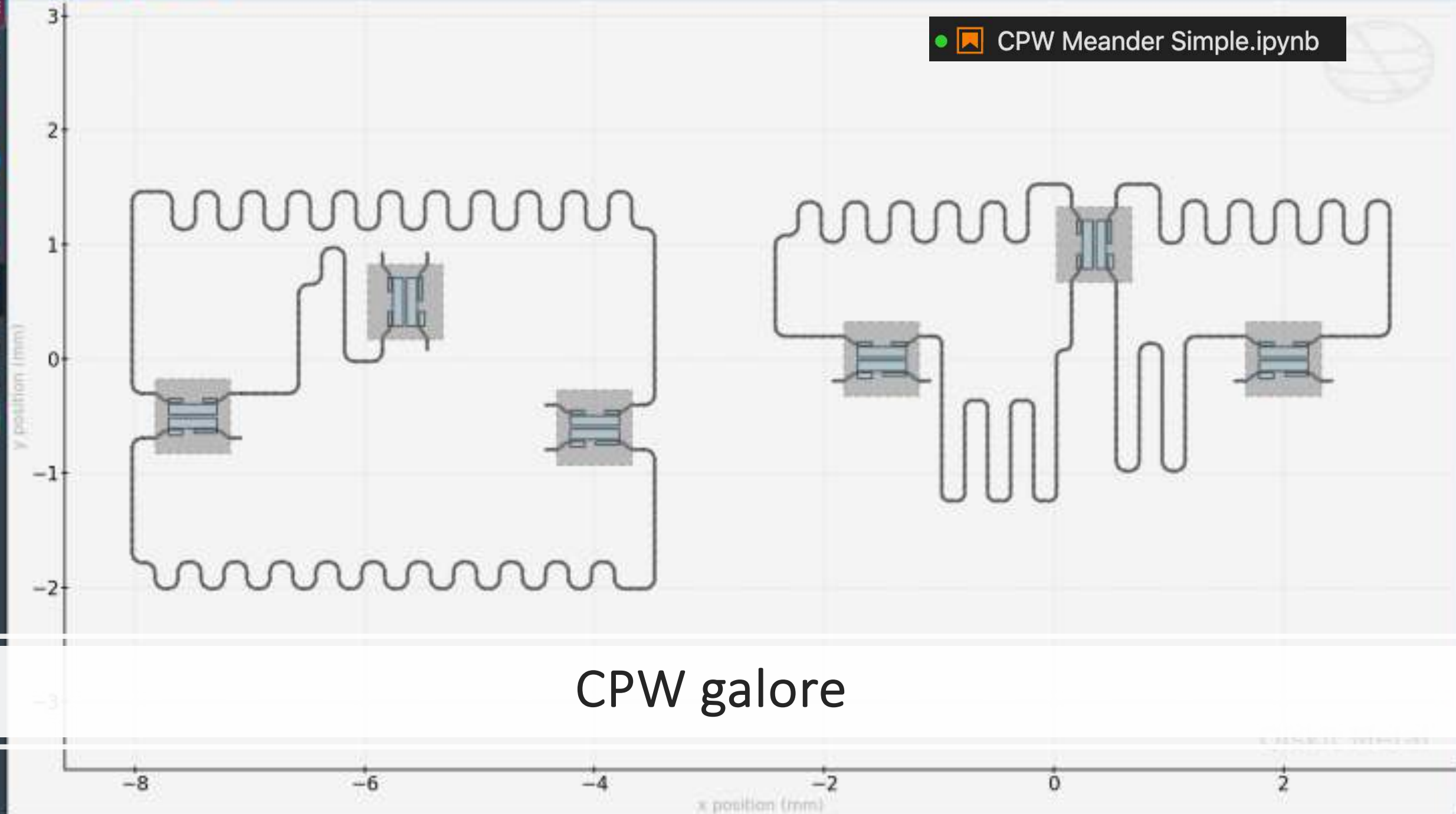
CPW Hybrid Auto and AStar.ipynb

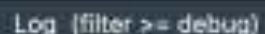


Demo Notebook for Hybrid CPW Algorithm with Anchors

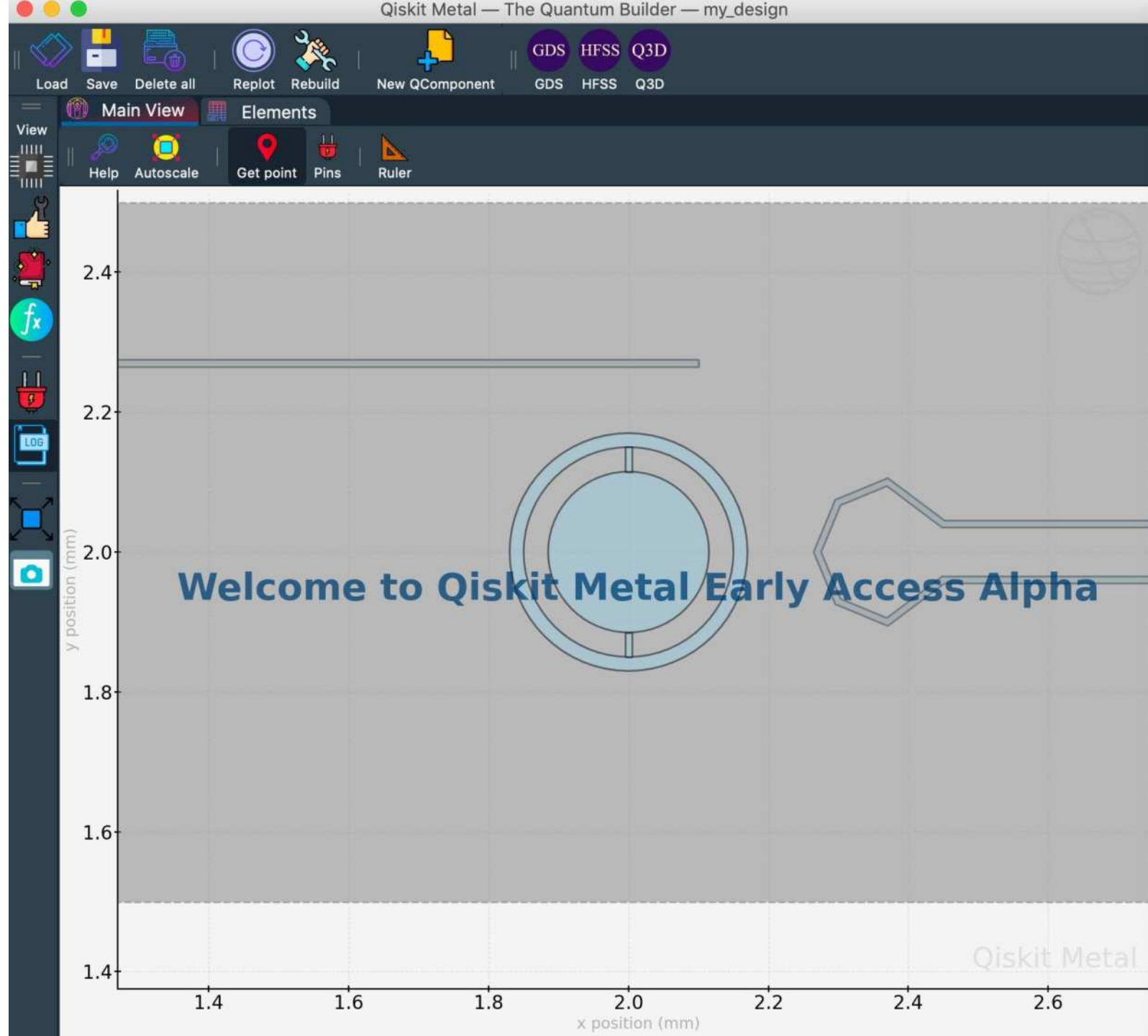
CPW Hybrid Auto and AStar.ipynb



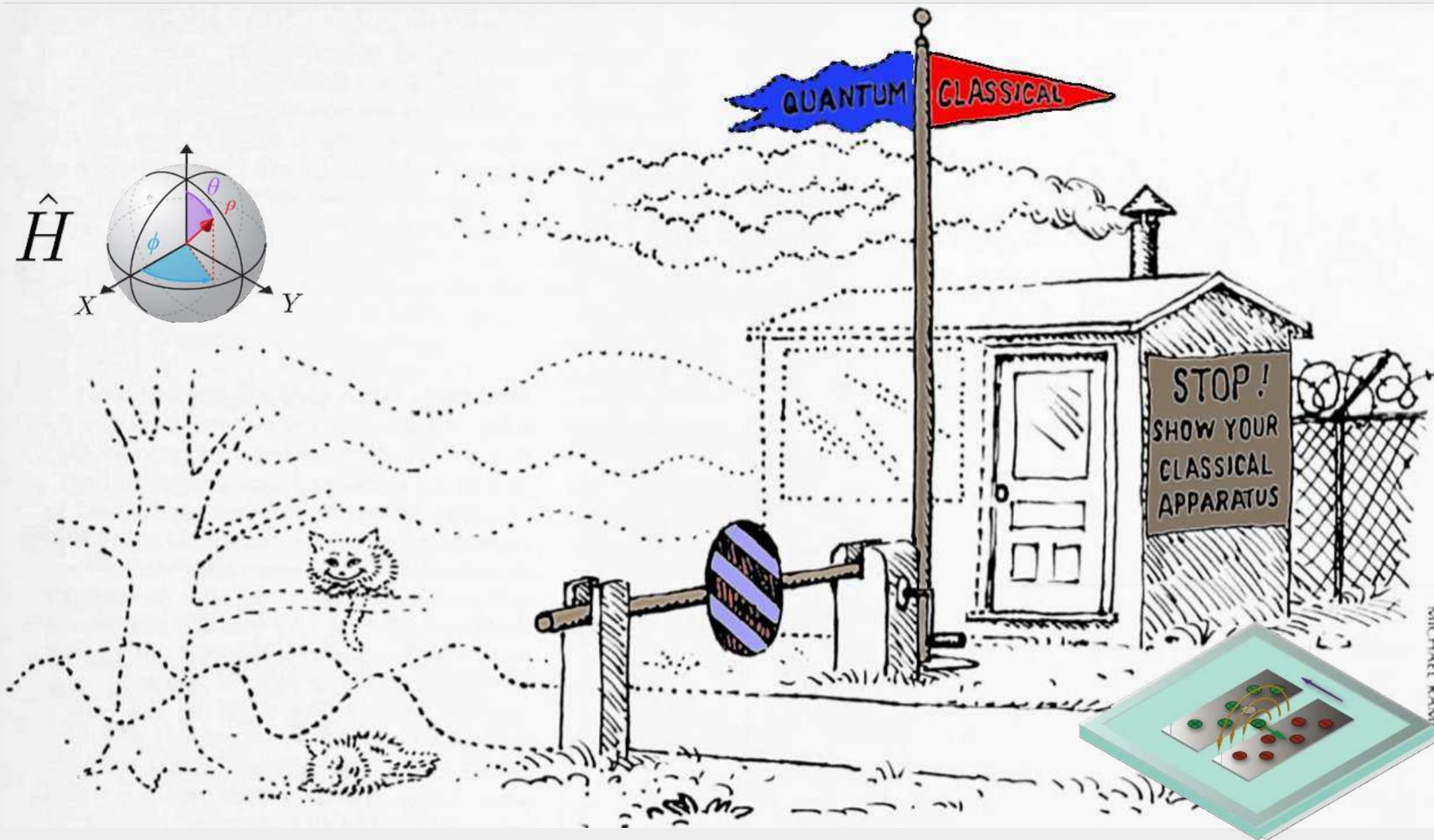




Concentric
transmon

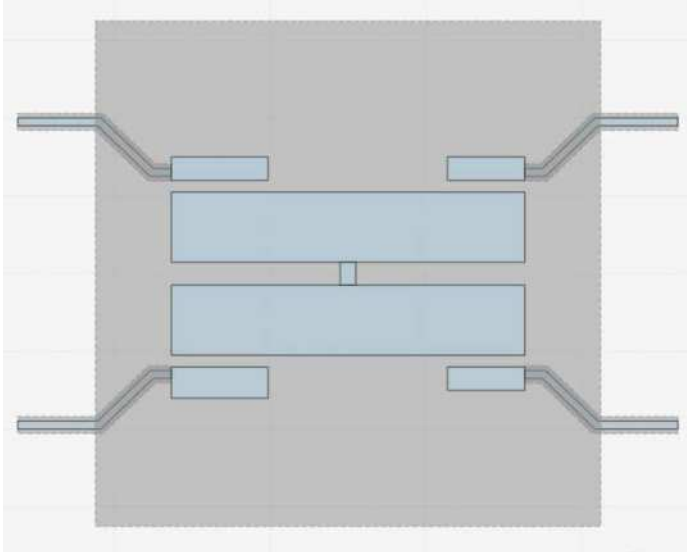


Analysis

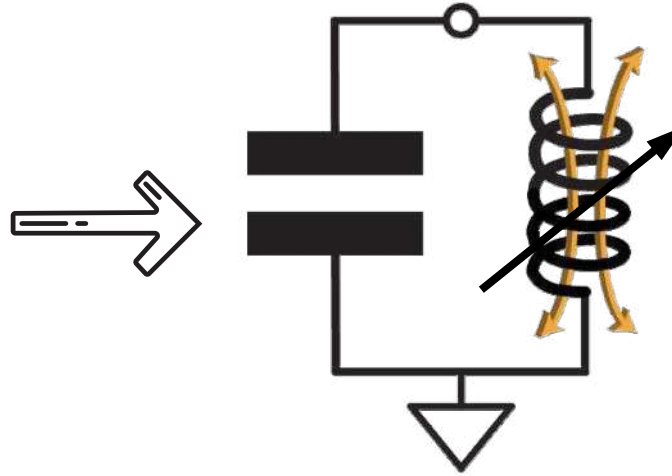


Drawing: Zurek, Physics Today (1991)

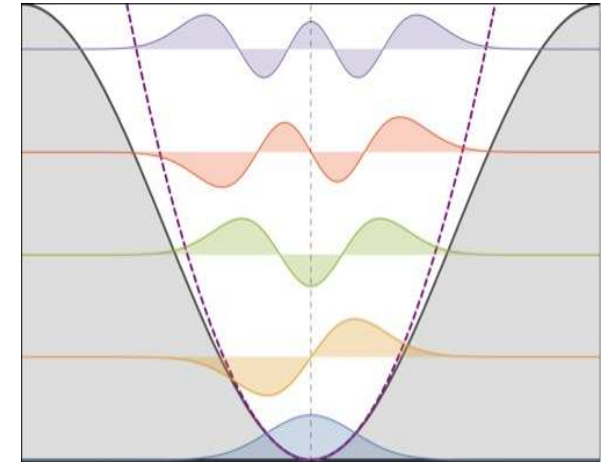
Rendering sub-component into simulation tool



Layout



Electromagnetic analysis



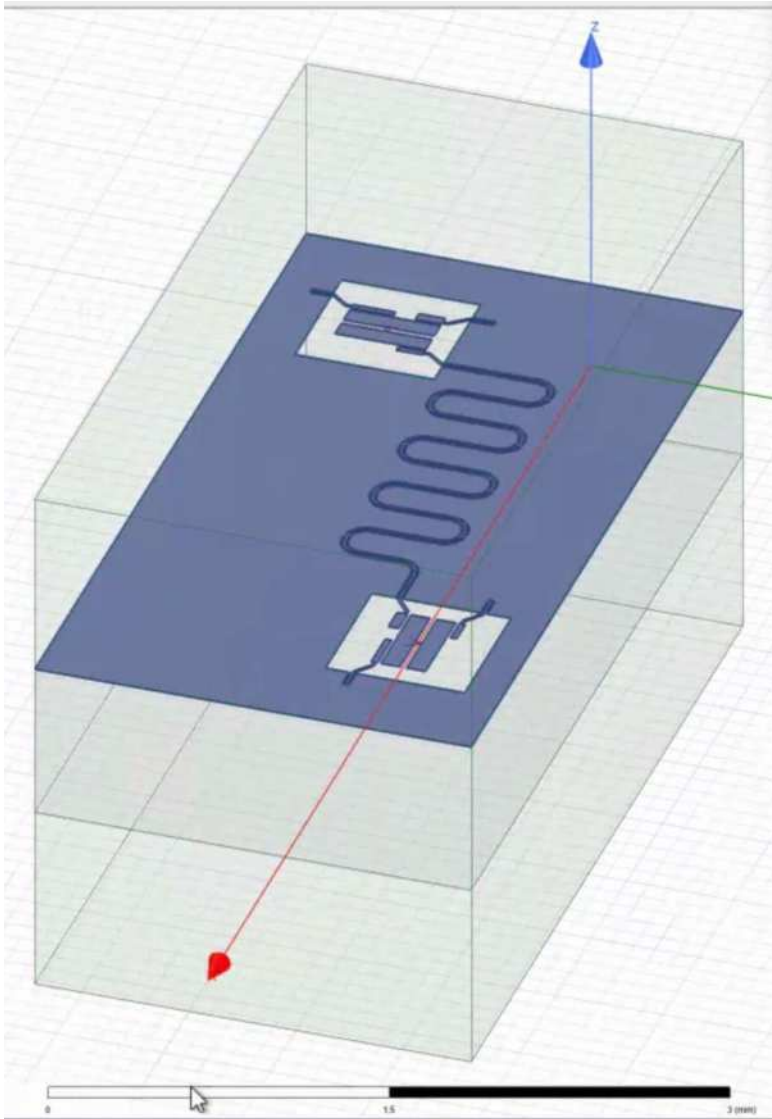
Quantum Hamiltonian analysis



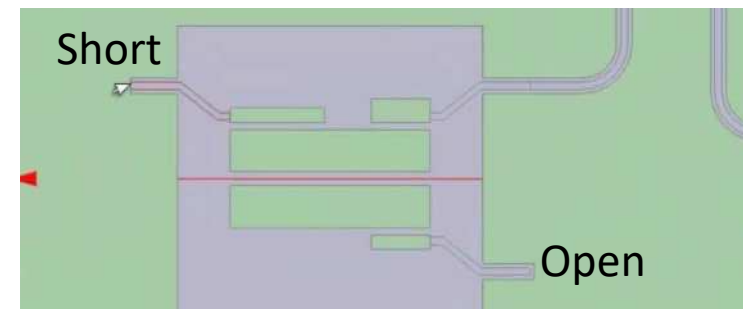
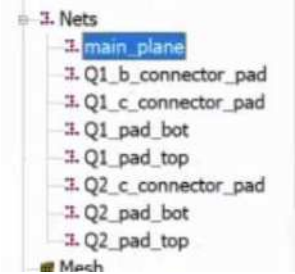
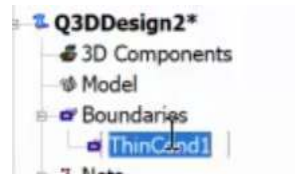
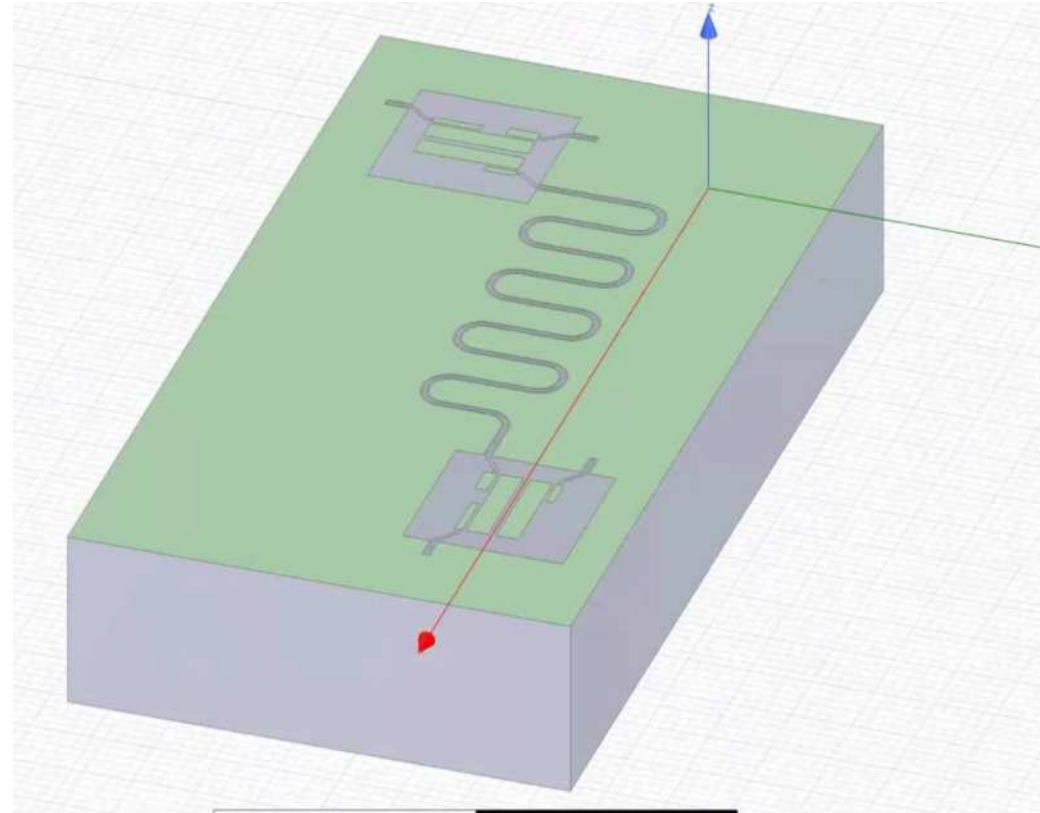
Qiskit Metal render & electromagnetic analysis

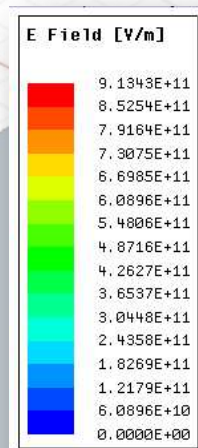
```
design.render(...)
```

HFSS



Q3D





Best practices

Native optimized curves

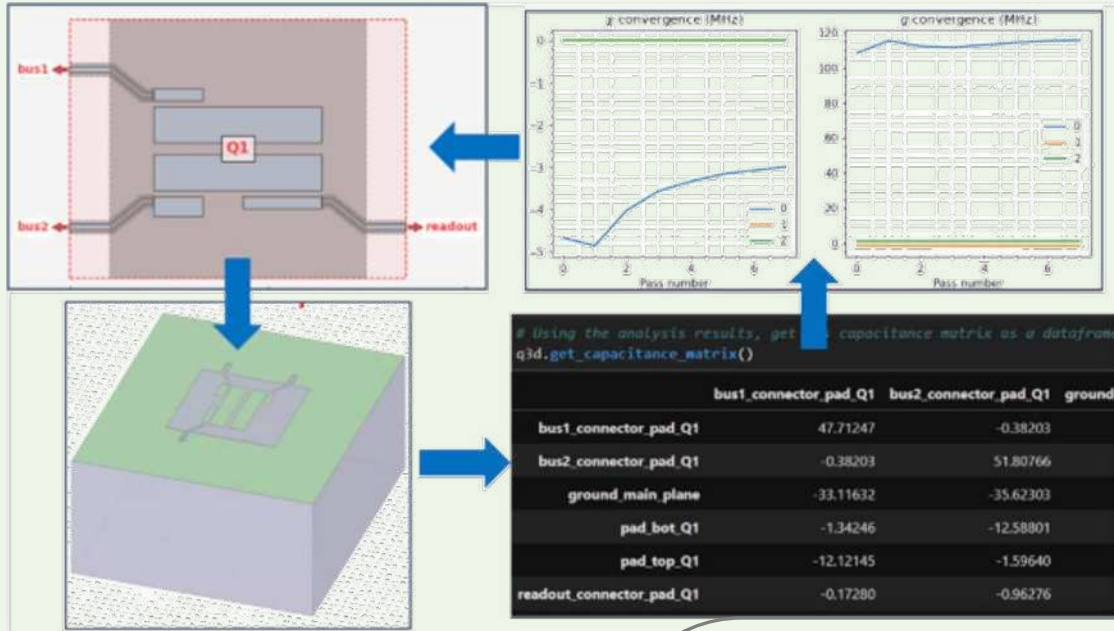
Smaller mesh RAM

0 0.35 0.7 (mm)

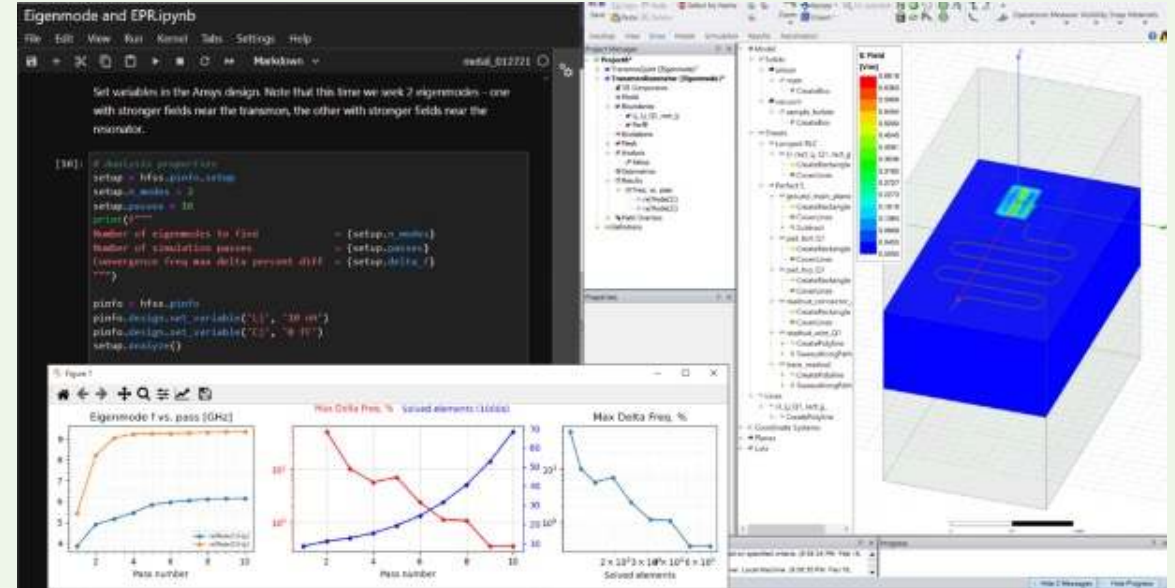
0 1 2 (mm)

Quantum analysis library

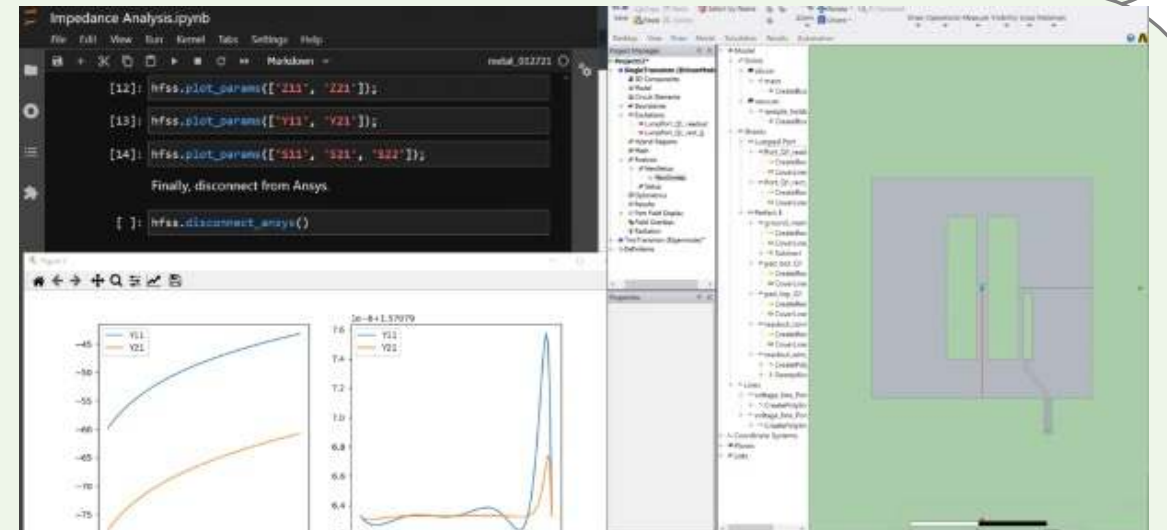
Capacitive



Eigenmode

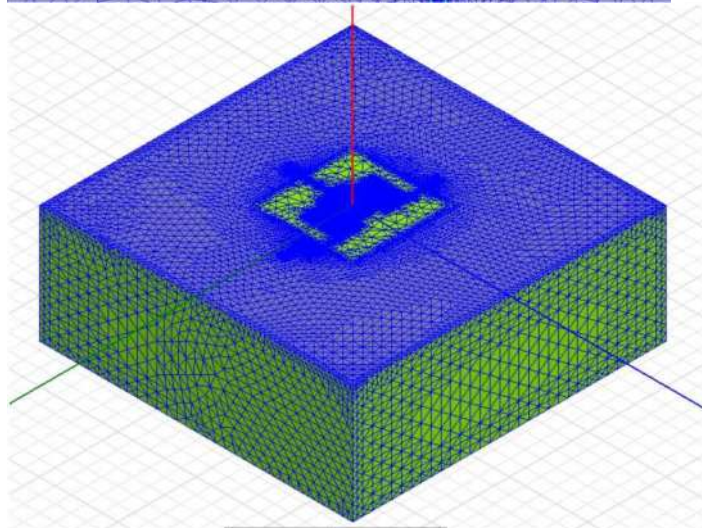
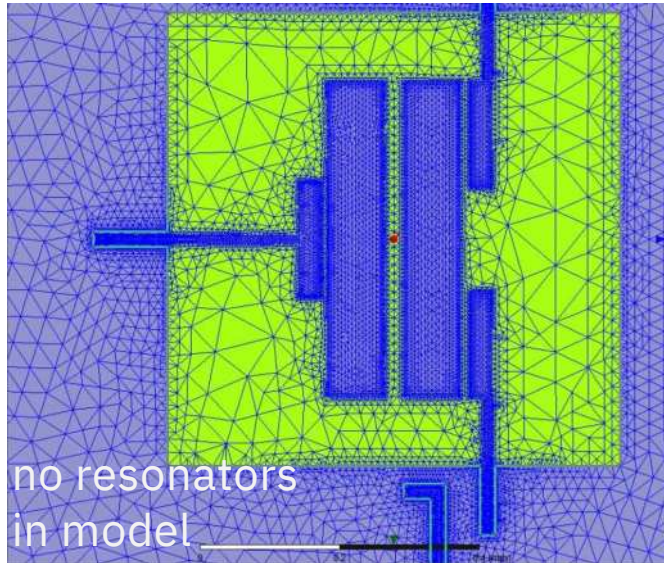


S, Z, Y Impedance Scattering

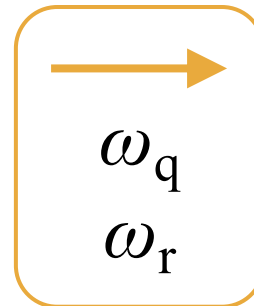
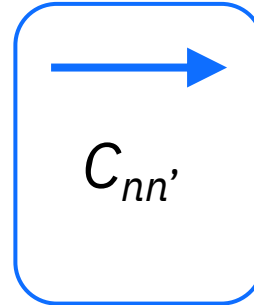


Lumped-oscillator model (LOM)

Quasi-static solver (Q3D)

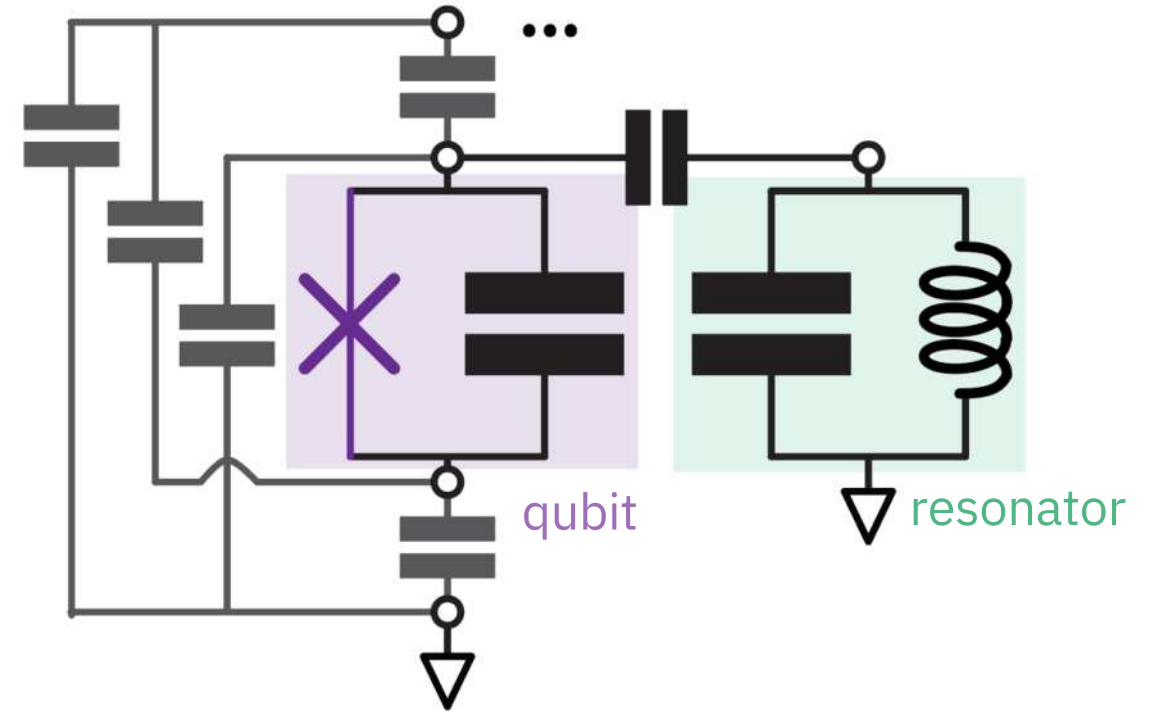


Provide



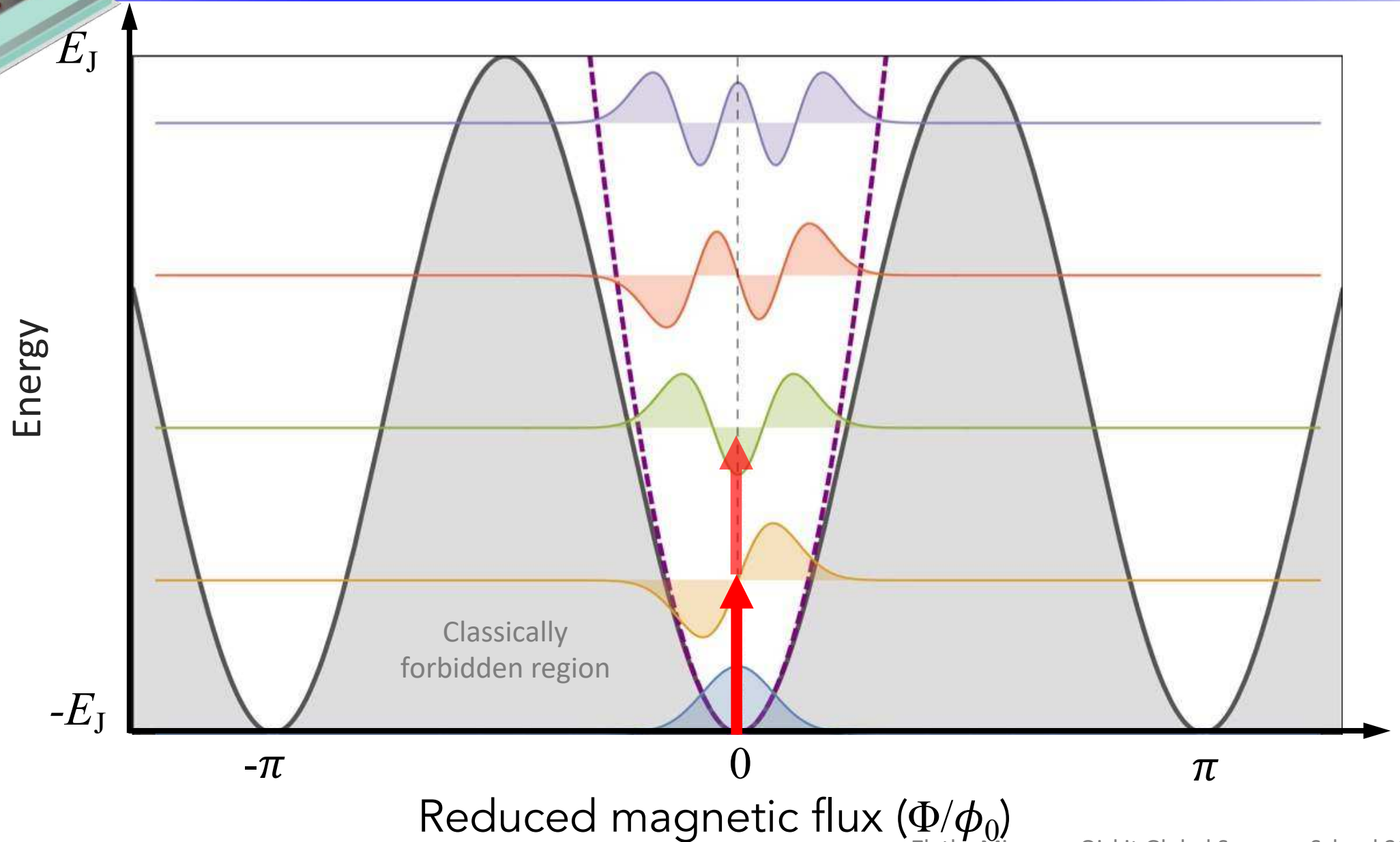
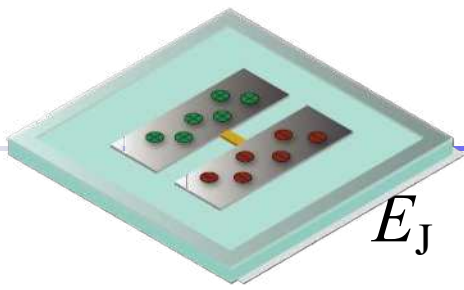
approximate
not suitable for 3D
some distributed effects beyond capture

Equivalent lumped model



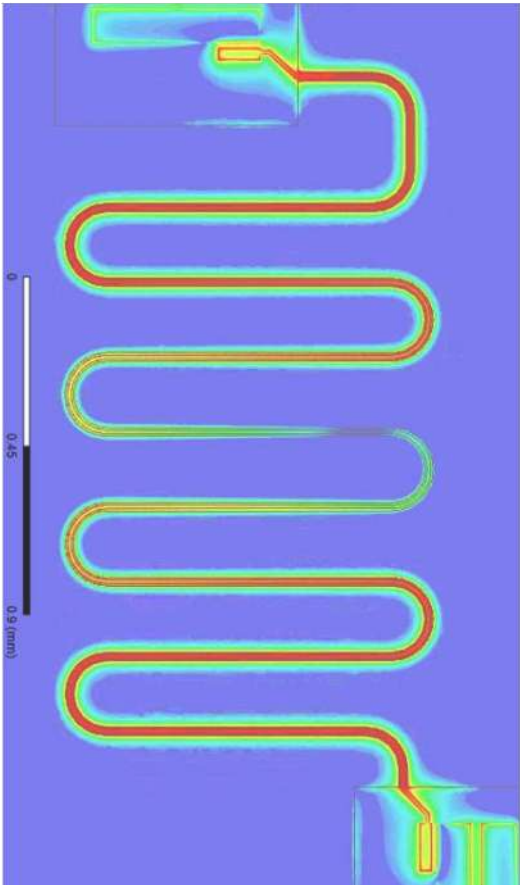
Yurke & Denker (1984); Devoret (1997); Burkard et al. (2004); Koch et al. (2007);... Minev et al., in prep (2021) ...

Exploring a transmon qubit

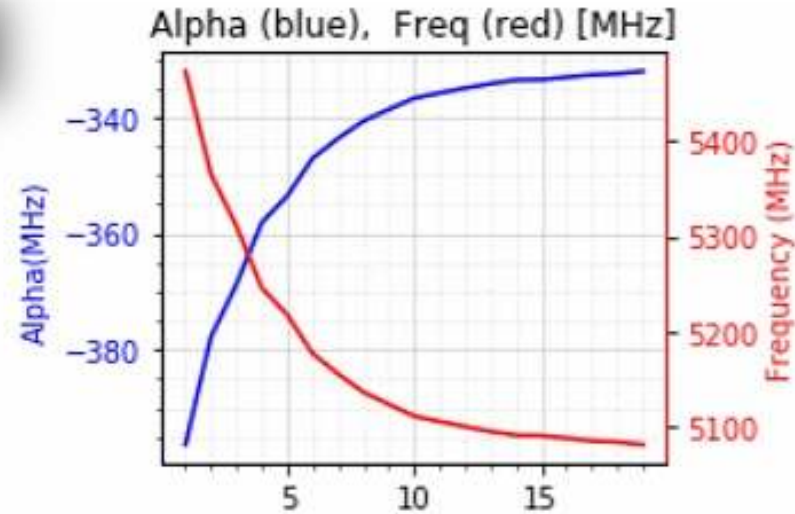


Automated analysis and reports

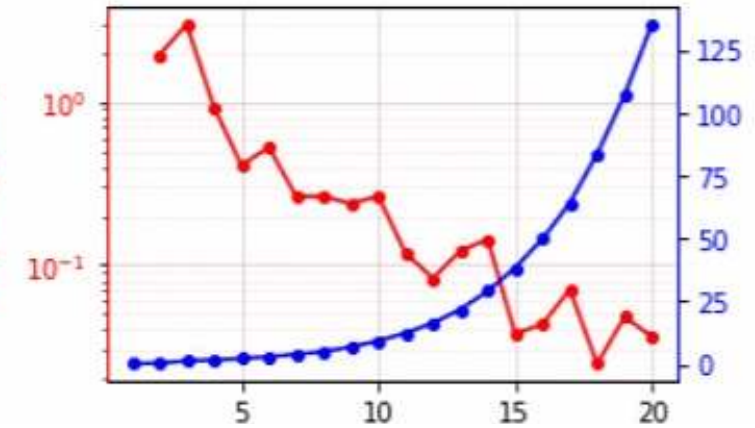
```
metal.analysis.lumped_model.analyze('Q1')
```



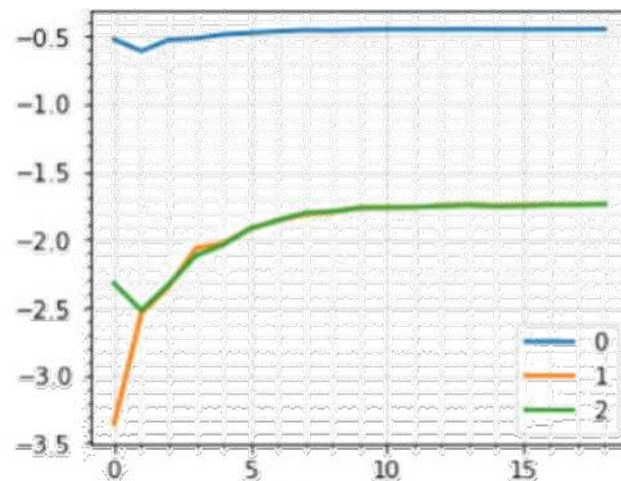
Qubit frequency & anharmonicity



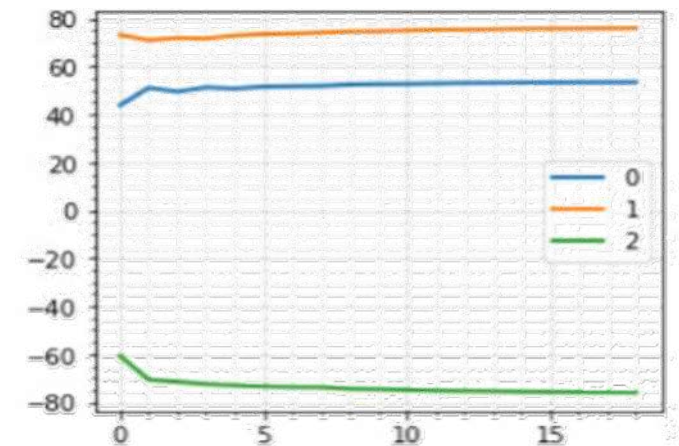
FE simulation convergence



cross-Kerr χ coupling (MHz)



Linear g coupling (MHz)



Analysis pass number

QComponents

Filter

Name	QComponent class	QComponent module	Build status	ID
Q1	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	1
Q2	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	2
Q3	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	3
Q4	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	4
cpw1	RouteMeander	qiskit_metal.components.interconnects.meandered	good	5
cpw2	RouteMeander	qiskit_metal.components.interconnects.meandered	good	6
cpw3	RouteMeander	qiskit_metal.components.interconnects.meandered	good	7
cpw4	RouteMeander	qiskit_metal.components.interconnects.meandered	good	8
OT01	OpenToGround	qiskit_metal.components.connectors.open_to_ground	good	9
OT02	OpenToGround	qiskit_metal.components.connectors.open_to_ground	good	10
OT03	OpenToGround	qiskit_metal.components.connectors.open_to_ground	good	11

Variables

Variable name	Value	Parsed value (in mm)
cpw_width	10 μ m	0.01
cpw_gap	6 μ m	0.006

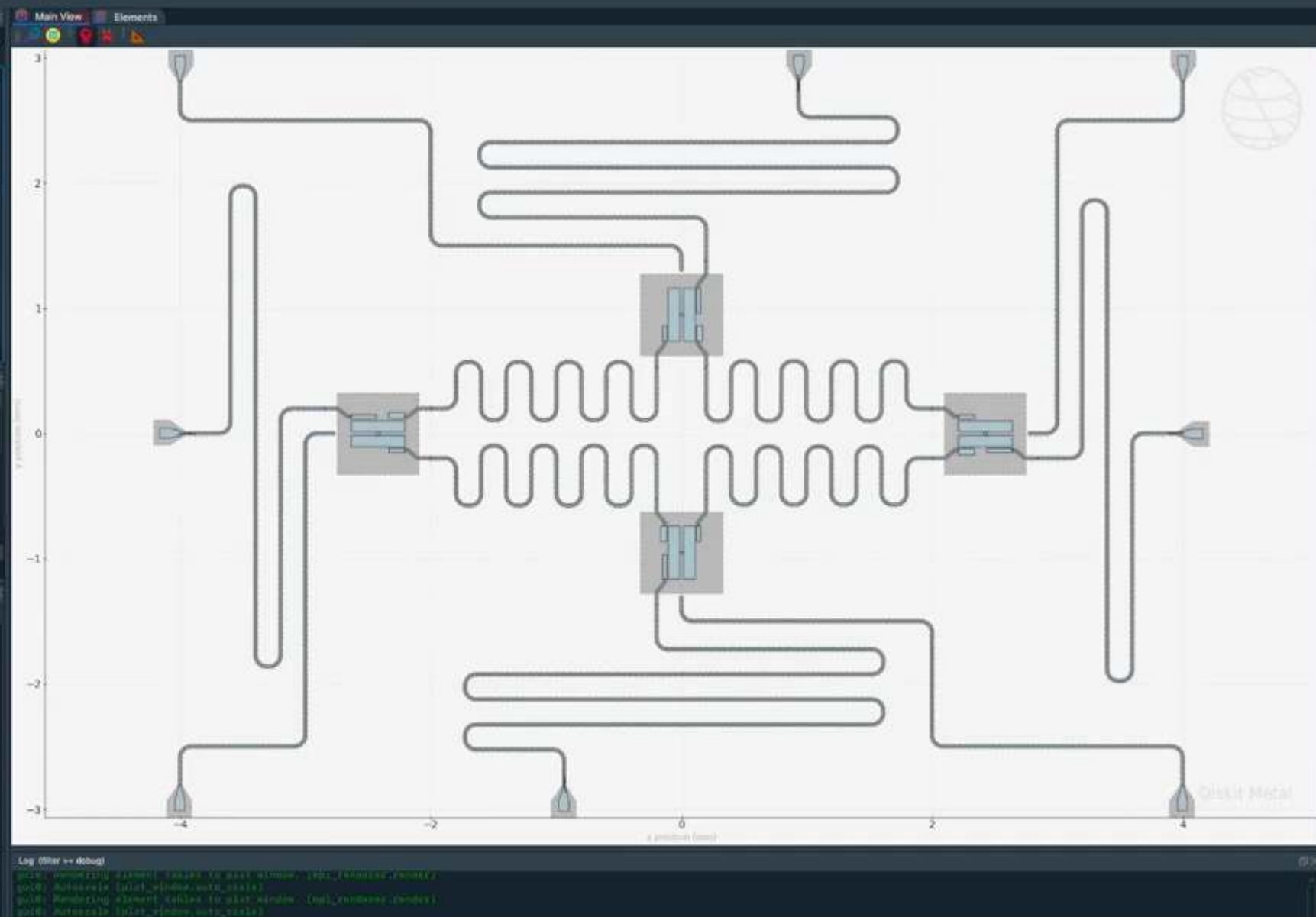
Add variable Delete variable

Library Pins Variables

Edit component

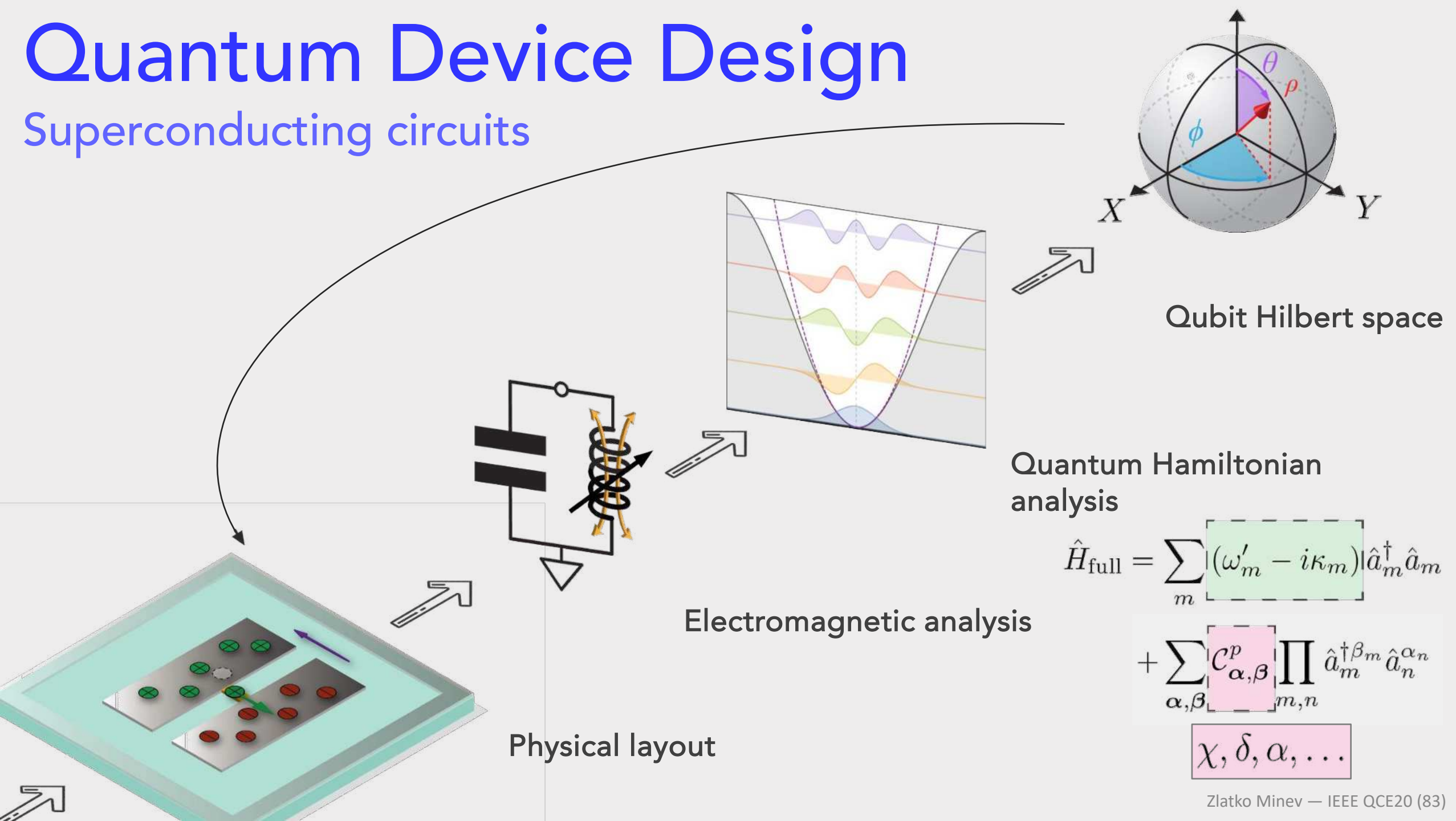
Name	Value	Parsed value
------	-------	--------------

Select a QComponent to edit
from the QComponents window



Quantum Device Design

Superconducting circuits



0.0.2

 Search Docs

[Installing Qiskit Metal](#)
[Metal Workflow](#)
[Frequently Asked Questions](#)

Contributor Guide

[Contributing to Qiskit Metal](#)
[Where Things Are](#)
[Reporting Bugs and Requesting Enhancements](#)
[Contributing Code](#)
[Contributing to Documentation](#)

Libraries

[Quantum devices](#)

API References

[Overview](#)
[QDesign](#)
[Analyse](#)
[Renderer](#)
[Toolbox](#)

API References Advanced

[GUI](#)
[Docs](#) > Qiskit Metal 0.0.2 documentation


Qiskit Metal 0.0.2 documentatio

HINT

You can open this documentation using

```
import qiskit_metal
qiskit_metal.open_docs()
```

ATTENTION

This is an early-access alpha version of Qiskit Metal. This folder will be expanded throughout the early-access period, based on both planned training and your feedback. Please let us know about anything you might want us to add or elaborate upon.

Qiskit for quantum hardware design (‘Qiskit Metal’) is a quantum device design and analysis SDK, library, and community. Call it quantum EDA (QEDA) and analysis.

About Metal

Designing quantum devices is the bedrock of the quantum ecosystem, but it is a difficult, multi-step process that connects traditionally disparate worlds.

Metal is automating and streamlining this process. Our vision is to develop a community-driven universal platform capable of orchestrating quantum chip development from concept to fabrication in a simple and open framework. Qiskit for quantum hardware design (*Qiskit Metal*) is: * Open source * Community-driven * Both a python API and a front-end visual GUI interface.

Metal & it’s vision (read full [Medium blog](#)):

We want to accelerate and lower the barrier to innovation on quantum devices. Today at the IEEE Quantum Week Conference, the team discussed their vision for this first-of-its-kind project. Led by quantum physicist Zlatko Minev and developed with other IBM Quantum team members, this project is meant for those interested in quantum hardware design: a suite of design automation tools that can be used to devise and analyze superconducting devices, with a focus on being able to

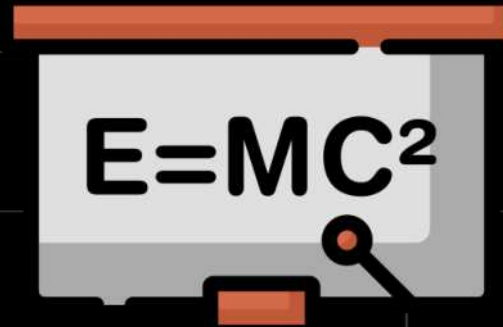
Qiskit Metal Quantum Hardware Design

Early-access program

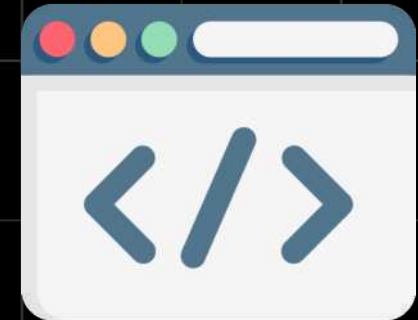
Experiment



Theory



Advocate

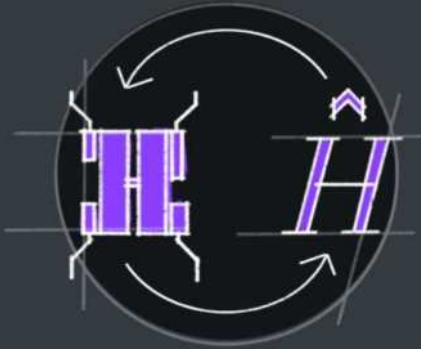


qiskit.org/metal
github.com/Qiskit/qiskit-metal
[#qiskit-metal-early-access](https://twitter.com/qiskit-metal-early-access)

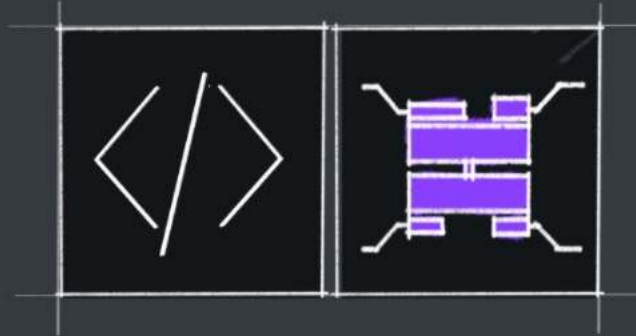
Zlatko Minev — IEEE QCE20 (85)

Why the Vision of Qiskit Metal

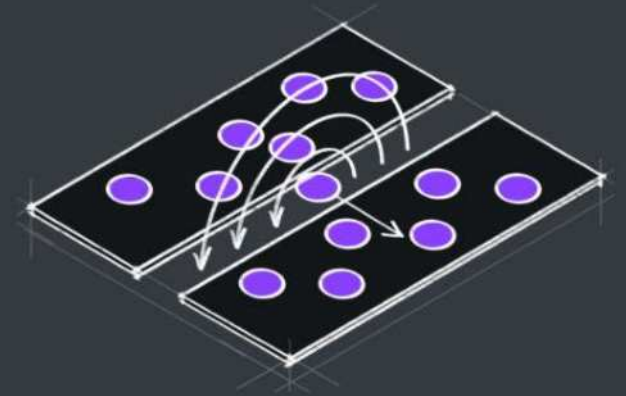
End-to-end automation



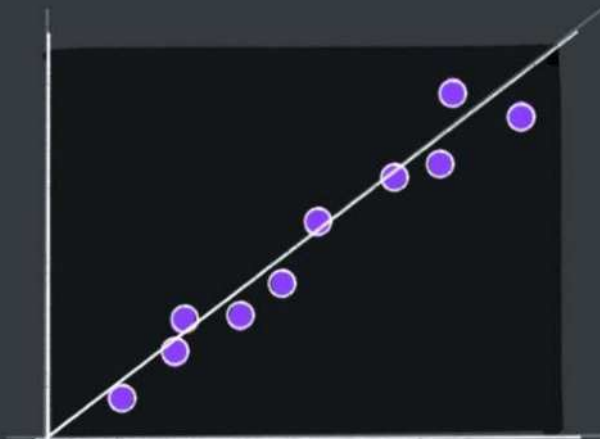
Flexible & extensible



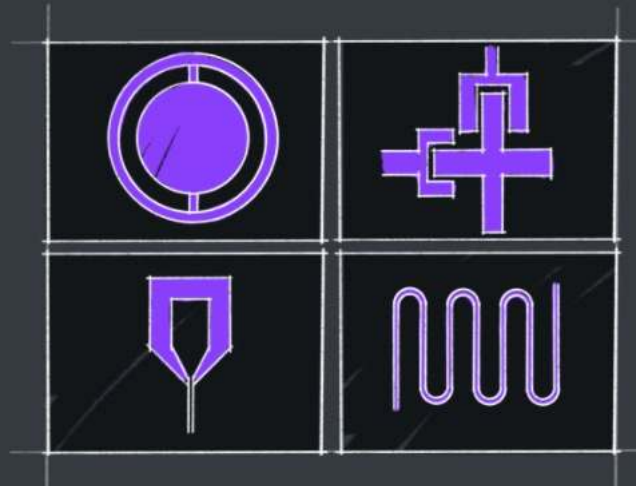
Light-weight interoperability



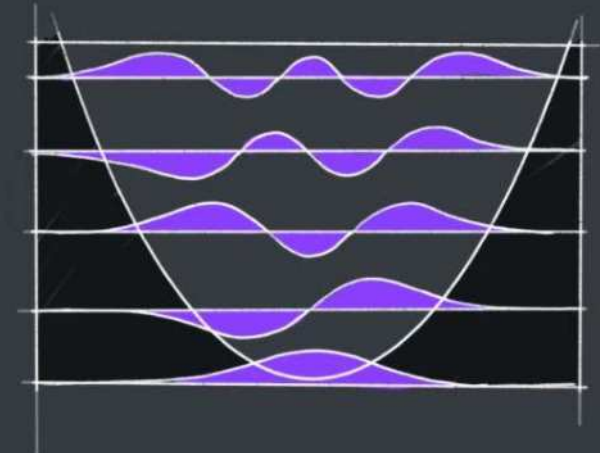
Experimentally tested



Library of components



Cutting edge resources



Building together

Call for community participation

Open source

Early access program

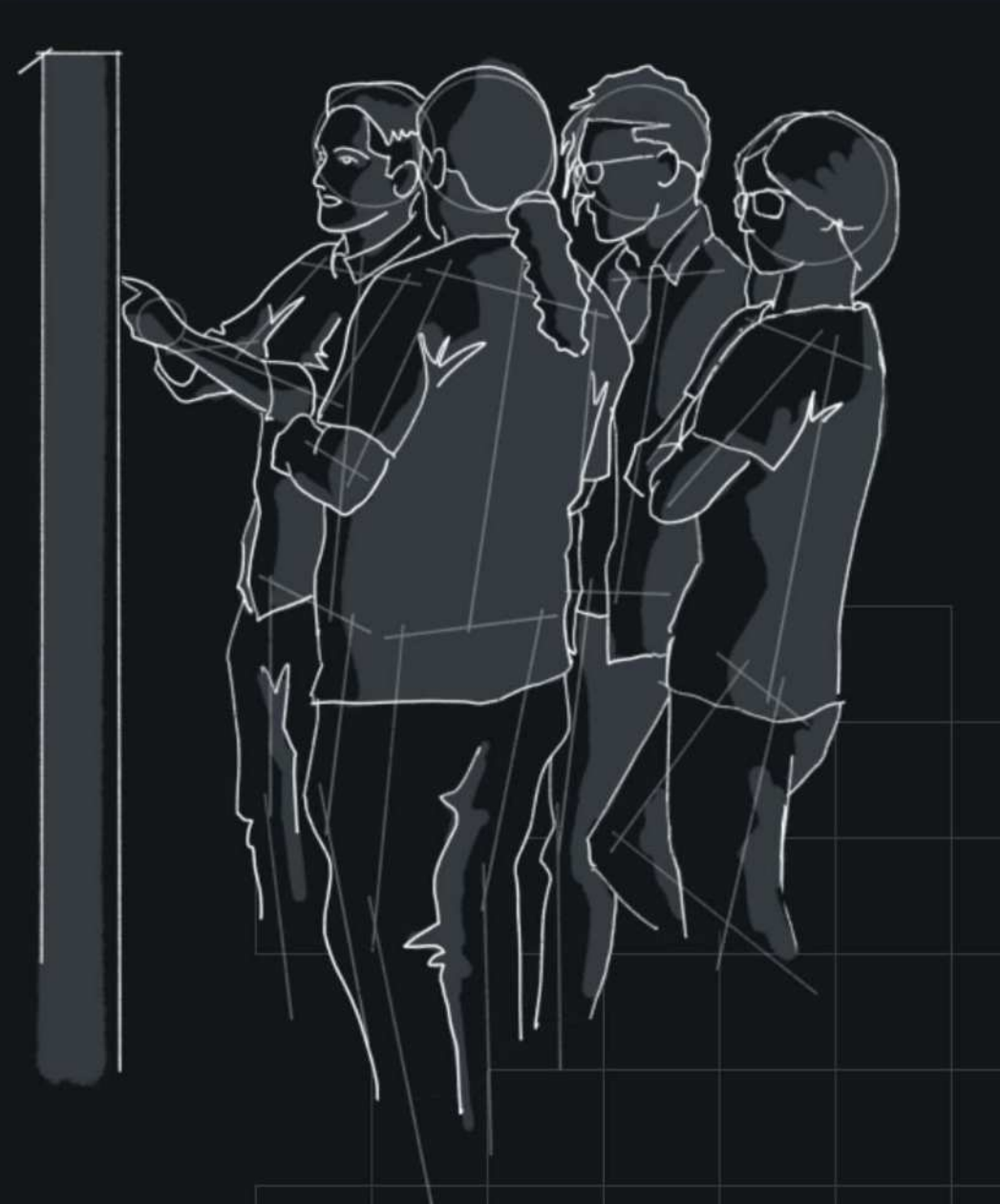
Education

See summer school lectures 16-21 by Z. Mineev from
Introduction to Quantum Computing and Quantum Hardware
and the *Qiskit Textbook*

qiskit.org/metal



@zlatko_minev



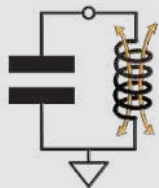
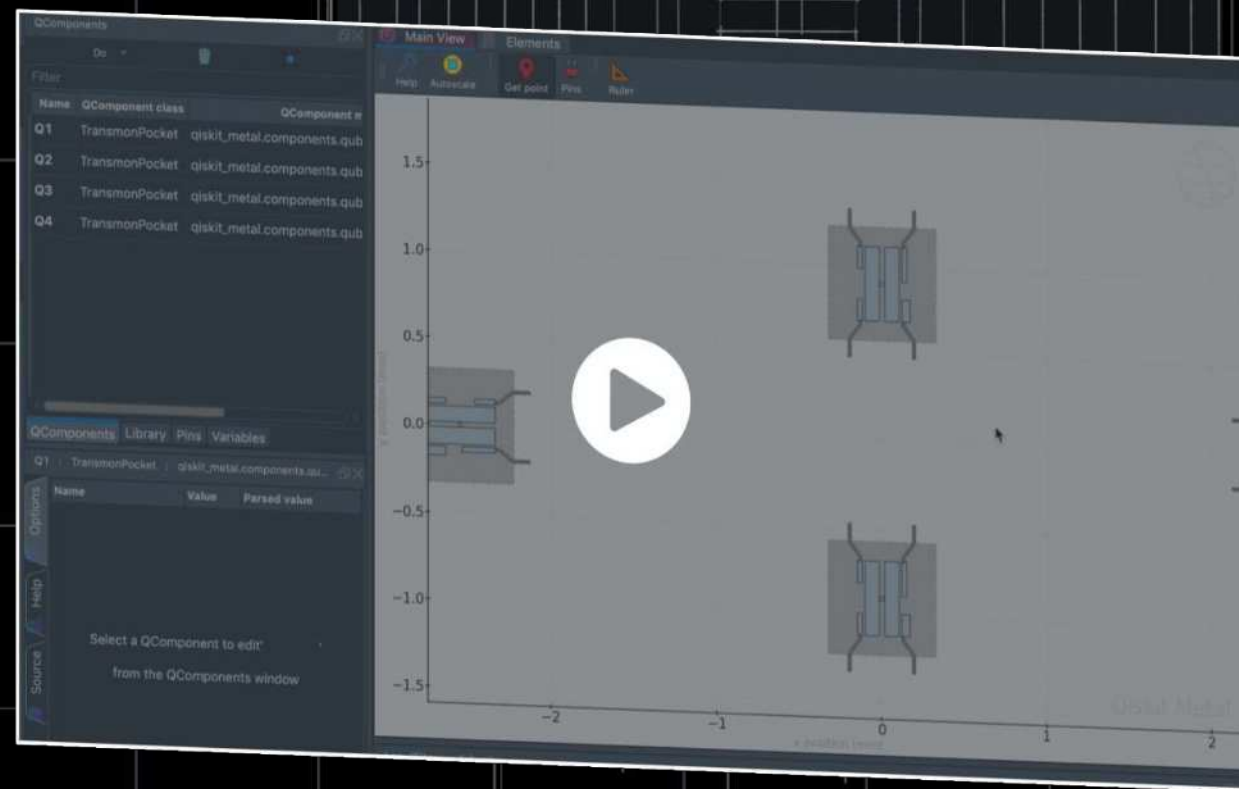
Qiskit | quantum device design

Project Metal

Open source

Learn more / request early access:

qiskit.org/metal



Zlatko K. Minev

IBM Quantum

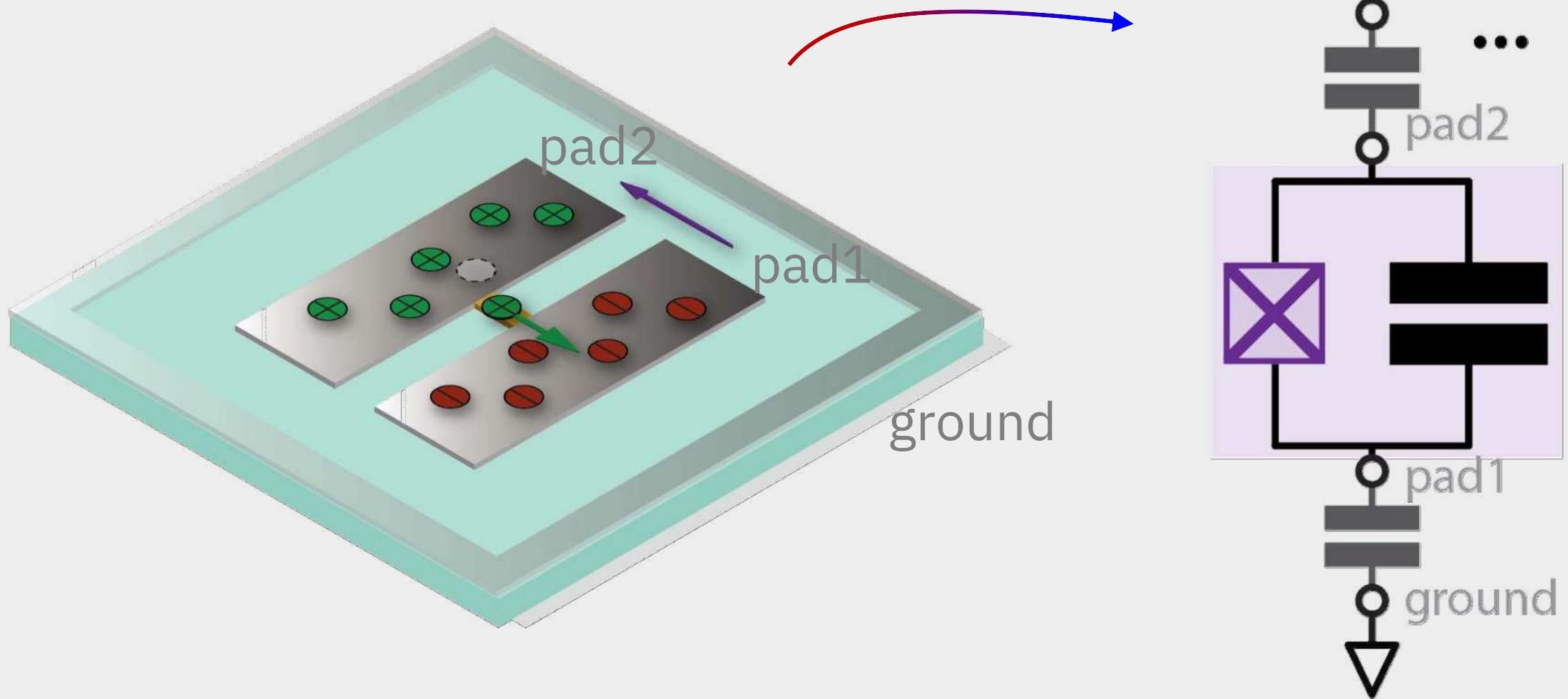


@zlatko_minev



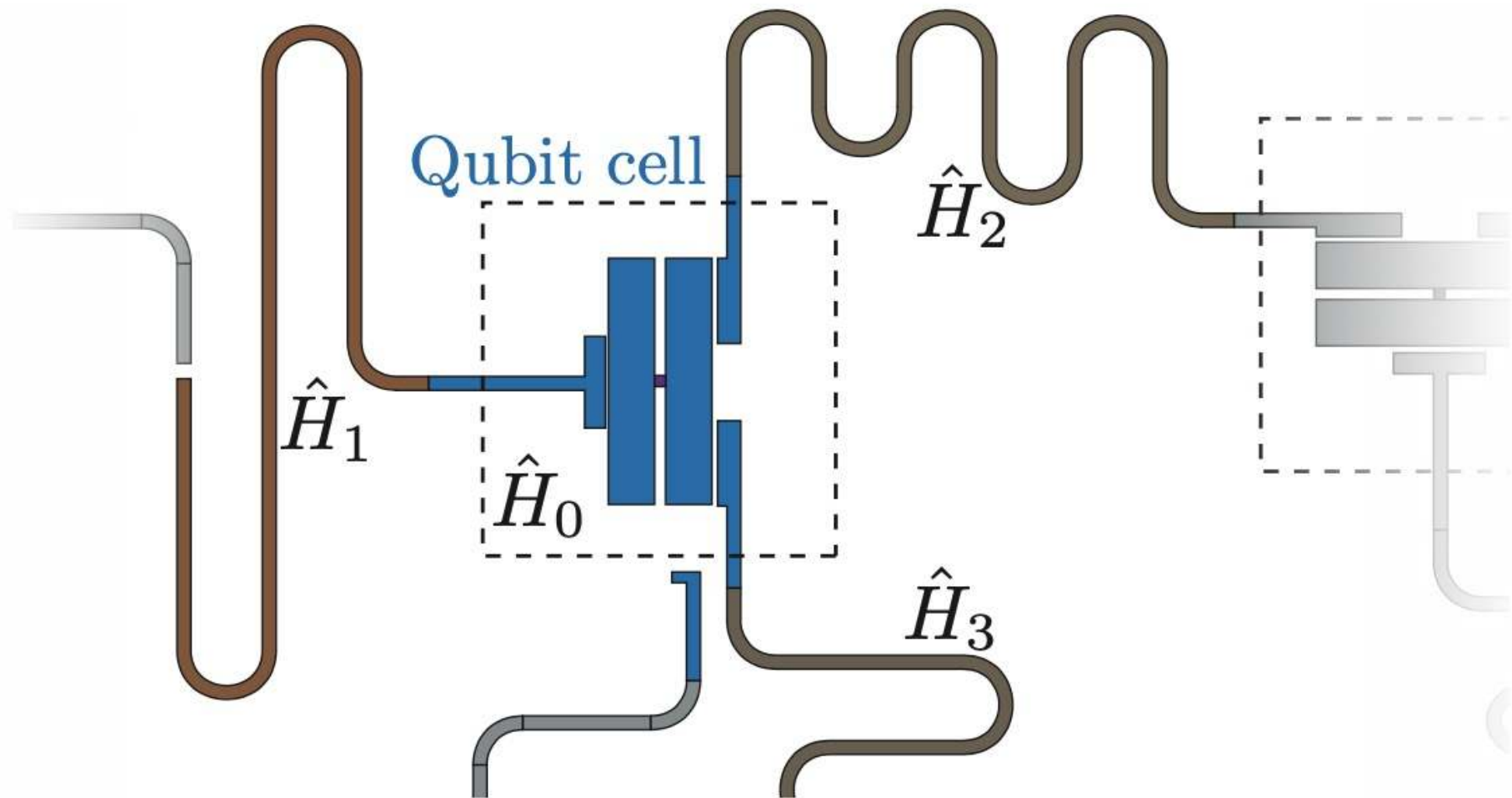
zlatko-minev.com

Floating transmon (lumped approximation)

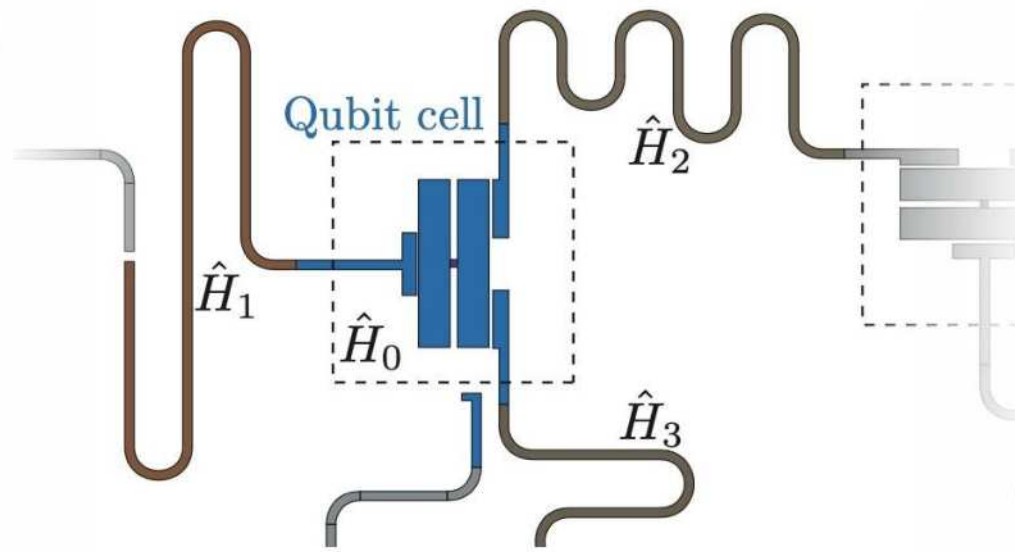


Quasi-lumped modular method

$$\hat{H}_{\text{full}} = \hat{H}_0 + \sum_{n=1}^K \hat{H}_n + \sum_{n=0}^K \sum_{m=n+1}^K \hat{H}_{nm}, \quad (1)$$



(a)

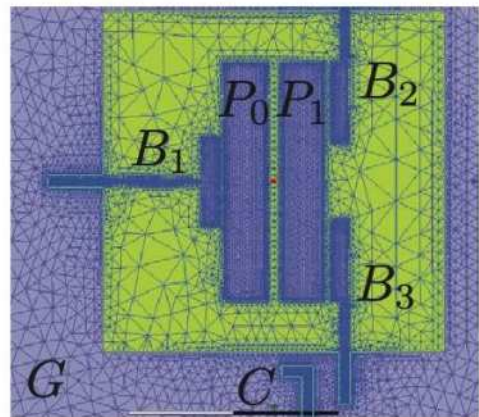


Circuit quantum electrodynamics (cQED) with modular quasi-lumped models

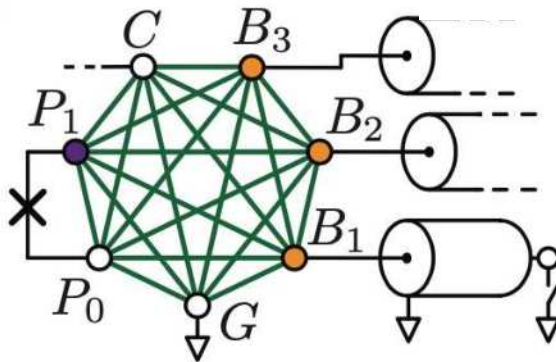
Zlatko K. Minev^{*} ..., and Jay M. Gambetta
 IBM T. J. Watson Research Center, Yorktown Heights, New York 10598, USA
 (Dated: February 9, 2021)

Modeling the quantum electrodynamics of physical circuits is a keystone problem in the process of realizing complex superconducting quantum phenomena and building a quantum computer. The remarkable growth of the field increasingly requires ever-more precise, widely-applicable, and modular modeling methods that can tie together the physical hardware to its quantum mechanical description to extract principal parameters of interest. Here, we build on previous work to present such a method based on interconnected cells, compactly represented as either lumped or distributed. We treat each lumped-model cell and its coupling to neighboring cells without further approximation, topologically in a generic way. The model construction natively accounts for large and potentially non-perturbative loading and coupling-induced effects. We verify the method experimentally on a large inter-connected quantum device and find a two-times improvement in the experimental agreement of dispersive non-linear couplings.

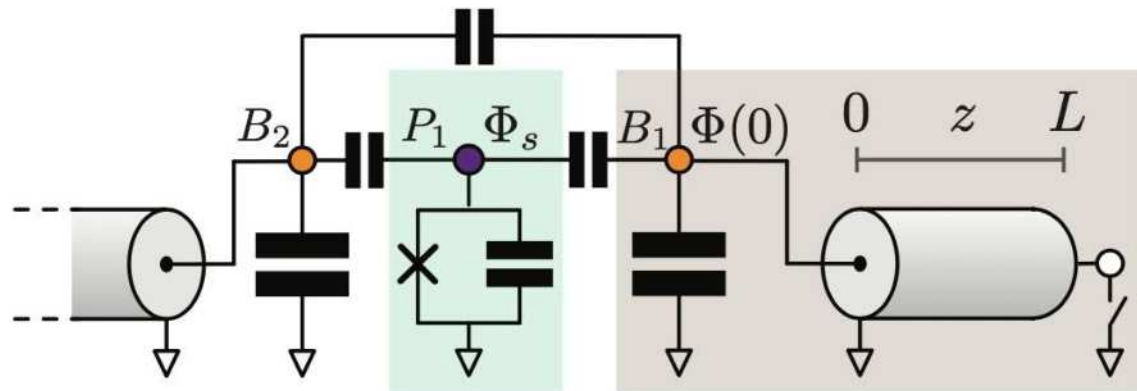
(b)



(c)



(d)



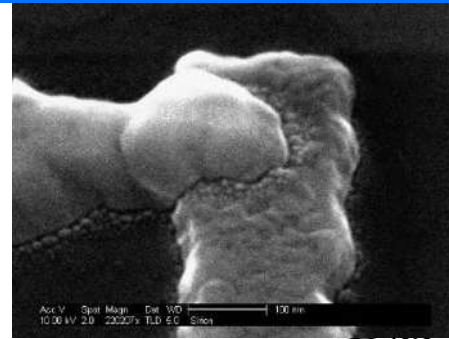
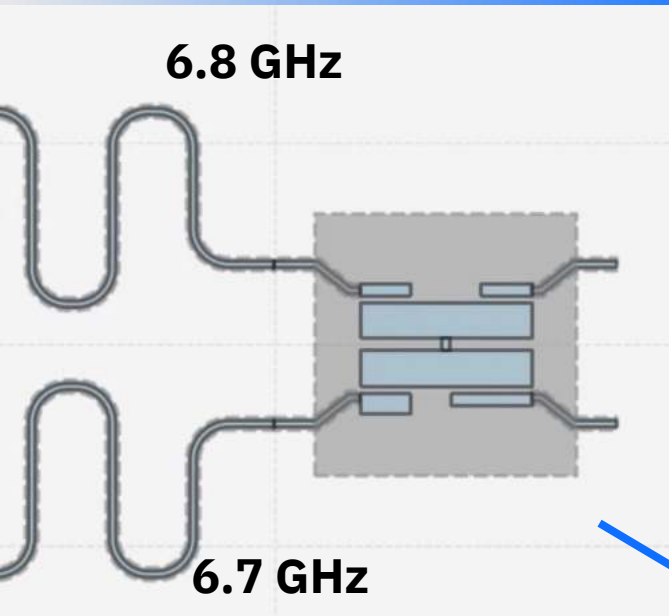
$$\hat{H}_{\text{full}} = \hat{H}_0 + \sum_{n=1}^K \hat{H}_n + \sum_{n=0}^K \sum_{m=n+1}^K \hat{H}_{nm}, \quad (1)$$

$$\hat{H}_{nm} = \hat{Q}_n \hat{Q}_m / C_{nm}^{\text{eff}} + \hat{\Phi}_n \hat{\Phi}_m / L_{nm}^{\text{eff}}, \quad (2)$$

$$\hat{H}_{nm} = \hat{Q}_n \hat{Q}_m / C_{nm}^{\text{eff}} = \hbar g_{nm} \hat{A}_n \hat{B}_m,$$

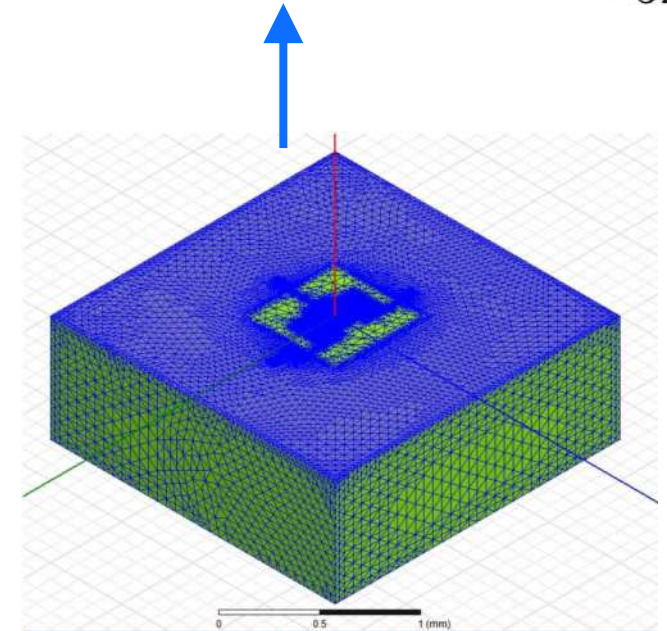
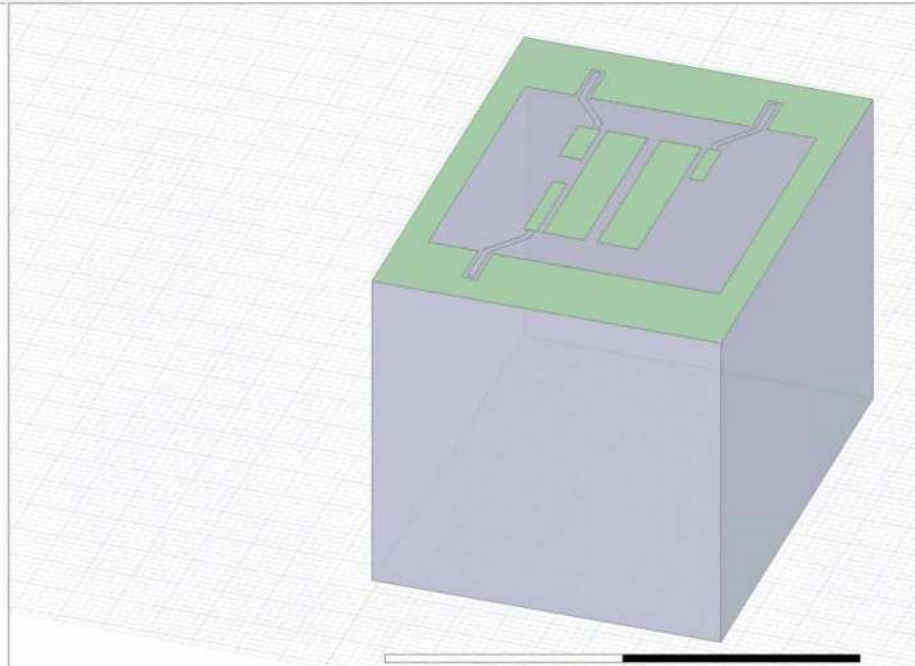
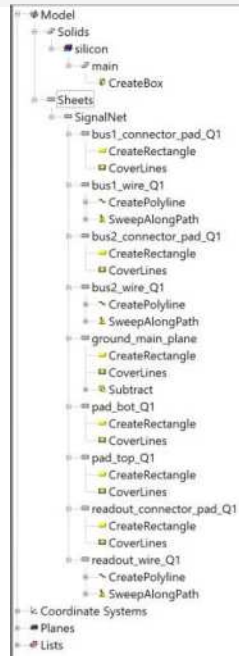
Yurke & Denker (1984), Devoret (1997),
 Burkard et al. (2004), Koch et al. (2007), ...

Metal -> Sim model/BC -> Analyze -> Capacitance matrix



2 fF

$$\begin{matrix} n_0 \\ n_1 \\ n_2 \\ n_3 \\ n_4 \\ n_5 \end{matrix} \begin{pmatrix} C_{0\Sigma} & -C_{01} & -C_{02} & -C_{03} & -C_{04} & -C_{05} \\ & C_{1\Sigma} & -C_{12} & -C_{13} & -C_{14} & -C_{15} \\ & & C_{2\Sigma} & -C_{23} & -C_{24} & -C_{25} \\ & & & C_{3\Sigma} & -C_{34} & -C_{35} \\ & & & & C_{4\Sigma} & -C_{45} \\ & & & & & C_{5\Sigma} \end{pmatrix},$$

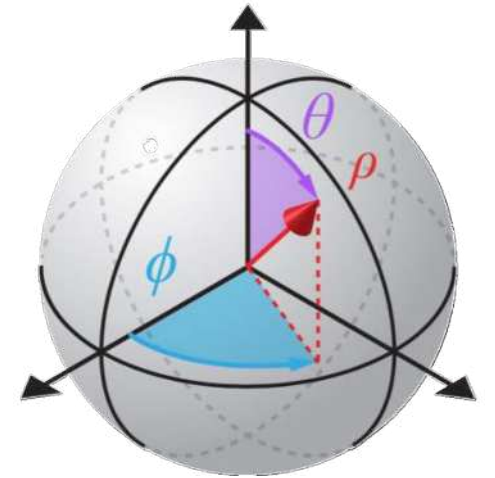
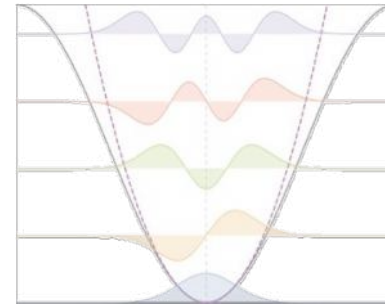
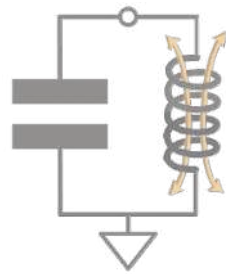
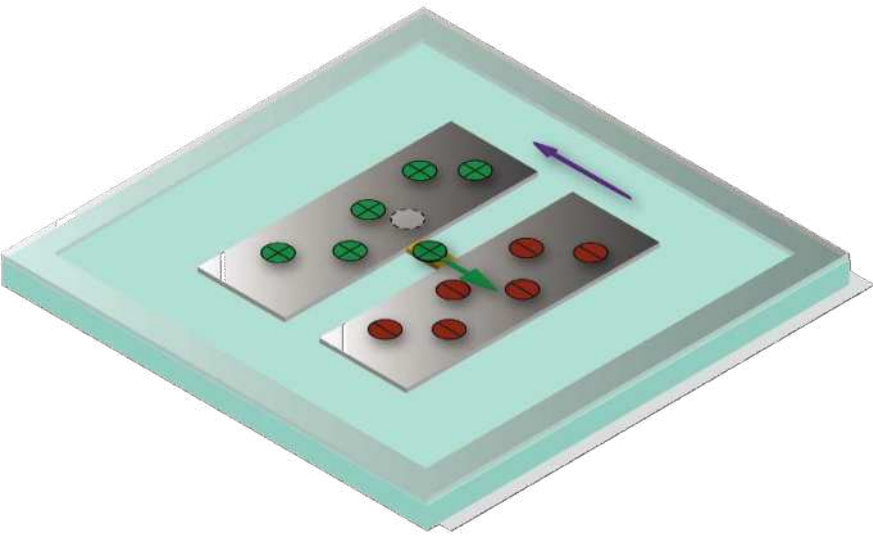


Minev & Gambetta, in prep (2020)

Zlatko Minev, IBM Quantum (92

To design quantum hardware

Energy-participation-ratio (EPR)
and Qiskit Metal



Zlatko K. Mineev

IBM Quantum

IBM T.J. Watson Research Center, Yorktown Heights, NY



@zlatko_mineev



zlatko-mineev.com

Thank you

Quantum chip design flow

Concept



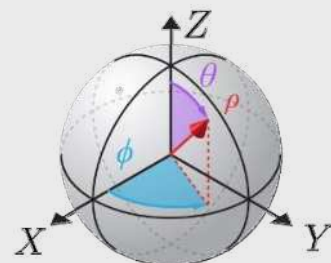
Quantum chip design flow

Complexity, Information, Accuracy

Concept



Hamiltonian



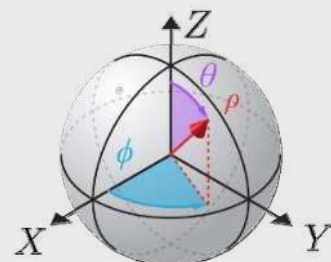
Quantum chip design flow

Complexity, Information, Accuracy

Concept



Hamiltonian



Layout



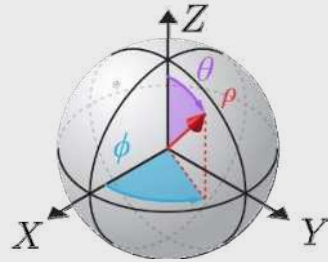
Quantum chip design flow

Complexity, Information, Accuracy

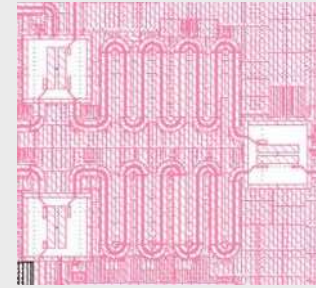
Concept



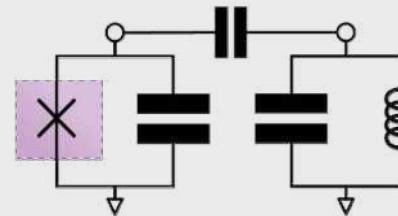
Hamiltonian



Layout



Electromagnetic
Analysis



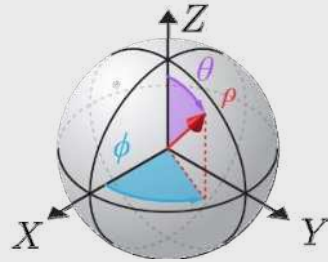
Quantum chip design flow

Complexity, Information, Accuracy

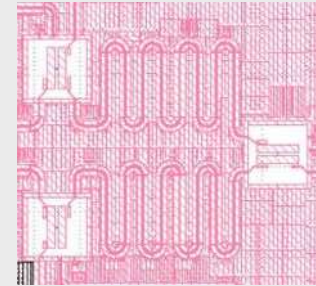
Concept



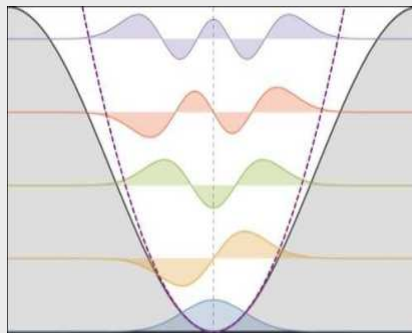
Hamiltonian



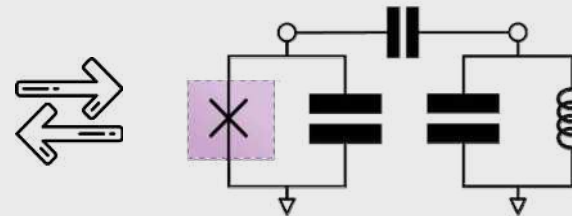
Layout



Quantum Analysis



Electromagnetic
Analysis



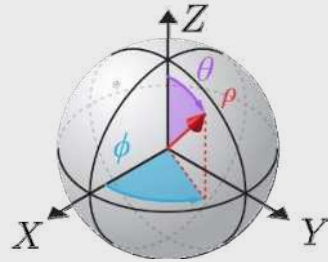
Quantum chip design flow

Complexity, Information, Accuracy
Risk, Cost, Time, Resources

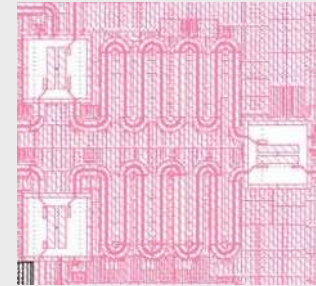
Concept



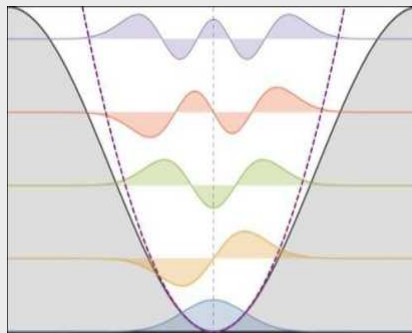
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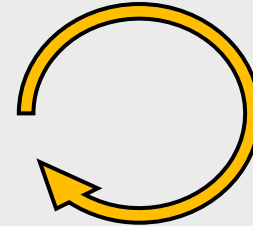
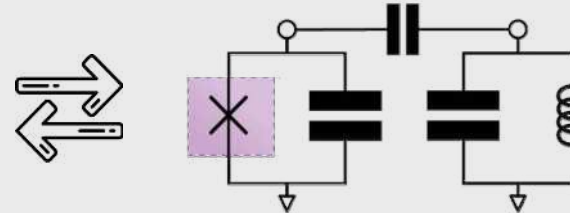
Layout



Quantum Analysis



Electromagnetic
Analysis



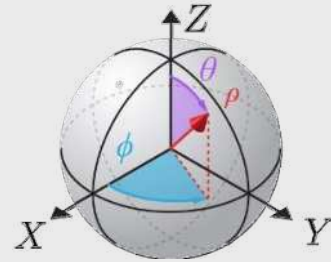
Quantum chip design flow

Complexity, Information, Accuracy
Risk, Cost, Time, Resources

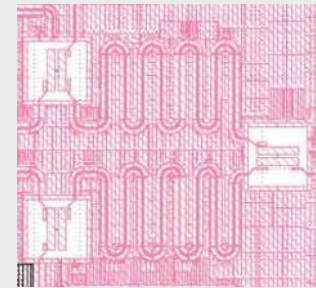
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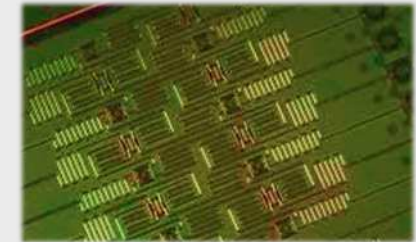
Hamiltonian



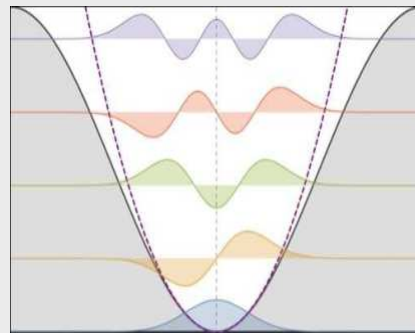
Layout



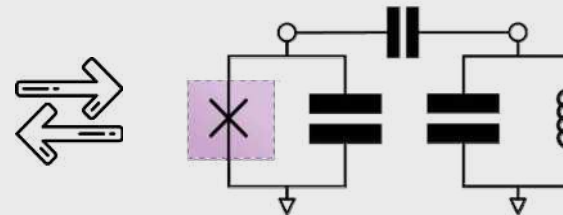
Fabrication



Quantum Analysis



Electromagnetic
Analysis



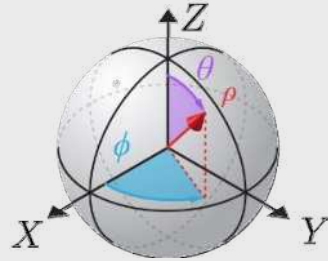
Quantum chip design flow

Complexity, Information, Accuracy
Risk, Cost, Time, Resources

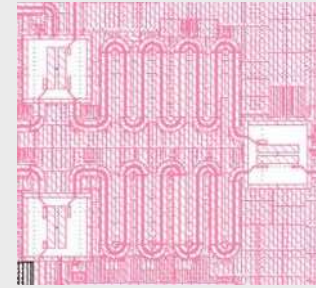
Concept



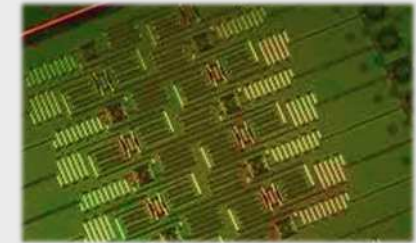
Hamiltonian



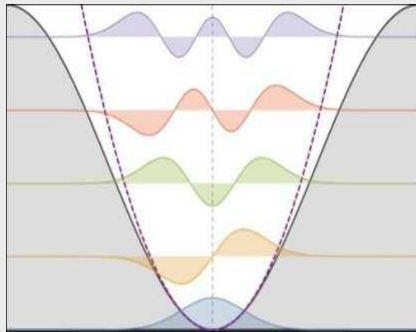
Layout



Fabrication

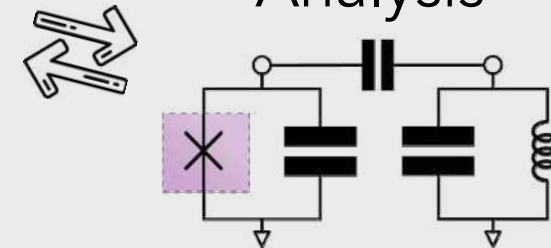


Quantum Analysis

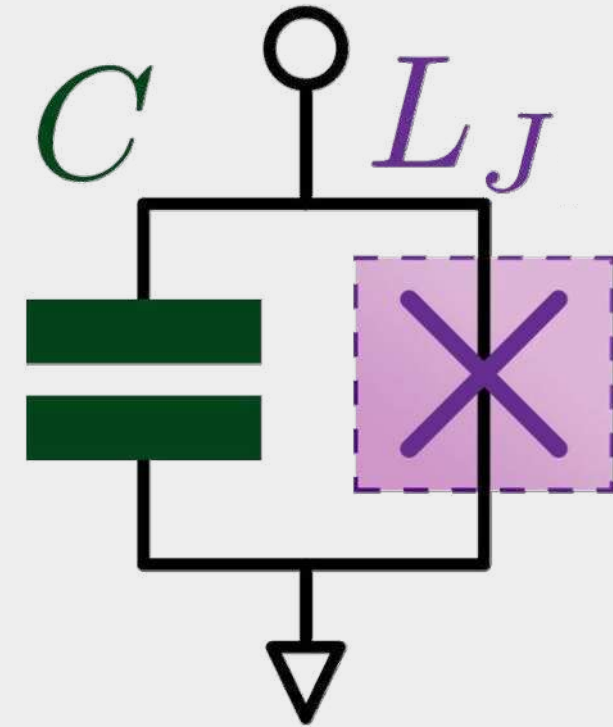
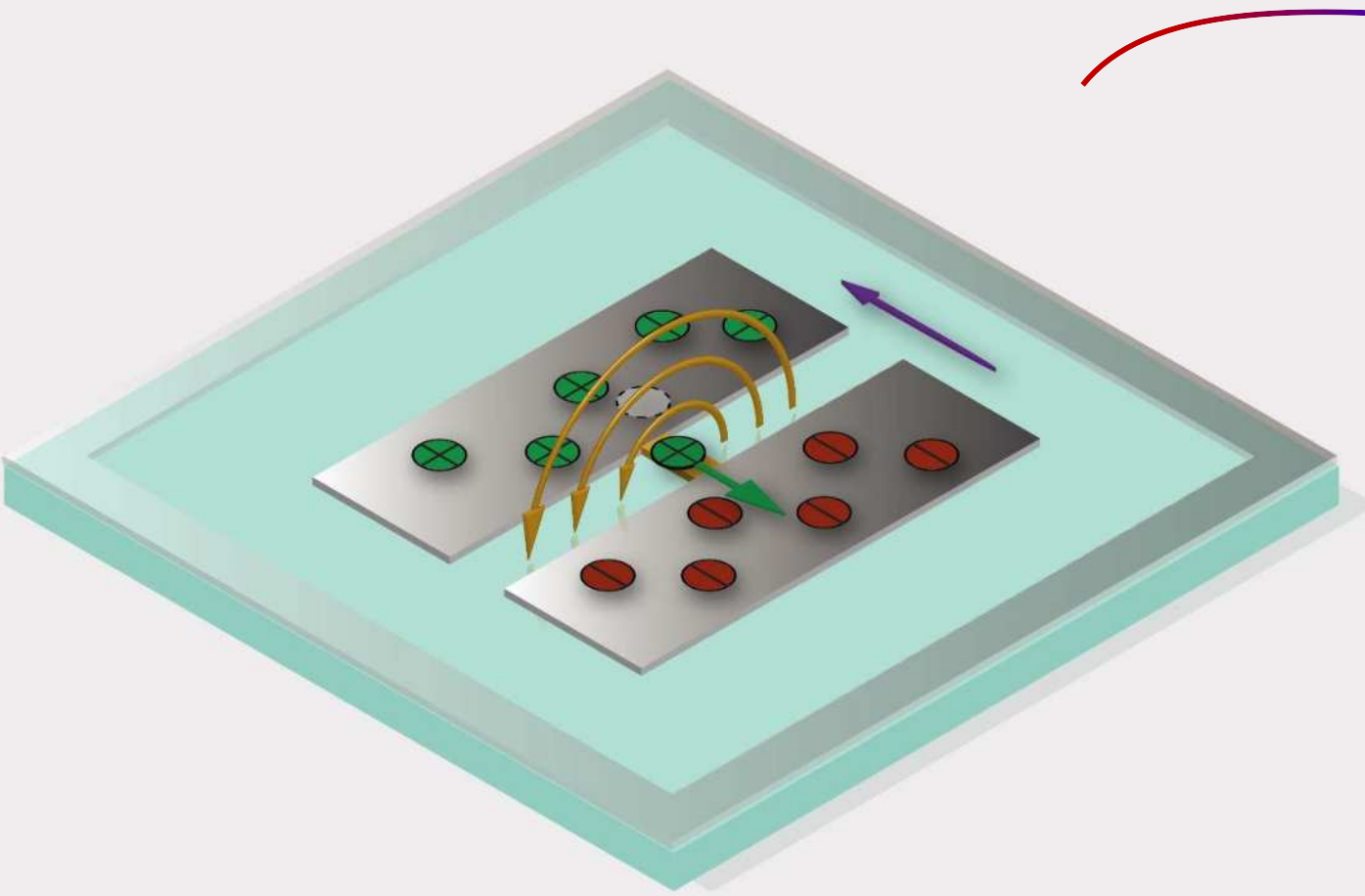


Project
Metal

Electromagnetic
Analysis



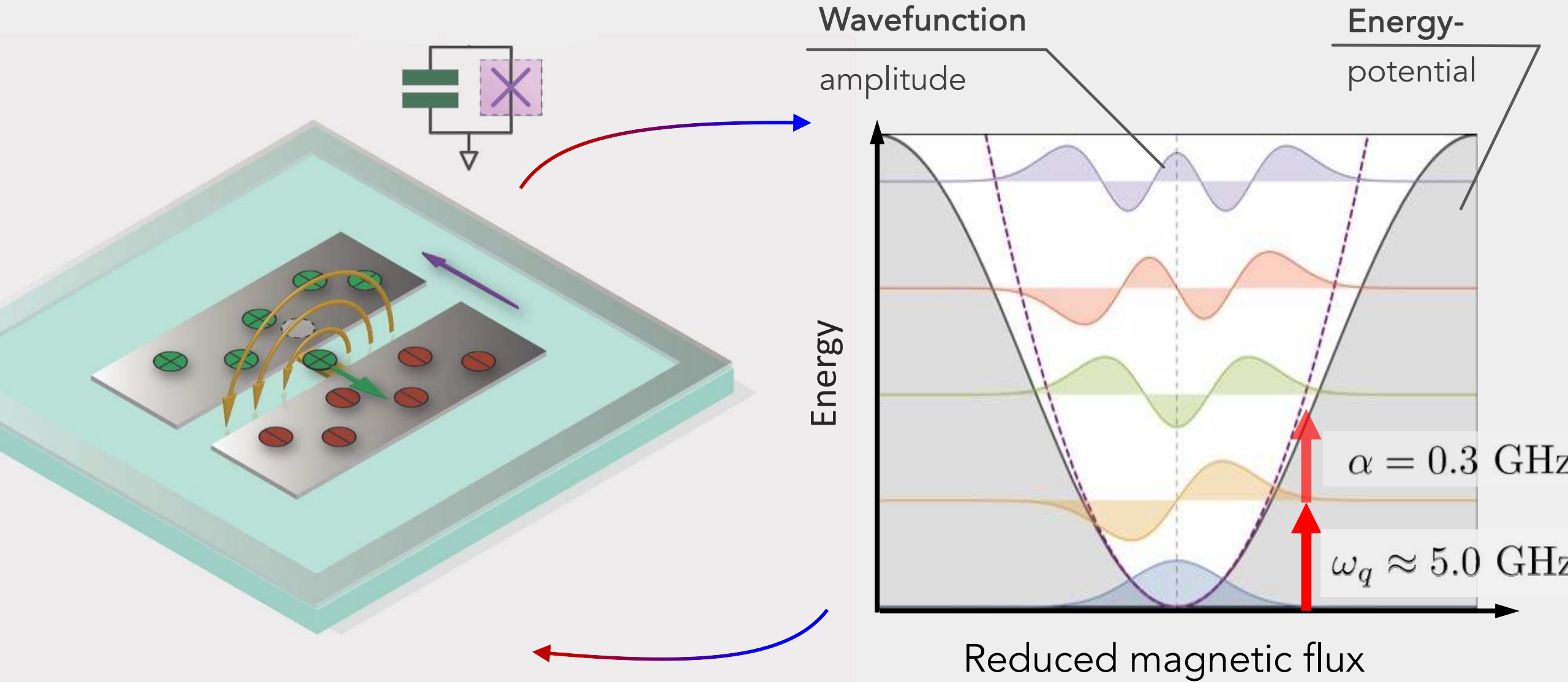
Transmon qubit: layout, electromagnetics, wavefunctions



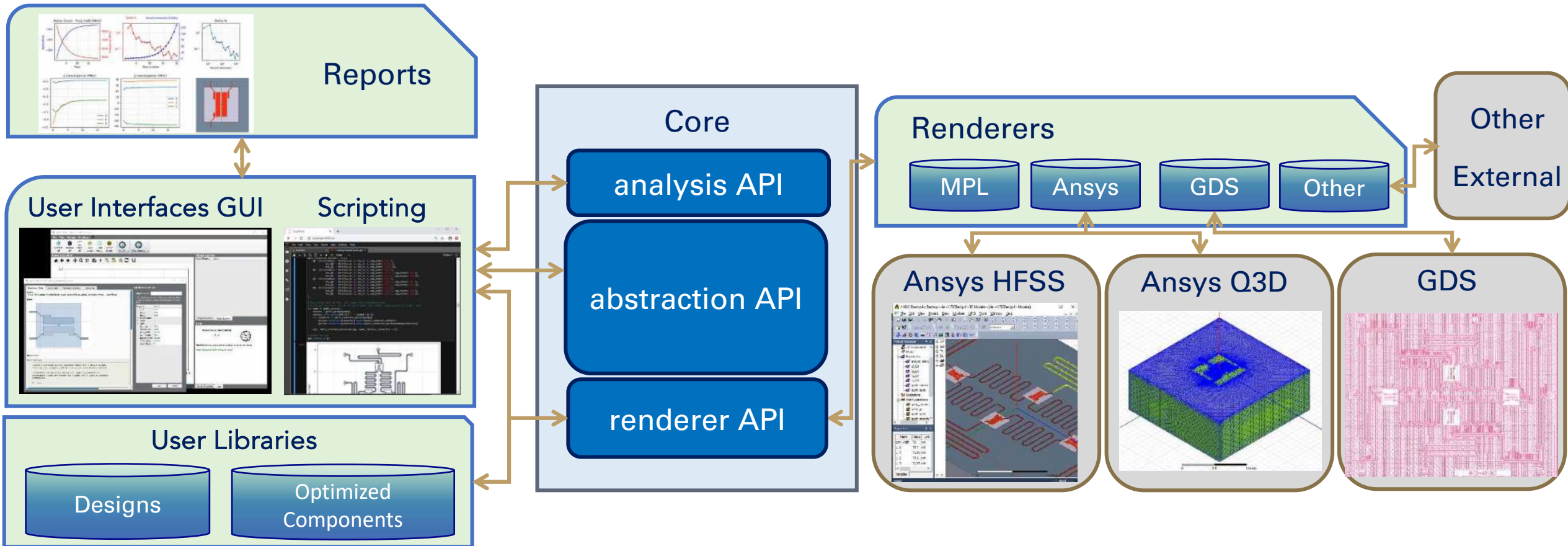
$$L_J = 14 \text{ nH} \quad \omega_0 = \sqrt{\frac{1}{LC}} = 5.3 \text{ GHz}$$

$$C_J = 65 \text{ fF} \quad Z = \sqrt{\frac{L}{C}} \approx 450 \text{ } \Omega$$

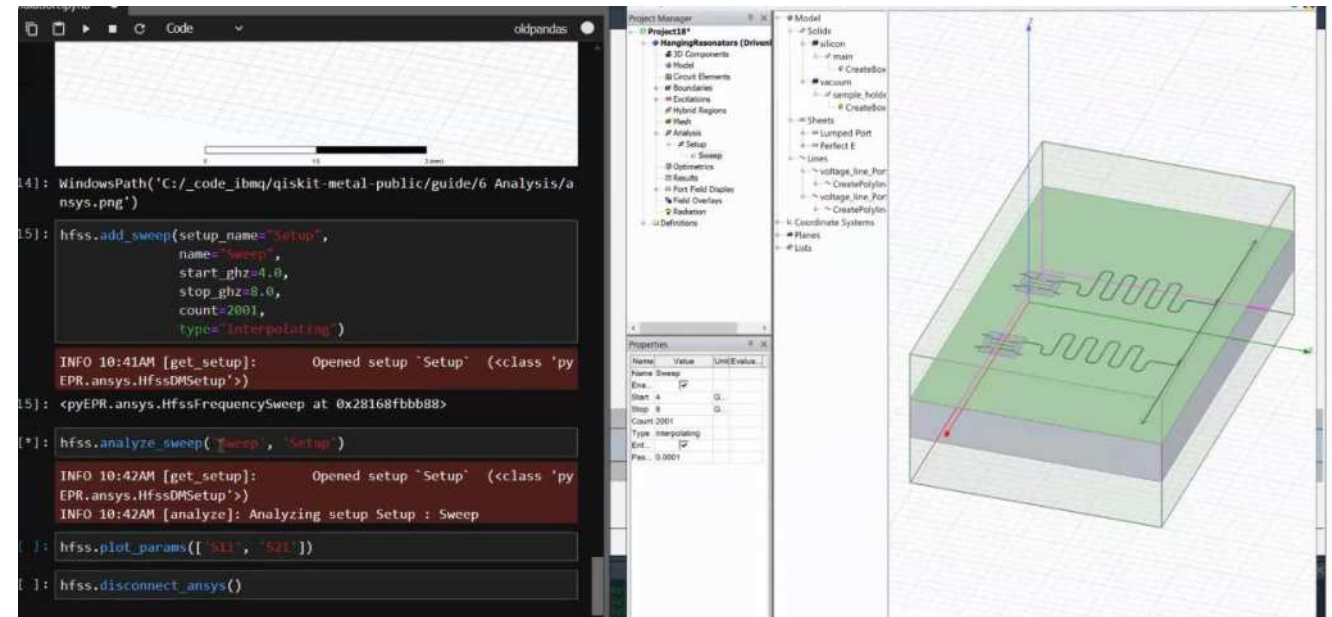
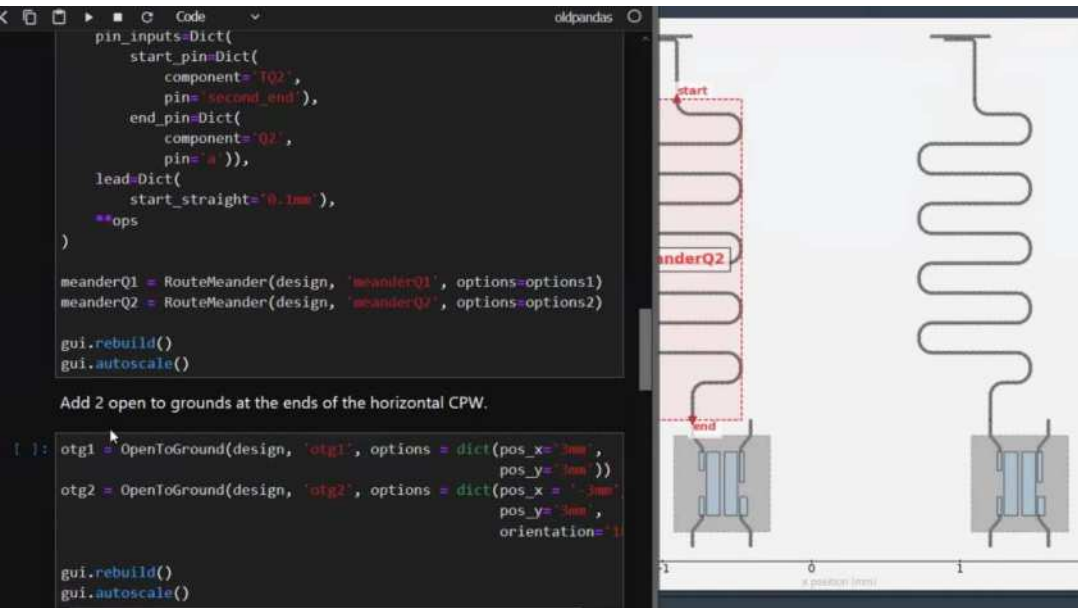
Transmon qubit: layout, electromagnetics, wavefunctions

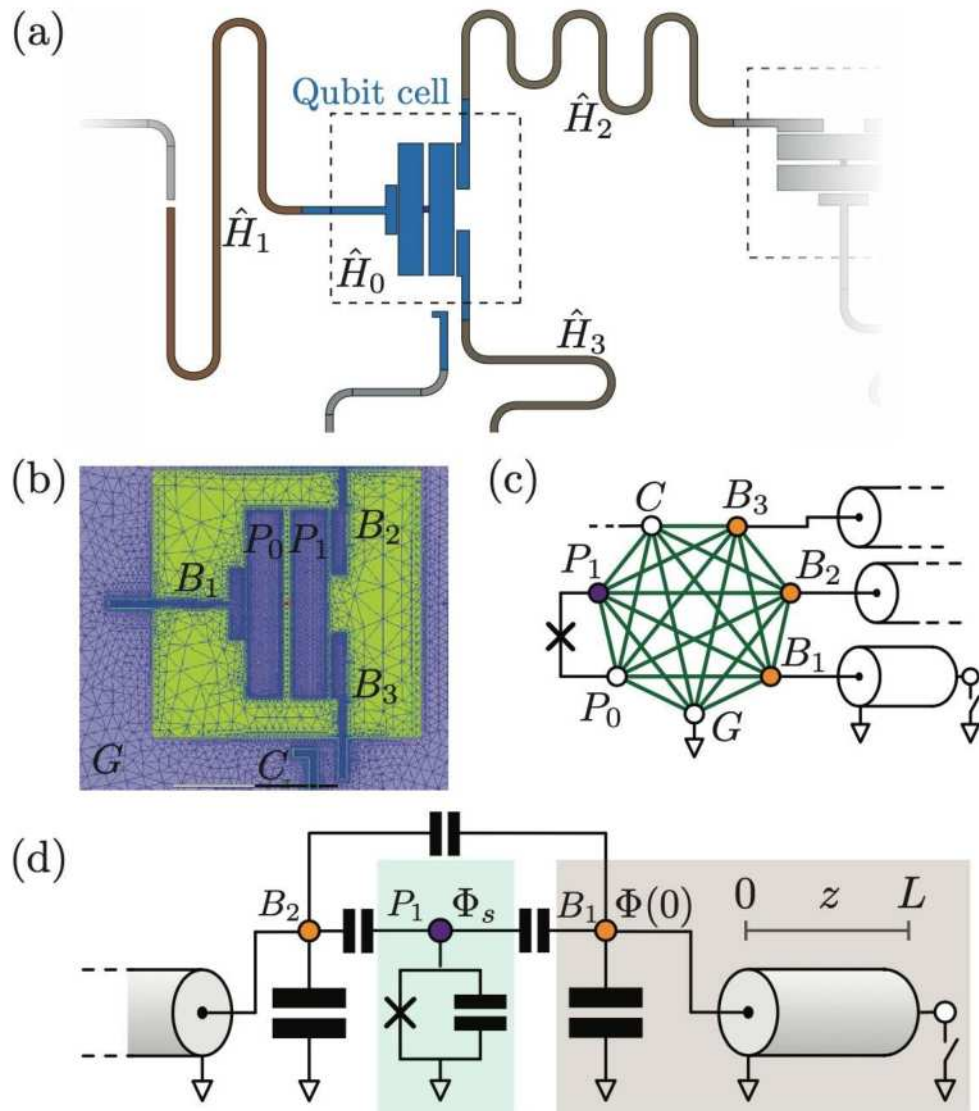


Qiskit Metal Container Diagram



Hanger



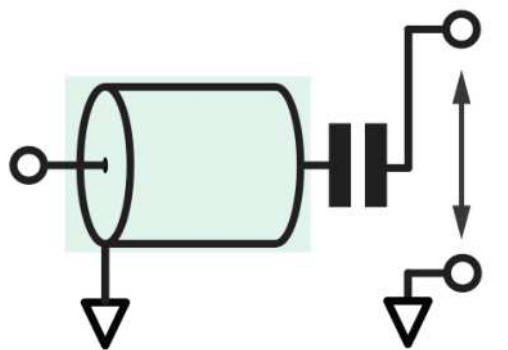
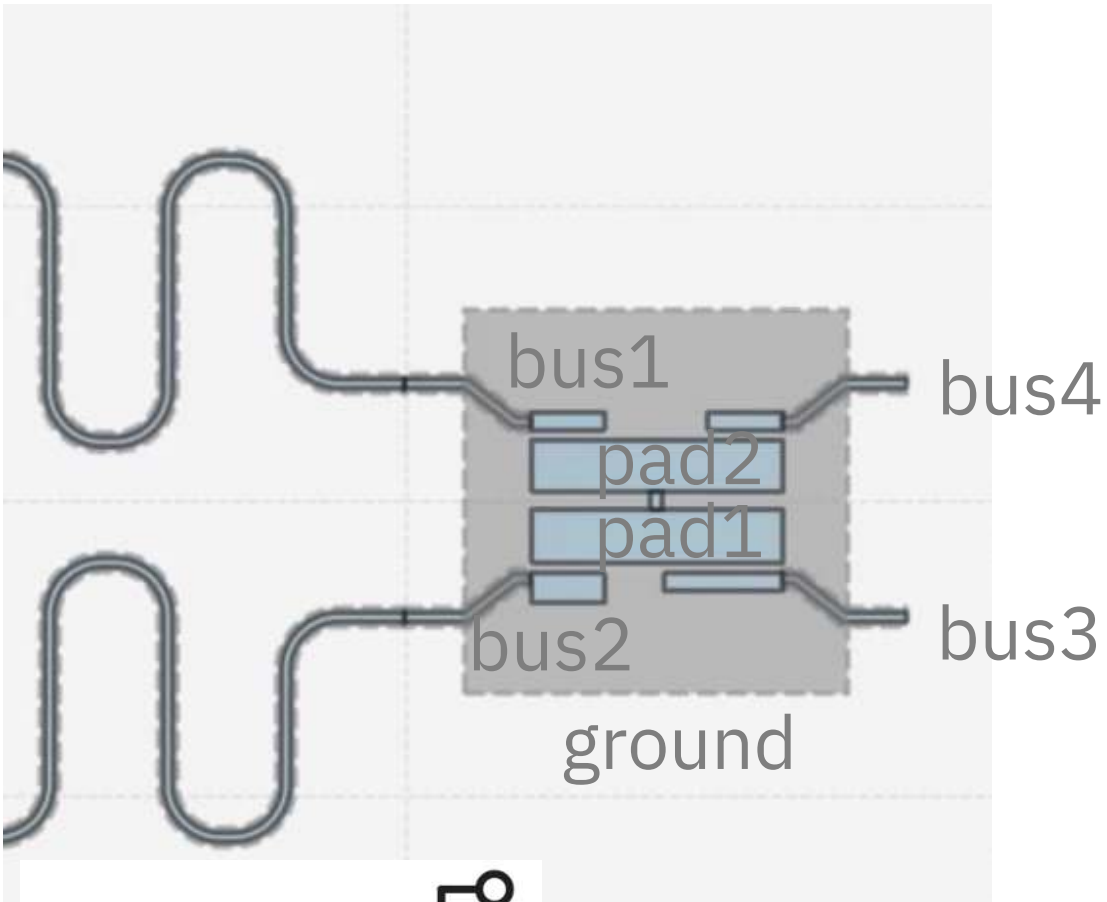


$$\hat{H}_{\text{full}} = \hat{H}_0 + \sum_{n=1}^K \hat{H}_n + \sum_{n=0}^K \sum_{m=n+1}^K \hat{H}_{nm} , \quad (1)$$

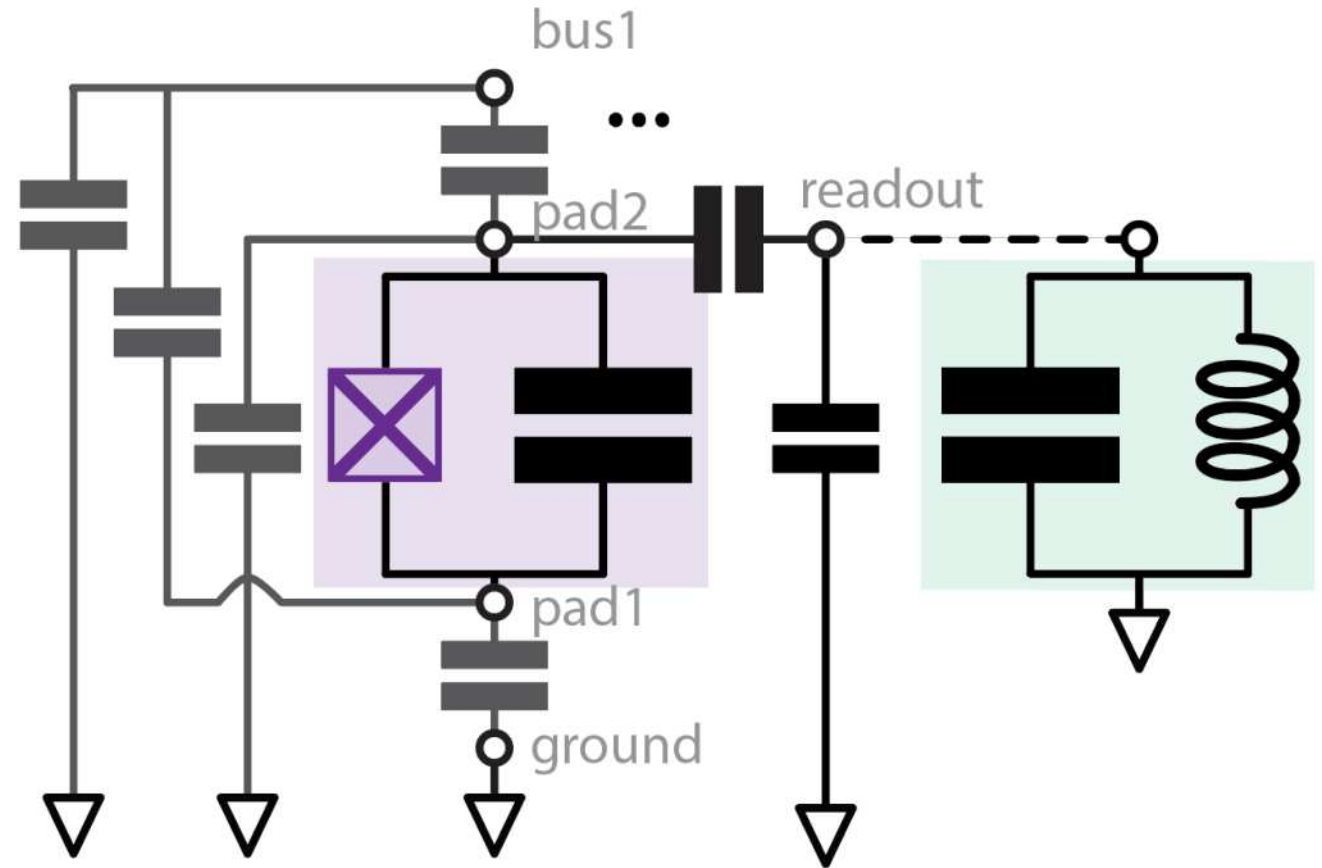
$$\hat{H}_{nm} = \hat{Q}_n \hat{Q}_m / C_{nm}^{\text{eff}} + \hat{\Phi}_n \hat{\Phi}_m / L_{nm}^{\text{eff}} , \quad (2)$$

$$\hat{H}_{nm} = \hat{Q}_n \hat{Q}_m / C_{nm}^{\text{eff}} = \hbar g_{nm} \hat{A}_n \hat{B}_m ,$$

Transmon coupled to other systems

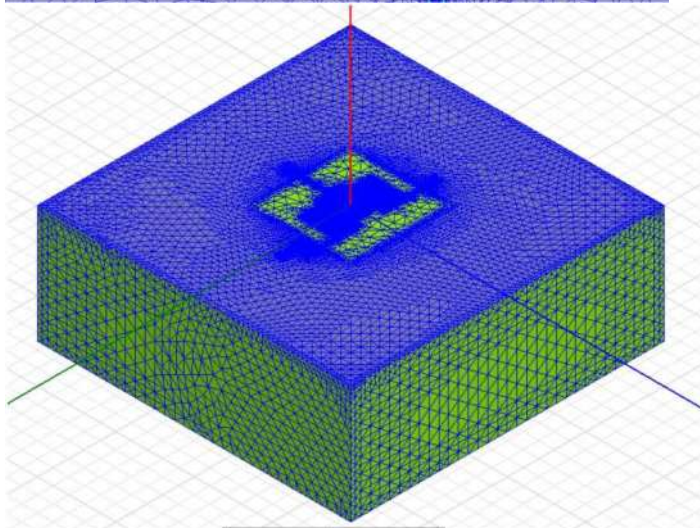
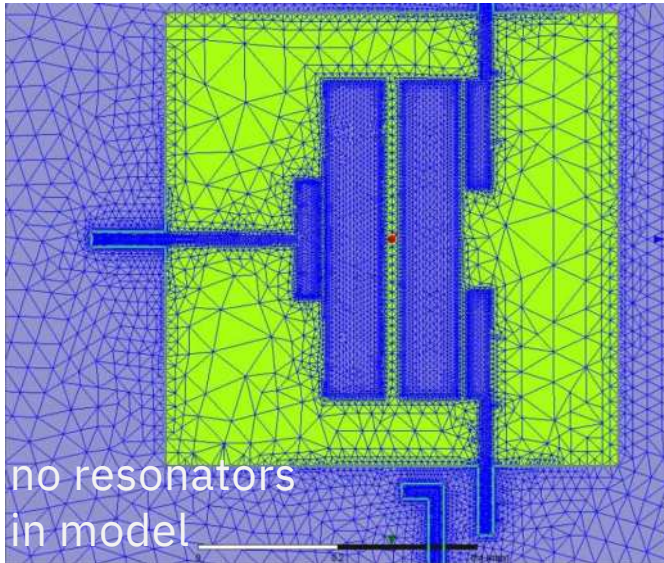


$$\hat{H}_{\text{full}} = \hat{H}_{\text{sys}}(\hat{a}) + \hat{H}_{\text{ext}}(\hat{b}_n) + \hat{H}_{\text{int}}(\hat{a}, \hat{b}_n)$$

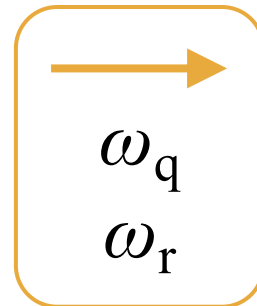
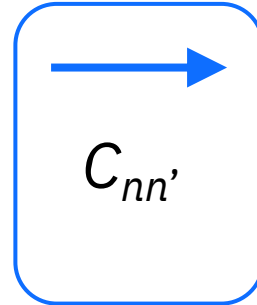


Lumped-oscillator model (LOM)

Quasi-static solver (Q3D)

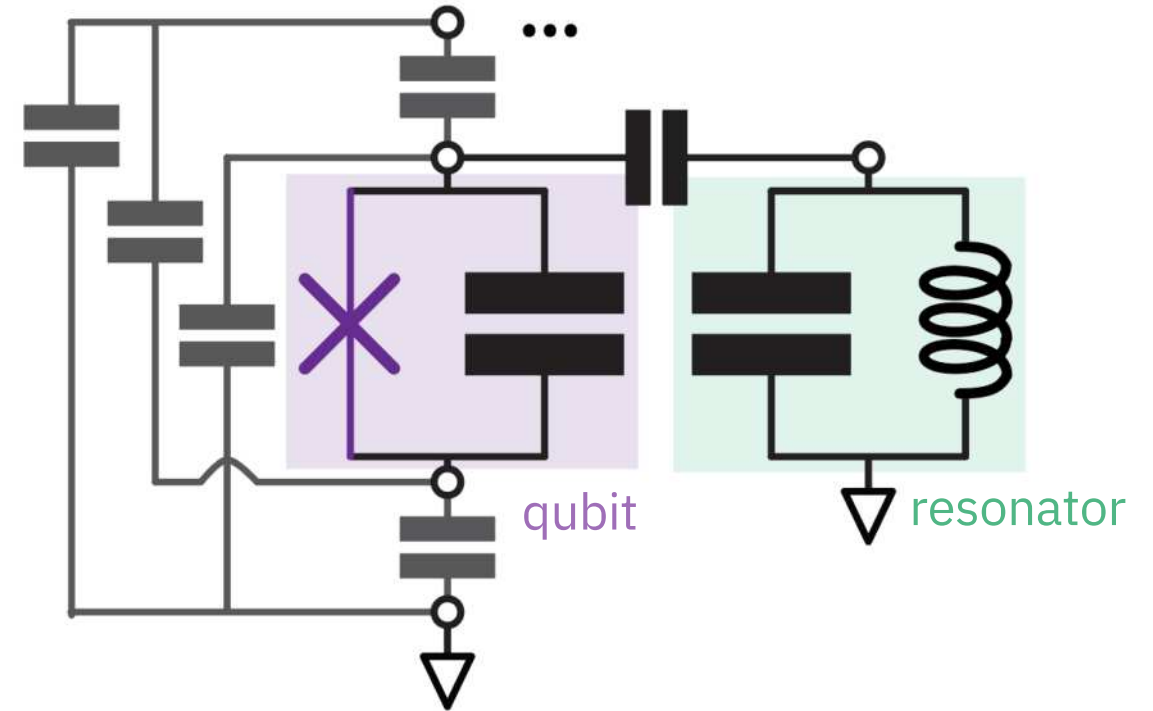


Provide



approximate
not suitable for 3D
some distributed effects beyond capture

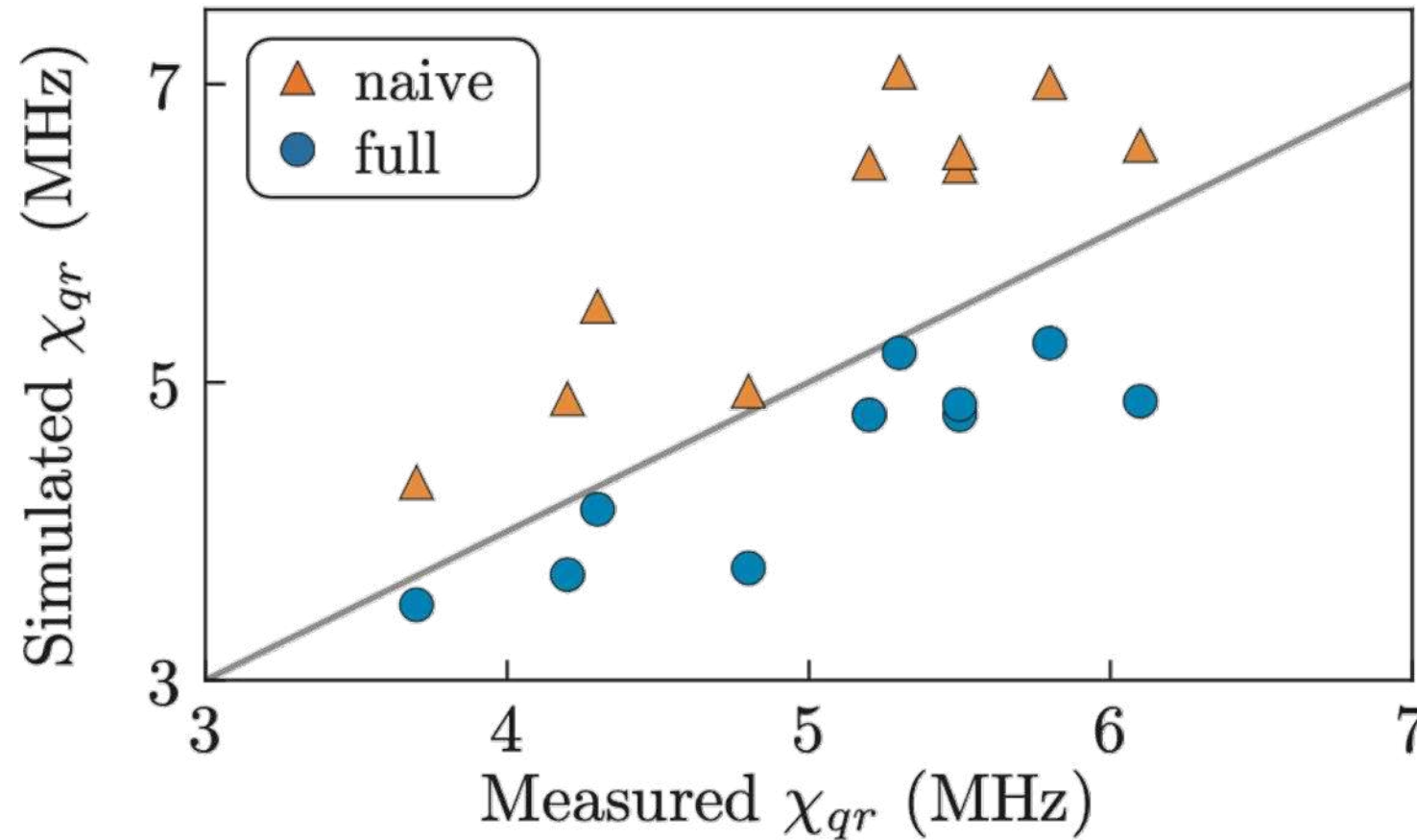
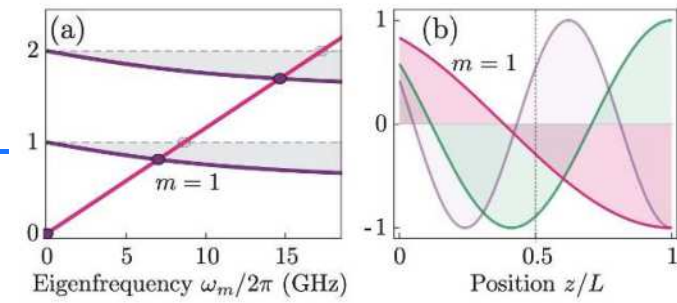
Equivalent lumped model



Minev & Gambetta, in prep (2020)

Yurke & Denker (1984), Devoret (1997),
Burkard et al. (2004), Koch et al. (2007), ...

Improved precision and agreement



Non-linear interaction with EPR

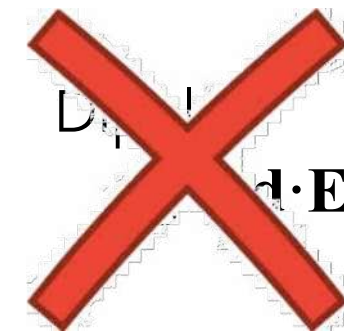
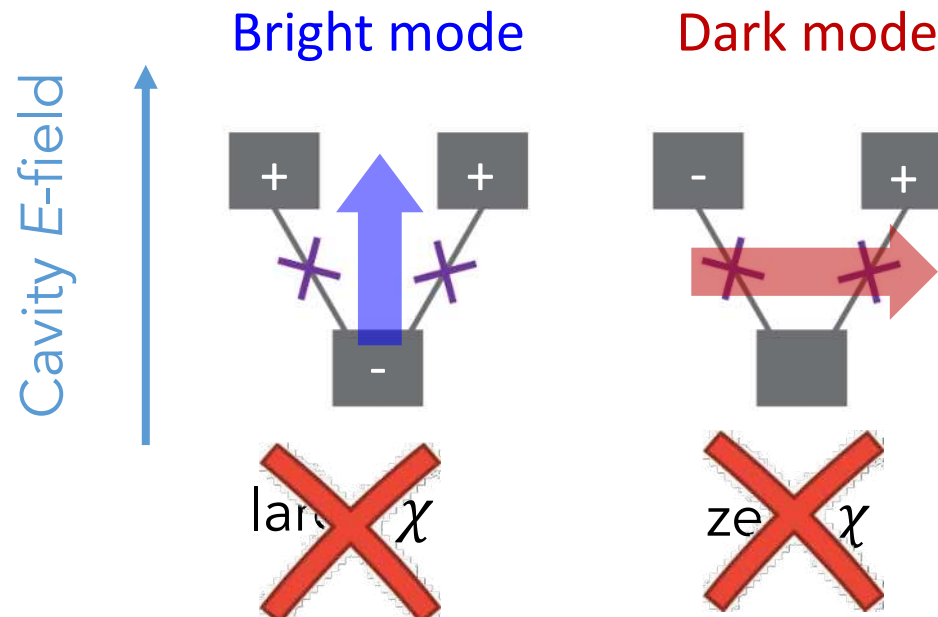
Non-linear interactions in terms of energy overlap

$$\chi_{mn} = \sum_{j=1}^J \frac{\hbar \omega_m \omega_n}{4E_j} p_{mj} p_{nj}$$

Energy participation overlap

(only variable subject to significant variation)

Intuition with dipole and EPR



same p_{mj} for both
 χ s are **equal**

For device, see

Gambetta *et al.*, PRL (2011), Srinivasan *et al.*, PRL (2011), Dumur *et al.*, PRB (2015), Zhang *et al.*, Nature JQI (2017), Mineev *et al.*, arXiv (2018)

Physical insights with EPR

Non-linear interactions in terms of energy overlap

$$\chi_{mn} = \sum_{j=1}^J \frac{\hbar \omega_m \omega_n}{4E_j} p_{mj} p_{nj}$$

Energy participation overlap
(only variable subject to significant variation)

Fundamental constraints

What circuits/parameters can and cannot be realized?

$$\sum_{m=1}^M p_{mj} = 1 \quad 0 \leq \sum_{j=1}^J p_{mj} \leq 1 \quad \sum_{m=1}^M s_{mj} s_{mj'} \sqrt{p_{mj} p_{mj'}} = 0$$

Arbitrary strength non-linear interaction can be calculated

Charge dispersion

$$\omega_0 = 6.50 \text{ GHz}$$

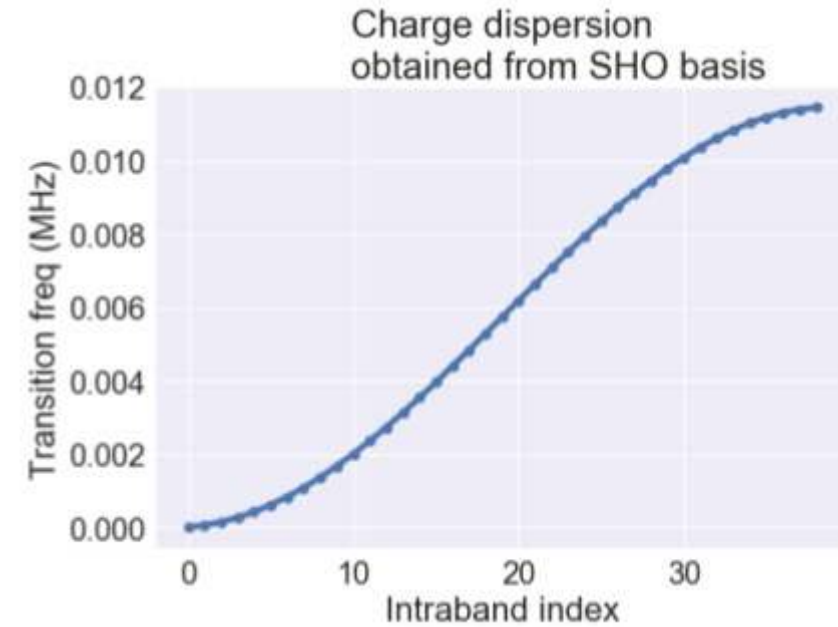
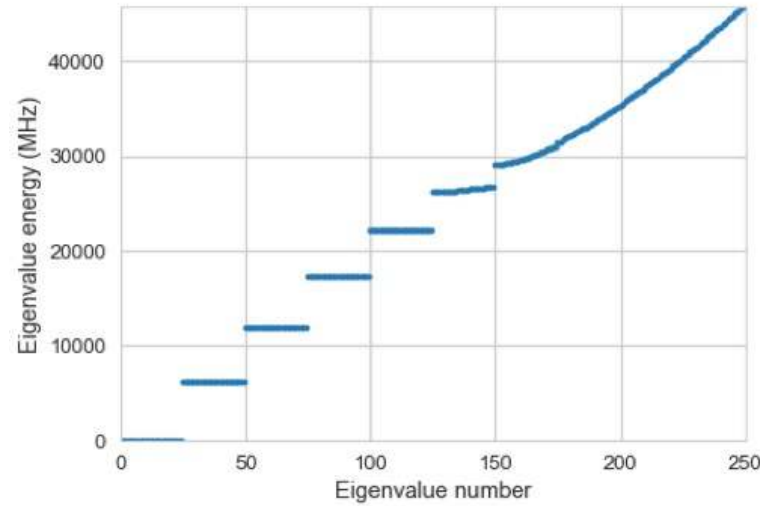
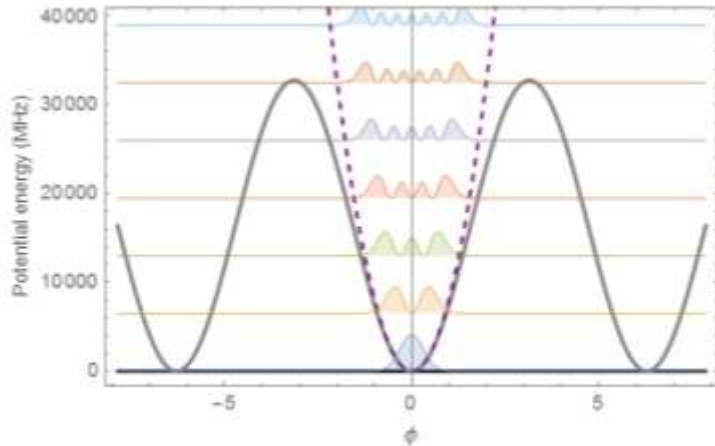
$$\alpha^{(4)} = 322.00 \text{ MHz}$$

$$\phi_{\text{ZPF}} = 0.45$$

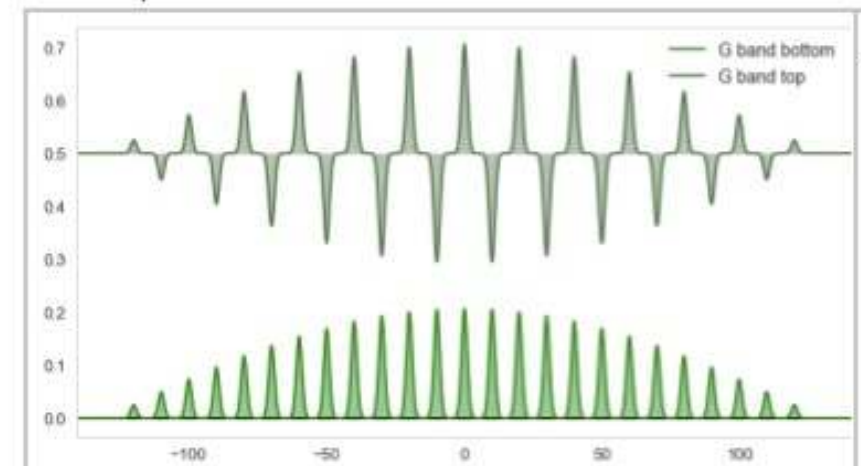
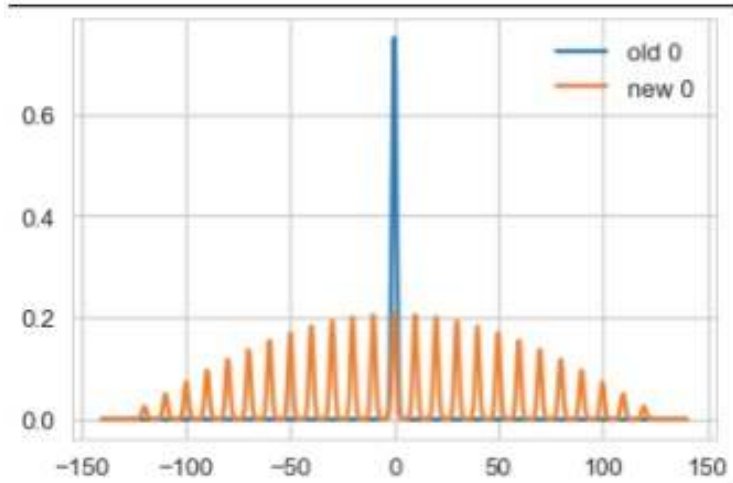
$$E_J = 16.40 \text{ GHz}$$

$$E_C = 0.32 \text{ GHz} \quad (E_J/E_C = 50.90)$$

$$\text{Dispersion} = -12.90 \text{ KHz}$$



$|\psi|^2$



ϕ